

31. USB 2.0 HS Function Module (USBf)

31.1 Overview

The USB module of this LSI is a dual-role device that has USB 2.0 host and function modules.

This LSI has a single USB port for two functions for host and function. These functions can be switched by the common control register (COMMCTRL).

Since operation as a host controller or a function controller are exclusive of each other, dynamic switching between the types of operation is not possible.

This section describes function controller operation.

The USBf module supports high-speed transfer and full-speed transfer defined by the USB 2.0 specification.

This LSI supports all transfer types that are defined in the USB specification. It incorporates an 8 KB buffer memory for data transfer and can use up to 10 pipes. Pipes 1 to 9 can be assigned an arbitrary endpoint number according to the user system.

Table 31.1 Specifications of the USB Module

Item	Description
For High-Speed USB	Built-in USB function controller
For all types of USB transfer	- Support for all types of USB transfer: - Control transfer - Bulk transfer - Interrupt transfer (high-bandwidth transfers not supported) - Isochronous transfer (high-bandwidth transfers not supported)
Bus interface	- Two DMA interfaces available - The dedicated AHB master interface for each DMA channel - High-speed data transfer for access to the internal FIFOs at 60 MB/second (when the bus width is 32 bits)
Pipe configuration	8 KB of buffer memory for USB communications - Up to 10 pipes can be selected (including the default control pipe) - Programmable pipe configuration - Any endpoint number can be assigned to pipes 1 to 9 - Transfer conditions that can be set for each pipe are listed below. - Pipe 0: Control transfer, single buffer fixed at 64 bytes - Pipes 1 and 2: Bulk transfer or isochronous transfer can be selected, continuous transfer mode, programmable buffer size (specifiable as up to 2 KB, double buffer is also specifiable) - Pipes 3 to 5: Bulk transfer, continuous transfer mode, programmable buffer size (specifiable as up to 2 KB, double buffer is also specifiable) - Pipes 6 to 9: Interrupt transfer, single buffer fixed at 64 bytes
Features of function controller operation	- High-speed (480 Mbps) and full-speed transfer (12 Mbps) are supported - Automatic recognition of high-speed operation or full-speed operation based on automatic response to the reset handshake - Control transfer stage monitoring - Monitoring of device state - Automatic response to SET_ADDRESS requests - NAK response interrupt function (NRDY) - SOF interpolation - L1 state is not supported
Other functions	- Judging the end of transfer on the basis of a transaction count - Adjustable timing for BRDY interrupt event notification (BFRE) - Automatic clearing of the buffer memory after the data for the pipe specified in the Dx FIFO port has been read (DCLRM) - NAK setting for response PID generated by end of transfer (SHTNAK)

Figure 31.1 is a block diagram of the USB module.

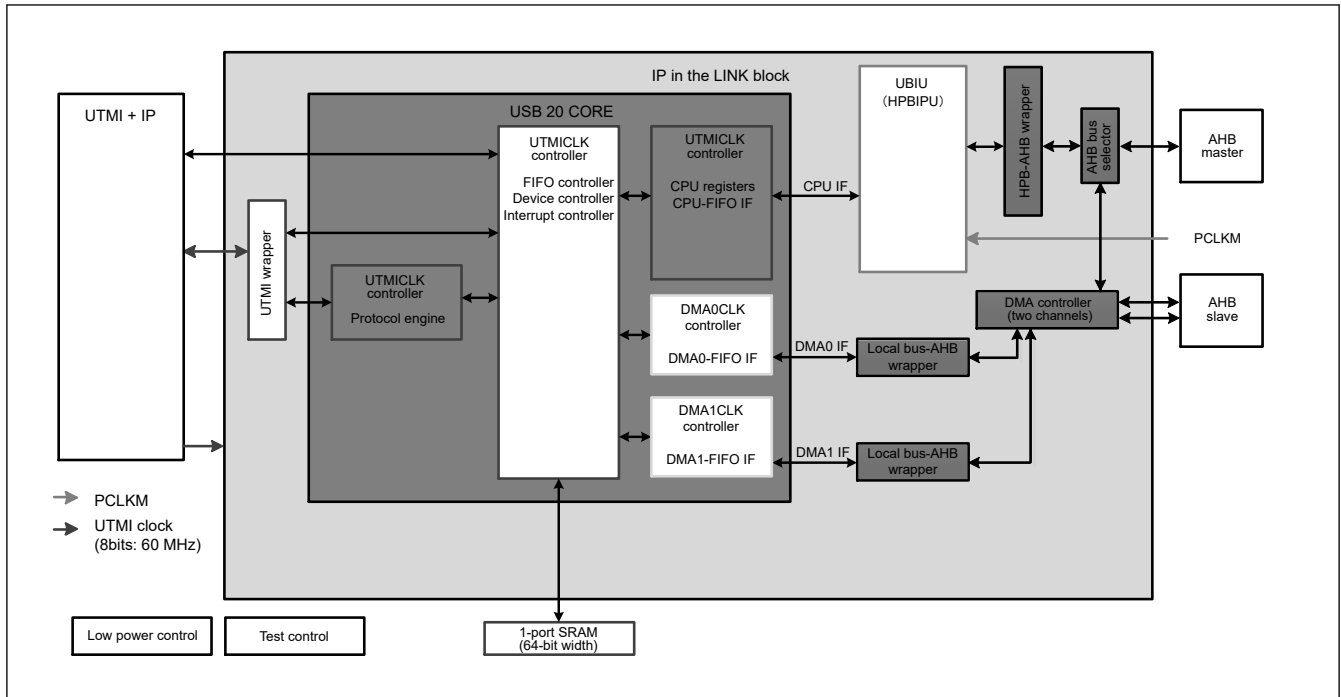


Figure 31.1 Block diagram of the USB module

Data transfer between this controller and the host controller connected to the USB bus uses buffer memory assigned to each pipe. For communication between this controller and the host controller, this controller converts the data stored in buffer memory into USB data packets and outputs them to the USB bus serially, and also inputs data packets from the USB bus and stores them in buffer memory.

31.1.1 Identification of the USB Transfer Speed

The hardware can automatically identify the USB transfer speed.

31.1.2 Bus Interface

(1) How to access the FIFO buffer memory

The following two types of access are available for the FIFO buffer memory for USB data transfers. To read from or write to the FIFO buffer memory, access (read or write) the FIFO ports from the CPU (DMAC).

Two channels of DMAC are provided.

1. CPU access
Specify a FIFO port address and write data to the FIFO buffer memory or read data from the FIFO buffer memory.
2. DMA access
Specify the dedicated DMAC and write data to the FIFO buffer memory or read data from the FIFO buffer memory.

USB data transfer is done in little endian. The byte endian swap function is available for FIFO port access; for 16 bit or 32 bit access, the endian can be switched through register settings.

31.1.3 USB Event

This controller sends an interrupt to the ICU to notify an event in USB operation.

Sending interrupts for notification can be enabled or disabled separately for each interrupt type and source through software settings.

31.1.4 USB Data Transfer

This controller performs all types of USB data transfer: control transfer, bulk transfer, interrupt transfer, and isochronous transfer. The following numbers of pipes are available for each transfer type.

- One pipe dedicated to control transfer
- Four pipes dedicated to interrupt transfer
- Three pipes dedicated to bulk transfer
- Two pipes selectively used for bulk transfer or isochronous transfer

Each pipe should be set up as necessary for the desired USB transfer in accordance with the user system; for example, transfer type, endpoint number, or maximum packet size.

This controller has an 8 KB buffer memory. For pipes dedicated for bulk transfer or selective pipes for bulk transfer or isochronous transfer, buffer memory assignment, buffer operating mode setting, or other necessary settings should be made in accordance with the user system. By setting the buffer operating mode, such as double buffer structure or continuous data packet transfer mode, fast data transfer is achieved with fewer interrupts.

CPU and DMA controller access to the buffer memory is made through three FIFO port registers.

31.1.5 DMA Transfer

This controller provides two channels of the DMA interface having the following functions.

- DMA transaction mode: Fetching in both register and link modes are supported
- Transfer size: A transfer size from 1 to 128 bytes can be selected separately for the transfer source and transfer destination
- Skip (scatter/gather) function: The access size and skip size can be specified separately for the transfer source and destination
- Suspend function: A running DMA transaction can be suspended temporarily
- Interval function: The interval of DMA transfers can be specified to control the bus occupancy

31.2 Register Descriptions

31.2.1 System Configuration Control

31.2.1.1 SYSCFG0 : System Configuration Control Register 0

Base address: USBF = 0x8020_1000

Offset address: 0x00

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	CNEN	HSE	—	DRPD	DPRP U	—	—	—	USBE
Value after reset:	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0

Bit	Symbol	Function	R/W
0	USBE	USB Block Operation Enable Enables or disables operation of the USB block. 0: USB block operation is disabled. 1: USB block operation is enabled.	R/W
3:1	—	These bits are read as 0. The write value should be 0.	R/W
4	DPRPU	D+ Line Resistor Control Enables or disables pulling up the D+ line in function controller operation. 0: Pulling up the line is disabled. 1: Pulling up the line is enabled.	R/W
5	DRPD	D+/D- Line Resistor Control Always set 0 at initialization.	R/W
6	—	This bit is read as 0. The write value should be 0.	R/W
7	HSE	High-Speed Operation Enable Enables or disables high-speed operation. 0: High-speed operation is disabled (full-speed). 1: High-speed operation is enabled (the controller detects the communication speed)	R/W

Bit	Symbol	Function	R/W
8	CNEN	Single-End Receiver Operation Enable Enables or disables the single-end receiver to operate. 0: Single-end receiver operation is disabled. 1: Single-end receiver operation is enabled.	R/W
15:9	—	These bits are read as 0. The write value should be 0.	R/W

Note: When a USB bus reset is detected, this register value is undefined.

Writing to this register is possible even while the UTMI clock (output of the clock signal from the PLL in the USB) is stopped. Note however that any values that are set while the UTMI clock is stopped are reflected after oscillation of the UTMI clock is restarted.

USB_E bit (USB Block Operation Enable)

The setting of the USB_E bit enables or disables operation of the USB block.

When the USB_E bit is modified from 1 to 0, this controller initializes the bits listed in [Table 31.2](#).

Table 31.2 Registers initialized by writing USB_E = 0

Register Name	Bit Name
SYSSTS0	LNST
DVSTCTR0	RHST
INTSTS0	DVSQ
USBADDR	USBADDR
USBREQ	bRequest bmRequestType
USBVAL	wValue
USBINDX	wIndex
USBLENG	wLength

The value of this bit must be changed when SUSPM = 1 and after the UTMI clock oscillation is started.

DRPD and DPRPU bit (D+/D- Line Resistor Control)

Settings related to USB data bus resistance are given in [Table 31.3](#), USB Data Bus Resistance Control. Use the DRPD and DPRPU bits to select USB data bus resistance.

Table 31.3 USB data bus resistance control

Settings		USB data bus resistance control		
DRPD	DPRPU	D- Line	D+ Line	Remarks
0	0	Open	Open	
0	1	Open	Pull-up	Make these settings when the module is used as a function controller.
1	0	Pull-down	Pull-down	Setting prohibited
1	1	Pull-down	Pull-up	Setting prohibited

D+ pull-up resistance control for function controller operation (DPRPU)

When this bit set to 1, this controller pulls up the D+ line to 3.3V and can notify the USB host that the function module is attached.

The controller cancels pulling-up of the D+ line by modifying this bit from 1 to 0, and the state for the USB host can be shown as detached.

HSE bit (High-Speed Operation Enable)

Setting this bit to 1 enables high-speed operation. Setting the HSE bit to 1 lets this controller perform high-speed or fullspeed operation based on the results of reset handshake.

Setting the HSE bit to 0 lets this controller perform full-speed operation.

On the other hand, setting the HSE bit to 1 lets this controller perform the reset handshake protocol and then perform high-speed or full-speed operation automatically according to the results.

This bit can be modified when DPRPU = 0.

CNEN bit (Single-End Receiver Operation Enable)

Setting this bit to 1 enables the single-end receiver to operate. This bit is used to prevent inrush current damage due to the floating state of the single-end receiver when it is detached. Furthermore, this bit makes it possible to see the LNST bits.

Set this bit to 1 when VBUS is detected by a VBUS interrupt, and clear this bit to 0 when VBUS is removed.

31.2.1.2 SYSCFG1 : System Configuration Control Register 1

Base address: USBF = 0x8020_1000

Offset address: 0x02

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	AWAIT[5:0]						—	—	BWAIT[5:0]					
Value after reset:	0	0	0	0	1	1	1	1	0	0	0	0	1	1	1	1

Bit	Symbol	Function	R/W
5:0	BWAIT[5:0]	CPU Bus Access Wait Specification Specify the number of wait cycles for access to this controller. 0x00: 0 wait cycles (2 access cycles) ⋮ 0x02: 2 wait cycles (4 access cycles) ⋮ 0x04: 4 wait cycles (6 access cycles) ⋮ 0x0F: 15 wait cycles (17 access cycles) (value after reset) ⋮ 0x3F: 63 wait cycles (65 access cycles)	R/W
7:6	—	These bits are read as 0. The write value should be 0.	R/W
13:8	AWAIT[5:0]	AHB-DMA Bridge Bus Access Wait Specification Specify the number of wait cycles for access to this controller. 0x00: 0 wait cycles (2 access cycles) ⋮ 0x02: 2 wait cycles (4 access cycles) ⋮ 0x04: 4 wait cycles (6 access cycles) ⋮ 0x0F: 15 wait cycles (17 access cycles) (value after reset) ⋮ 0x3F: 63 wait cycles (65 access cycles)	R/W
15:14	—	These bits are read as 0. The write value should be 0.	R/W

Note: When a USB bus reset is detected, this register value is undefined.

BWAIT[5:0] bits (CPU Bus Access Wait Specification)

For continuous access to the registers at addresses of this controller beginning at 0x80201004, at least 67 ns must be secured.

To satisfy this restriction, control wait cycles using the frequency of Bus clock (PCLKM). The value after a reset is the 17 clock cycles, so select an appropriate value no greater than this.

This setting is the same as that for waiting in access to a FIFO port register. The maximum speeds of access to the FIFO ports are as follows:

MBW = 10b (32-bit width): Max 60 MB/sec

MBW = 01b (16-bit width): Max 30 MB/sec

MBW = 00b (8-bit width): Max 15 MB/sec

AWAIT[5:0] bits (AHB-DMA Bridge Bus Access Wait Specification)

These bits specify the number of wait cycles for access to the AHB-DFIFO0/AHB-DFIFO1 bus.

Restrictions for access cycles are the same as those for the CPU bus access wait specification bits (BWAIT).

31.2.1.3 SYSSTS0 : System Configuration Status Register

Base address: USBF = 0x8020_1000

Offset address: 0x04

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LNST[1:0]	
Value after reset:	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	LNST[1:0]	USB Data Line Status Monitor Indicate the state of the USB line.	R
15:2	—	These bits are read as 0.	R

Note: When a USB bus reset is detected, this register value is undefined.

LNST[1:0] bits (USB Data Line Status Monitor)

The states of USB data bus line of this controller is shown in [Table 31.4](#). The controller monitors the states of USB data bus lines (D+ and D- lines) in the LNST bits of the SYSSTS0 register.

Reference to the LNST bits must be made following attach processing (DPRPU = 1) after USBE has been set to 1.

Table 31.4 States of USB data bus line

LNST[1]	LNST[0]	Full-speed operations	High-speed operations	Chirp operations* ¹
0	0	SE0	Squelch* ²	Squelch* ²
0	1	J state	Unsquench* ³	Chirp J* ⁴
1	0	K state	Invalid	Chirp K* ⁵
1	1	SE1	Invalid	Invalid

Note 1. Chirp: Reset handshake protocol being executed in high-speed operations enabled state (HSE = 1)

Note 2. Squelch: SE0 or idle state

Note 3. Unsquench: High-speed J state or high-speed K state

Note 4. Chirp J: Chirp J-state

Note 5. Chirp K: Chirp K-state

31.2.2 USB Signal Control**31.2.2.1 DVSTCTR0 : Device State Control Register 0**

Base address: USBF = 0x8020_1000

Offset address: 0x08

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	WKUP	—	—	—	—	—	RHST[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	RHST[2:0]	Reset Handshake Indicate the state of reset handshake.	R
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
8	WKUP	Wakeup Output Enables or disables the remote wakeup (resume signal output). Only 1 can be written to this bit. 0: Remote wakeup signal is not output. 1: Remote wakeup signal is output.	R/W
15:9	—	These bits are read as 0. The write value should be 0.	R/W

Note: When a USB bus reset is detected, The WKUP is set to 0 and other bits are undefined.

RHST[2:0] bits (Reset Handshake)

These bits indicate the result of reset handshake. Table 31.5 lists the results of reset handshake.

Table 31.5 State of Reset Handshake

Bus state	RHST bit value
Powered or no connection	000b
Reset handshake in progress	100b
Full-speed connection	010b
High-speed connection	011b

When the HSE bit is set to 1, the RHST bits indicate 100b when this controller detects a USB bus reset. Then, these bits indicate 011b when this controller outputs Chirp-K and detects Chirp-JK from the USB host three times. If the connection speed is not fixed to high speed within 2.5 ms after Chirp-K output, these bits indicate 010b.

When the HSE bit is set to 0, these bits indicate 010b when the controller detects a USB bus reset.

A DVST interrupt is generated when the value of the RHST bits is 010b or 011b after the USB bus reset is detected by the controller.

WKUP bit (Wakeup Output)

With the WKUP bit set to 1, this controller outputs the remote wakeup signal to the USB bus.

This controller controls the output time of a remote wakeup signal. When this bit is set to 1 by software, the controller clears this bit to 0 after outputting the 10 ms K-state.

According to the USB Specifications, the USB bus idle state must be kept for 5 ms or longer before a remote wakeup signal is sent. If this controller writes 1 to this bit right after detection of the suspended state, the K-state will be output after 2 ms.

Do not write 1 to this bit unless the device is in the suspended state (bits DVSQ = 1xxb) and the USB host enables the remote wakeup signal.

When this bit is set to 1, the internal clock must not be stopped even in the suspended state (write 1 to this bit while the SUSPM bit = 1).

31.2.3 Test Mode

31.2.3.1 TESTMODE : USB Test Mode Register

Base address: USBF = 0x8020_1000

Offset address: 0x0C

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	UTST[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	UTST[3:0]	Test Mode *See the detailed description below.	R/W

Bit	Symbol	Function	R/W
15:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: When a USB bus reset is detected, this register value is undefined.

UTST[3:0] bits (Test Mode)

Writing values to these bits allows this controller to output a USB test signal during high-speed operations. [Table 31.6](#) shows the test mode operations of the controller.

Table 31.6 Test mode operations

Test mode	UTST settings
Normal operation	0000b
Test_J	0001b
Test_K	0010b
Test_SE0_NAK	0011b
Test_Packet	0100b
Test_Force_Enable	—
Reserved	0101b to 0111b

Write to these bits according to the SetFeature request from the USB host during high-speed communication.

This controller does not enter the suspended state while these bits are 0001b to 0100b.

To perform normal USB communications after the test mode is set, apply a hardware reset.

31.2.4 FIFO Ports

31.2.4.1 CFIFO, DnFIFO : FIFO Port Register (n = 0, 1)

Base address: USBF = 0x8020_1000

Offset address: 0x014 (CFIFO/CFIFOL/CFIFOLL)
 0x016 (CFIFOH)
 0x017 (CFIFOHH)
 0x018 + 0x4 × n (DnFIFO/DnFIFOL/DnFIFOLL)
 0x01A + 0x4 × n (DnFIFOH)
 0x01B + 0x4 × n (DnFIFOHH)

Bit position: 31

0

Bit field:

FIFOPORT[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	FIFOPORT[31:0]	FIFO Port Accessing these bits allows reading the received data from the FIFO buffer or writing the transmit data to the FIFO buffer.	R/W

Note: When a USB bus reset is detected, this register value is undefined.

The transmission/reception buffer of this controller is made up of a FIFO structure (FIFO buffer). Use the FIFO port register for access to the FIFO buffer. There are three FIFO ports: CFIFO, D0FIFO, and D1FIFO ports. Also, there is a DxFIFO continuous transfer port for continuous transfer. Each FIFO port consists of a port register (CFIFO, D0FIFO, or D1FIFO) that handles reading of data from the FIFO buffer memory and writing of data to the FIFO buffer memory, a port select register (CFIFOSEL, D0FIFOSEL, or D1FIFOSEL) that is used to select the pipe assigned to the FIFO port, and a port control register (CFIFOCTR, D0FIFOCTR, or D1FIFOCTR).

The FIFO ports have the following restrictions.

1. Access to an FIFO buffer for DCP should be through the CFIFO port.

- Access to an FIFO buffer by DMA transfer when DFACC = 00b (cycle stealing mode) should be through the DxFIFO port.
- Access to an FIFO buffer by DMA transfer when DFACC = 01b or 10b (16 byte or 32 byte continuous access mode) should be through the DxFIFO continuous transfer port.
- The DxFIFO ports can also be accessed by the CPU.
- When using functions specific to the FIFO port, the pipe number (selected pipe) specified by the CURPIPE bits cannot be changed (when the DMA transfer function is used, etc.).
- Registers containing the FIFO port do not affect the other FIFO ports.
- Do not assign the same pipe to separate FIFO ports.
- In the FIFO buffer state, there are two types of access rights: one assigned to the CPU, and the other to SIE. Access from the CPU is not possible when SIE has the rights to access the FIFO buffer.

FIFOPORT[31:0] bits (FIFO Port)

This controller accesses the FIFO buffer assigned to the pipe number written to the CURPIPE bits of the corresponding select registers (CFIFOSEL, D0FIFOSEL, and D1FIFOSEL) through access to any of these registers by software.

This register can only be accessed when the FRDY bit of each control register (CFIFOCTR, D0FIFOCTR, or D1FIFOCTR) indicates 1 (or while DnFIFOSEL.DREQE = 1).

The valid bits of this register depend on the settings of the corresponding MBW and BIGEND bits.

Table 31.7 to Table 31.9 list the valid bits.

Table 31.7 Endian operation in 32 bit access (MBW = 10)

BIGEND	b31 to b24	b23 to b16	b15 to b8	b7 to b0
0	N + 3 address	N + 2 address	N + 1 address	N + 0 address
1	N + 0 address	N + 1 address	N + 2 address	N + 3 address

Table 31.8 Endian operation in 16 bit access (MBW = 01)

BIGEND	b31 to b24	b23 to b16	b15 to b8	b7 to b0
0	Writing: Invalid Reading: Prohibited*1		N + 1 address	N + 0 address
1	N + 0 address	N + 1 address	Writing: Invalid Reading: Prohibited*1	

Note 1. Reading data from the invalid bits in word or byte units is prohibited.

Table 31.9 Endian operation in 8 bit Access (MBW = 00)

BIGEND	b31 to b24	b23 to b16	b15 to b8	b7 to b0
0	Writing: Invalid Reading: Invalid*1			N + 0 address
1	N + 0 address	Writing: Invalid Reading: Prohibited*1		

Note 1. Reading data from the invalid bits in word or byte units is prohibited.

31.2.4.2 CFIFOSEL : CFIFO Port Select Register

Base address: USBF = 0x8020_1000

Offset address: 0x20

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	RCNT	REW	—	—	MBW[1:0]	—	BIGEND	—	—	ISEL	—	CURPIPE[3:0]				

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
3:0	CURPIPE[3:0]	FIFO Port Access Pipe Specification Specify the pipe number to access the CFIFO port. 0x0: DCP 0x1: Pipe 1 0x2: Pipe 2 ⋮ 0x8: Pipe 8 0x9: Pipe 9 Others: Setting prohibited	R/W
4	—	This bit is read as 0. The write value should be 0.	R/W
5	ISEL	FIFO Port Access Direction when DCP is Selected Specifies access direction of the FIFO port when DCP is selected in CURPIPE bits. 0: Reading from the buffer memory 1: Writing to the buffer memory	R/W
7:6	—	These bits are read as 0. The write value should be 0.	R/W
8	BIGEND	FIFO Port Endian Control Specifies the CFIFO port byte endian. 0: Little endian 1: Big endian	R/W
9	—	This bit is read as 0. The write value should be 0.	R/W
11:10	MBW[1:0]	CFIFO Port Access Bit Width Specify the bit width for accessing the CFIFO port. 0 0: 8 bit width 0 1: 16 bit width 1 0: 32 bit width 1 1: Setting prohibited	R/W
13:12	—	These bits are read as 0. The write value should be 0.	R/W
14	REW	Buffer Pointer Rewind Specifies 1 when rewinding the buffer pointer. This bit is read as 0. 0: The buffer pointer is not rewind. 1: The buffer pointer is rewind.	R/W
15	RCNT	Read Count Mode Specifies the read mode for the value in the DTLN bits in CFIFOCTR. 0: The DTLN bits are cleared when all the received data has been read from the CFIFO. 1: The DTLN bits are decremented each time the received data is read from the CFIFO.	R/W

Note: When a USB bus reset is detected, this register value is undefined.

CURPIPE[3:0] bits (FIFO Port Access Pipe Specification)

Write the pipe number for the data to be read or written through the CFIFO port.

When modifying this bit field, write this bit first and then read this bit. Only when the written value matches the read one, proceed to the next process.

Do not specify the same pipe in the CURPIPE bits of the CFIFOSEL, D0FIFOSEL, and D1FIFOSEL registers.

Even if the setting of these bits is modified during access to the FIFO buffer, the state of the FIFO buffer is retained, with continued access proceeding after these bits are re-set to the value before the modification.

ISEL bit (FIFO Port Access Direction when DCP is Selected)

To change this bit when the specified pipe is DCP, first write the data to this bit and then read it. Proceed to the next process after checking if the written value matches with the read value.

When the settings of this bit are modified during access to the FIFO buffer, access up to then is saved. Access to the buffer can be continued after rewriting the settings.

Settings in this bit should be made at the same as that of the CURPIPE bits.

BIGEND bit (FIFO Port Endian Control)

In this bit, set the byte endian for the CFIFO port.

For details, see the description of FIFO Port Bits (CFIFO, D0FIFO, and D1FIFO).

MBW[1:0] bits (CFIFO Port Access Bit Width)

In this bit field, set the bit width for accessing the CFIFO port.

When the pipe specified in the CURPIPE bits is in the receiving direction, if reading is started after setting this bit field, do not modify the value of the MBW bits until all data is read.

When the specified pipe is in the receiving direction, modify the value of the CURPIPE bit once and then set the CURPIPE and MBW bits at the same time.

For the procedure for changing the CURPIPE bit, see the description for the same.

When the specified pipe is in the transmitting direction, to switch the bit width from 8 bits to 16 or 32 bits or from 16 bits to 32 bits is not allowed while writing to the buffer memory is in progress.

With the 16 bit or 32 bit width setting, writing to odd bytes is possible by exercising byte access control.

REW bit (Buffer Pointer Rewind)

When the selected pipe is in the receiving direction, setting the REW bit to 1 while the FIFO buffer is being read allows re-reading the FIFO buffer from the first data (in double-buffer operation, re-reading the currently-read FIFO buffer plane from the first data is allowed).

Do not set REW to 1 simultaneously with modifying the CURPIPE bits. Before setting REW to 1, be sure to check that FRDY is 1.

To re-write to the FIFO buffer again from the first data for the pipe in the transmitting direction, use the BCLR bit.

RCNT bit (Read Count Mode)

When 0 is written to this bit, if all received data of the FIFO buffer assigned to the pipe specified in the CURPIPE bits (selected pipe) is read (when the data is read on one side of a double buffer), this controller clears the DTLN bits in the CFIFOCTR register to 0.

When 1 is written to this bit, the controller decrements the value of the DTLN bits in the CFIFOCTR register whenever the received data is read from the FIFO buffer assigned to the specified pipe.

31.2.4.3 DnFIFOSEL : DnFIFO Port Select Register (n = 0, 1)

Base address: USBF = 0x8020_1000

Offset address: 0x028 + 0x4 × n

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	RCNT	REW	DCLR M	DREQ E	MBW[1:0]	—	BIGEN D	—	—	—	—	—	CURPIPE[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	CURPIPE[3:0]	FIFO Port Access Pipe Specification 0x0: Not specified 0x1: Pipe 1 0x2: Pipe 2 : 0x8: Pipe 8 0x9: Pipe 9 Others: Setting prohibited	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W
8	BIGEND	FIFO Port Endian Control Specifies the DnFIFO port byte endian. 0: Little endian 1: Big endian	R/W

Bit	Symbol	Function	R/W
9	—	This bit is read as 0. The write value should be 0.	R/W
11:10	MBW[1:0]	FIFO Port Access Bit Width Specify the bit width for accessing the FIFO port. 0 0: 8 bit width 0 1: 16 bit width 1 0: 32 bit width 1 1: Setting prohibited	R/W
12	DREQE	DMA Request Enable This bit is used to notify for DMAC. 0: DMA request is disabled 1: DMA request is enabled	R/W
13	DCLRM	Auto Buffer Memory Clear Mode after Specified Pipe Data is Read Enables or disables automatic clearing of the buffer memory after data has been read out using the selected pipe. 0: Auto buffer clear mode is disabled 1: Auto buffer clear mode is enabled	R/W
14	REW	Buffer Pointer Rewind Specifies 1 when rewinding the buffer pointer. This bit is read as 0. 0: The buffer pointer is not rewind 1: The buffer pointer is rewind	R/W
15	RCNT	Read Count Mode Specifies the read mode for the value in the DTLN bits in Dx_FIFOCR. 0: The DTLN bits are cleared when all the received data has been read. 1: The DTLN bits are decremented each time the received data is read.	R/W

Note: When a USB bus reset is detected, this register value is undefined.

CURPIPE[3:0] bits (FIFO Port Access Pipe Specification)

Write the pipe number for the data to be read or written through the DxFIFO port.

When modifying this bit field, first write a value to this bit field and then read it. Check if the write value matches the read value and then proceed to the next process.

Do not specify the same pipe in the CURPIPE bits of the CFIFOSEL, D0FIFOSEL, and D1FIFOSEL registers.

Even if the setting of these bits is modified during access to the FIFO buffer, the state of the FIFO buffer is retained, with continued access proceeding after these bits are re-set to the value before the modification.

BIGEND bit (FIFO Port Endian Control)

Write the byte endian for the DxFIFO port in this bit.

For details, see the description of FIFO Port Bits (CFIFO, D0FIFO, and D1FIFO).

MBW[1:0] bits (FIFO Port Access Bit Width)

Write the bit width for accessing the DxFIFO port in this bit.

When the pipe specified in the CURPIPE bits is in the receiving direction, if reading is started after setting this bit field, do not modify the value of the MBW bits until all data is read.

When the specified pipe is in the receiving direction, set the CURPIPE and MBW bits at the same time.

For the procedure for changing the CURPIPE bit, see the description for the same.

For details, see the description of CURPIPE[3:0] Bits (FIFO Port Access Pipe Specification Bits).

DREQE bit (DMA Request Enable)

Use this bit to enable or disable DMA request.

When enabling DMA request, set this bit to 1 after setting the CURPIPE bits.

When modifying the setting of the CURPIPE bits, do so after setting this bit to 0.

DCLRM bit (Auto Buffer Memory Clear Mode after Specified Pipe Data is Read)

After reading data in the specified pipe, disable or enable automatic clearing of the FIFO buffer. When 1 is written to this bit, this controller sets the BCLR bit to 1 for a FIFO buffer if a zero-length packet is received when the FIFO buffer assigned to the specified pipe is empty, or when a short packet is received and the data is completely read while BFRE = 1.

When using this controller with the BRDYM bit set to 1, be sure to set this bit to 0.

REW bit (Buffer Pointer Rewind)

When the selected pipe is in the receiving direction, setting the REW bit to 1 while the FIFO buffer is being read allows re-reading the FIFO buffer from the first data (in double-buffer operation, re-reading the currently-read FIFO buffer plane from the first data is allowed).

Do not set the REW bit to 1 simultaneously with modifying the CURPIPE bits. Before setting the REW bit to 1, be sure to check that the FRDY bit is 1.

To re-write to the FIFO buffer again from the first data for the pipe in the transmitting direction, use the BCLR bit.

RCNT bit (Read Count Mode)

When 0 is written to this bit, if all received data of the FIFO buffer assigned to the pipe specified in the CURPIPE bits (selected pipe) is read (when the data is read on one side of a double buffer), this controller clears the DTLN bits in the DxFIFOCTR register to 0.

When 1 is written to this bit, the controller decrements the value of the DTLN bits in the DxFIFOCTR register whenever the received data is read from the FIFO buffer assigned to the specified pipe.

31.2.4.4 CFIFOCTR, DnFIFOCTR : FIFO Port Control Register (n = 0, 1)

Base address: USBF = 0x8020_1000

Offset address: 0x022 (CFIFOCTR)
0x02A + 0x4 × n (DnFIFOCTR)

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	BVAL	BCLR	FRDY	—	DTLN[11:0]											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
11:0	DTLN[11:0]	Receive Data Length Indicate the length of the receive data.	R
12	—	This bit is read as 0. The write value should be 0.	R/W
13	FRDY	FIFO Port Ready Indicates whether the FIFO port can be accessed. 0: FIFO port is not accessible. 1: FIFO port is accessible.	R
14	BCLR	CPU Buffer Clear Specifies 1 to clear the FIFO buffer on the CPU side for the selected pipe. Only 1 can be written to this bit. This bit is read as 0. 0: Invalid 1: Clears the CPU buffer memory on the CPU side.	R/W
15	BVAL	Buffer Memory Valid Flag Specifies 1 when writing has ended in the CPU-side FIFO buffer for the pipe specified in CURPIPE (selected pipe). Only 1 can be written to this bit. 0: Invalid 1: Writing ended	R/W

Note: When a USB bus reset is detected, this register value is undefined.

DTLN[11:0] bits (Receive Data Length)

The DTLN bits indicate the length of the receive data. While the FIFO buffer is being read, the DTLN bits indicate different values depending on the RCNT bit value as described below.

1. When RCNT = 0:
The length of received data is set in these bits, and the value is retained until all received data has been read from a single FIFO buffer plane.
While BFRE is 1, these bits retain the length of the receive data until BCLR is set to 1 even after all data has been read.
2. When RCNT = 1:
This controller decrements the value indicated by these bits each time data is read from the FIFO buffer.
(The value is decremented by 1 when MBW is 00b, by 2 when MBW is 01b, and by 4 when MBW is 10b.)
This controller sets these bits to 0 when all data has been read from one FIFO buffer plane. However, in double-buffer operation, if data has been received in one FIFO buffer plane before all data has been read from the other plane, this controller sets these bits to indicate the length of the receive data in the latter plane when all data has been read from the former plane.
When reading these bits during FIFO buffer reading while RCNT = 1, note that these bits are updated within 150 ns after a read cycle for the FIFO port.

FRDY bit (FIFO Port Ready)

This bit indicates whether the FIFO port can be accessed from the CPU (DMAC).

In the following cases, the controller sets FRDY to 1, but data cannot be read via the FIFO port because there is no data to be read. In these cases, set BCLR to 1, clear the FIFO buffer, and then enable transmission and reception of the next data.

1. A zero-length packet is received when the FIFO buffer assigned to the selected pipe is empty.
2. A short packet is received and the data is completely read while BFRE is 1.

BCLR bit (CPU Buffer Clear)

When this bit is set to 1, this controller clears the FIFO buffer on the CPU for the selected pipe.

When double-buffer operation is set for the FIFO buffer assigned to the selected pipe, this controller clears only one plane of the FIFO buffer even when both planes are read-enabled.

When the selected pipe is the DCP, setting BCLR to 1 allows this controller to clear the FIFO buffer regardless of whether the FIFO buffer is on the CPU side or SIE side. When clearing the buffer on the SIE side, set the PID bits for the DCP to NAK before setting BCLR to 1.

When the selected pipe is not the DCP, writing 1 to this bit should be done while FRDY indicates 1. When checking the FRDY bit after executing BCLR, allow an interval of at least 80 ns after executing BCLR before referencing FRDY.

BVAL bit (Buffer Memory Valid Flag)

When the pipe specified in the CURPIPE bits (selected pipe) is for transmission, write 1 to this bit in the following cases. This controller sets the FIFO buffer from the CPU side to the SIE side, enabling transmission.

1. To transmit the short packet, set this bit to 1 after data has been written.
2. To transmit a zero-length packet, set this bit to 1 before writing data to FIFO.
3. Set this bit to 1 after the number of data bytes has been written for the pipe in continuous transfer mode, where the number is a natural integral multiple of the maximum packet size and less than the buffer size.

When the data of the maximum packet size has been written for the pipe in non-continuous transfer mode, this controller sets this bit to 1 and switches the FIFO buffer from the CPU side to the SIE side, enabling transmission.

When 1 is written to the BVAL and BCLR bits at the same time when the specified pipe is in the transmitting direction, the controller clears the old data and enables the transmission of a zero-length packet.

Writing 1 to this bit should be done while FRDY indicates 1. When checking the FRDY bit after executing BVAL, allow an interval of at least 80 ns after executing BVAL before referencing FRDY.

When the specified pipe is in the receiving direction, do not write 1 to this bit.

31.2.5 Enabling Interrupts

31.2.5.1 INTENB0 : Interrupt Enable Register 0

Base address: USBF = 0x8020_1000

Offset address: 0x30

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	VBSE	RSME	SOFE	DVSE	CTRE	BEMP E	NRDY E	BRDY E	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	—	These bits are read as 0. The write value should be 0.	R/W
8	BRDYE	Buffer Ready Interrupt Enable Enables or disables the USB interrupt output when the BRDY interrupt is detected. 0: Interrupt output disabled 1: Interrupt output enabled	R/W
9	NRDYE	Buffer Not Ready Response Interrupt Enable Enables or disables the USB interrupt output when the NRDY interrupt is detected. 0: Interrupt output disabled 1: Interrupt output enabled	R/W
10	BEMPE	Buffer Empty Interrupt Enable Enables or disables the USB interrupt output when the BEMP interrupt is detected. 0: Interrupt output disabled 1: Interrupt output enabled	R/W
11	CTRE	Control Transfer Stage Transition Interrupt Enable Enables or disables the USB interrupt output when the CTRT interrupt is detected. 0: Interrupt output disabled 1: Interrupt output enabled	R/W
12	DVSE	Device State Transition Interrupt Enable Enables or disables the USB interrupt output when the DVST interrupt is detected. 0: Interrupt output disabled 1: Interrupt output enabled	R/W
13	SOFE	Frame Number Update Interrupt Enable Enables or disables the USB interrupt output when the SOF interrupt is detected. 0: Interrupt output disabled 1: Interrupt output enabled	R/W
14	RSME	Resume Interrupt Enable Enables or disables the USB interrupt output when the RESM interrupt is detected. 0: Interrupt output disabled 1: Interrupt output enabled	R/W
15	VBSE	VBUS Interrupt Enable Enables or disables the USB interrupt output when the VBINT interrupt is detected. 0: Interrupt output disabled 1: Interrupt output enabled	R/W

Note: When a USB bus reset is detected, this register value is undefined.

31.2.5.2 INTENB1 : Interrupt Enable Register 1

Base address: USBF = 0x8020_1000

Offset address: 0x32

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	PDDE TINTE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PDDDETINTE	PDDDETINT Detection Interrupt Enable Enables or disables the USB interrupt output when a PDDDETINT detection interrupt is detected. 0: Interrupt output disabled 1: Interrupt output enabled	R/W
15:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: When a USB bus reset is detected, this register value is undefined.

31.2.5.3 BRDYENB : BRDY Interrupt Enable Register

Base address: USBF = 0x8020_1000

Offset address: 0x36

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	PIPEBRDYE[9:0]									
Value after reset:	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	PIPEBRDYE[9:0]	BRDY Interrupt Enable for Each Pipe Enable or disable the BRDY interrupt for each pipe. 0: Interrupt output disabled 1: Interrupt output enabled	R/W
15:10	—	The read values are undefined. The write value should be 0.	R/W

Note: When a USB bus reset is detected, this register value is undefined.

Note: The bit number corresponds to the pipe number.

PIPEBRDYE[9:0] bits (BRDY Interrupt Enable for Each Pipe)

On detecting the BRDY interrupt for the pipe corresponding to the bit in this register which has been set to 1 by software, this controller sets 1 to the corresponding PIPEBRDY bit in BRDYSTS and the BRDY bit in INTSTS0, and asserts the interrupt signal through USB_FI.

While at least one PIPEBRDY bit in BRDYSTS indicates 1, this controller asserts the interrupt signal through USB_FI when the corresponding interrupt enable bit of this register is changed from 0 to 1 by software.

31.2.5.4 NRDYENB : NRDY Interrupt Enable Register

Base address: USBF = 0x8020_1000

Offset address: 0x38

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	PIPENRDYE[9:0]									
Value after reset:	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	PIPENRDYE[9:0]	NRDY Interrupt Enable for Each Pipe Enable or disable the NRDY interrupt for each pipe. 0: Interrupt output disabled 1: Interrupt output enabled	R/W
15:10	—	The read values are undefined. The write value should be 0.	R/W

Note: When a USB bus reset is detected, this register value is undefined.

Note: The bit number corresponds to the pipe number.

PIPENRDYE[9:0] bits (NRDY Interrupt Enable for Each Pipe)

On detecting the NRDY interrupt for the pipe corresponding to the bit in this register which has been set to 1 by software, this controller sets 1 to the corresponding PIPENRDY bit in NRDYSTS and the NRDY bit in INTSTS0, and asserts the interrupt signal through the USB Function Interrupt (USB_FI).

While at least one PIPENRDY bit in NRDYSTS indicates 1, this controller asserts the interrupt signal through USB_FI when the corresponding interrupt enable bit of this register is changed from 0 to 1 by software.

31.2.5.5 BEMPENB : BEMP Interrupt Enable Register

Base address: USBF = 0x8020_1000

Offset address: 0x3A

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	PIPEBEMPE[9:0]									
Value after reset:	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	PIPEBEMPE[9:0]	BEMP Interrupt Enable for Each Pipe Enable or disable the BEMP interrupt for each pipe. 0: Interrupt output disabled 1: Interrupt output enabled	R/W
15:10	—	The read values are undefined. The write value should be 0.	R/W

Note:

- When a USB bus reset is detected, this register value is undefined.
- The bit number corresponds to the pipe number.

PIPEBEMPE[9:0] bits (BEMP Interrupt Enable for Each Pipe)

On detecting the BEMP interrupt for the pipe corresponding to the bit in this register which has been set to 1 by software, this controller sets 1 to the corresponding PIPEBEMP bit in BEMPSTS and the BEMP bit in INTSTS0, and asserts the interrupt signal through USB_FI.

While at least one PIPEBEMP bit in BEMPSTS indicates 1, this controller asserts the interrupt signal through USB_FI when the corresponding interrupt enable bit of this register is changed from 0 to 1 by software.

31.2.6 SOF Control Register**31.2.6.1 SOFCFG : SOF Pin Configuration Register**

Base address: USBF = 0x8020_1000

Offset address: 0x3C

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	BRDY M	INTL	EDGE STS	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	—	These bits are read as 0. The write value should be 0.	R/W
4	EDGESTS	Interrupt Edge Processing Status Indicates the state of processing in response to interrupts when edge-sensing has been selected. 0: Processing in response to an edge-sensed interrupt is not in progress. 1: Processing in response to an edge-sensed interrupt is in progress	R

Bit	Symbol	Function	R/W
5	INTL	Interrupt Output Sense Select ^{*2} Selects sense mode for USB interrupt output. 0: Edge sense 1: Level sense	R/W
6	BRDYM	PIPEBRDY Interrupt Status Clear Timing ^{*1} Specifies the timing for clearing the PIPEBRDY interrupt state. This bit can be set only in the initial setting (before communications). The setting cannot be changed once communication starts. 0: Software clears the state. 1: Hardware clears the state by reading from the FIFO buffer or by writing to the FIFO buffer.	R/W
15:7	—	These bits are read as 0. The write value should be 0.	R/W

Note: When a USB bus reset is detected, this register value is undefined.

Note 1. When setting the BRDYM bit to 1, set the INTL bit to 1 (level sense).

Note 2. With the INTL bit set to 0, to stop the system clock (SUSPM = 0) after clearing the interrupt state, write 0 to the SUSPM bit after checking that the EDGESTS bit is set to 0.

31.2.7 Interrupt Status

31.2.7.1 INTSTS0 : Interrupt Status Register 0

Base address: USBF = 0x8020_1000

Offset address: 0x40

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	VBINT	RESM	SOFR	DVST	CTRT	BEMP	NRDY	BRDY	VBST S	DVSQ[2:0]			VALID	CTSQ[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	x	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	CTSQ[2:0]	Control Transfer Stage Indicate the control transfer stage. 0 0 0: Idle or setup stage 0 0 1: Control read data stage 0 1 0: Control read status stage 0 1 1: Control write data stage 1 0 0: Control write status stage 1 0 1: Control write (no data) status stage 1 1 0: Control transfer sequence error 1 1 1: Reserved	R
3	VALID	USB Request Reception Indicates whether reception of the USB request has been detected. When writing to this bit, only 0 can be written. 0: Not detected 1: Setup packet reception	R/W
6:4	DVSQ[2:0]	Device State Indicate the device state. 0 0 0: Powered state 0 0 1: Default state 0 1 0: Address state 0 1 1: Configured state 1 x x: Suspended state	R
7	VBSTS	VBUS Input Status Indicates the input state of the USB_VBUSIN pin. 0: USB_VBUSIN pin is at the low level 1: USB_VBUSIN pin is at the high level	R

Bit	Symbol	Function	R/W
8	BRDY	BRDY Interrupt Status Indicates the BRDY interrupt state. 0: BRDY interrupts not generated 1: BRDY interrupts generated	R
9	NRDY	NRDY Interrupt Status Indicates the NRDY interrupt state. 0: NRDY interrupts not generated 1: NRDY interrupts generated	R
10	BEMP	BEMP Interrupt Status Indicates the BEMP interrupt state. 0: BEMP interrupts not generated 1: BEMP interrupts generated	R
11	CTRT	Control Transfer Stage Transition Interrupt Status Indicates the control transfer stage transition interrupt state. When writing to this bit, only 0 can be written. 0: Control transfer stage transition interrupts not generated 1: Control transfer stage transition interrupts generated	R/W ^{*1}
12	DVST	Device State Transition Interrupt Status Indicates the device state transition interrupt state. When writing to this bit, only 0 can be written. 0: Device state transition interrupts not generated 1: Device state transition interrupts generated	R/W ^{*1}
13	SOFR	Frame Number Update Interrupt Status Indicates the frame number update interrupt state. When writing to this bit, only 0 can be written. 0: SOF interrupts not generated 1: SOF interrupts generated	R/W ^{*1}
14	RESM	Resume Interrupt Status ^{*2} Indicates the resume detection interrupt state. When writing to this bit, only 0 can be written. 0: Resume interrupts not generated 1: Resume interrupts generated	R/W ^{*1}
15	VBINT	VBUS Change Detect Interrupt Status ^{*2} Indicates the VBUS change detection interrupt state. When writing to this bit, only 0 can be written. 0: VBUS interrupts not generated 1: VBUS interrupts generated	R/W ^{*1}

Note: When a USB bus reset is detected, The DVST is set to 1 and DVSQ[2:0] are set to 001b and other bits are undefined.

Note 1. To clear the state indicated by the VBINT, RESM, SOFR, DVST or CTRT bits, write 0 only to the bit to be cleared, and write 1 to the other bits. Do not write 0 to the status bit set to 0.

Note 2. The controller detects the change in state indicated by the VBINT and RESM bits of this register, even while the clock is being stopped (SUSPM = 0), and conveys the interrupt if the corresponding interrupt is enabled. When the clock is enabled, clear the state using software.

BRDY bit (BRDY Interrupt Status)

This controller sets a BRDY bit to 1 when, among the PIPEBRDY bits for which the corresponding PIPEBRDYE bits in BRDYENB have been set to 1, at least one PIPEBRDY bit in BRDYSTS is set to 1 (that is, when this controller detects the BRDY interrupt state in at least one pipe among the pipes for which software has enabled the output of BRDY interrupts).

For the conditions for the setting of PIPEBRDY bits, see the description of the BRDYSTS register.

When a BRDY bit has been set to 1, the controller clears the bit in response to software writing 0 to all PIPEBRDY bits corresponding to PIPEBRDYE bits which have been set to 1.

A BRDY bit will not be cleared to 0 in response to software only writing 0 to individual bits that have been set to 1.

NRDY bit (NRDY Interrupt Status)

This controller sets an NRDY bit to 1 when, among the PIPEBRDY bits for which the corresponding PIPENRDYE bits in NRDYENB have been set to 1, at least one PIPENRDY bit in BNRDYSTS is set to 1 (that is, when this controller detects the NRDY interrupt state in at least one pipe among the pipes for which software has enabled the output of NRDY interrupts).

For the conditions for the setting of PIPENRDY bits, see the description of the NRDYSTS register.

When an NRDY bit has been set to 1, the controller clears the bit in response to software writing 0 to all PIPENRDY bits corresponding to PIPENRDYE bits which have been set to 1.

An NRDY bit will not be cleared to 0 in response to software only writing 0 to individual bits that have been set to 1.

BEMP bit (BEMP Interrupt Status)

This controller sets a BEMP bit to 1 when, among the PIPEBRDY bits for which the corresponding PIPEBEMPE bits in BEMPENB have been set to 1, at least one PIPEBEMP bit in BEMPSTS is set to 1 (that is, when this controller detects the BEMP interrupt state in at least one pipe among the pipes for which software has enabled the output of BEMP interrupts).

For the conditions for the setting of PIPEBEMP bits, see the description of the BEMPSTS register.

When a BEMP bit has been set to 1, the controller clears the bit in response to software writing 0 to all PIPEBEMP bits corresponding to PIPEBEMPE bits which have been set to 1.

A BEMP bit will not be cleared to 0 in response to software only writing 0 to individual bits that have been set to 1.

CTRT bit (Control Transfer Stage Transition Interrupt Status)

If this controller detects the stage transition of control transfer, it updates the CTSQ value and sets 1 to this bit.

When this interrupt occurs, clear the state before the controller detects the next control transfer stage transition.

DVST bit (Device State Transition Interrupt Status)

If this controller detects a change in the device state, it updates the DVSQ value and sets 1 to this bit.

When this interrupt occurs, clear the state before the controller detects the next device state transition.

SOFR bit (Frame Number Update Interrupt Status)

The conditions when the controller sets 1 in this bit are below.

When updating the frame number, this controller sets 1 to this bit (this interrupt is detected every 1 ms).

The controller detects the SOFR interrupt by internal interpolation even if the SOF packet from the USB host is corrupted.

RESM bit (Resume Interrupt Status)

If this controller is in the suspended state (DVSQ = 1xxb) and the DP pin falling edge is detected, 1 is set to this bit.

VBINT bit (VBUS Change Detect Interrupt Status)

When this controller detects a change in the USB_VBUSIN pin input value (from high to low and from low to high), 1 is written to this bit. The controller writes the input value of the USB_VBUSIN pin to the VBSTS bit. When the VBINT interrupt occurs, use the software to execute a consistency check several times during reading the VBSTS bit, and remove the chattering effect.

31.2.7.2 INTSTS1 : Interrupt Status Register 1

Base address: USBF = 0x8020_1000

Offset address: 0x42

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	PDDETINT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PDDETINT	PDDET Detection Interrupt Status ^{*1} Indicates the PDDETINT interrupt status. 0: PDDET interrupts not generated 1: PDDET interrupts generated	R/W ^{*2}
15:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: When a USB bus reset is detected, The bit 12 is set to 1 and other bits are undefined.

Note 1. To clear the status indicated by the PDDETINT bit, write 0 to bit0 only and 1 to other bits. Do not write 0 to status bits that indicate 0.

Note 2. This controller detects changes in the status indicated by the PDDDETINT bit in this register even while the clock is stopped (SUSPM = 0), and reports corresponding interrupts if they are enabled. Perform a software status clear after the clock is enabled.

PDDDETINT bit (PDDDET Detection Interrupt Status)

This bit is set to 1 when a change (high to low or low to high) in EXH_PDDDETSTS (BC1/2-PD detection status signal) is detected. When a PDDDETINT interrupt is generated, read the PDDDETSTS bit several times by software to perform debounce processing.

31.2.7.3 BRDYSTS : BRDY Interrupt Status Register

Base address: USBF = 0x8020_1000

Offset address: 0x46

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	PIPEBRDY[9:0]									
Value after reset:	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	PIPEBRDY[9:0]	BRDY Interrupt Status for Each Pipe ^{*1} Indicate the BRDY interrupt state for each pipe. When writing to these bits, only 0 can be written. 0: Interrupts not generated 1: Interrupts generated	R/W ^{*2*3}
15:10	—	The read values are undefined. The write value should be 0.	R/W

Note: When a USB bus reset is detected, this register value is undefined.

Note 1. The bit number corresponds to the pipe number.

Note 2. When BRDYM = 0, to clear the state of each bit of this register, write 0 only to the bit to be cleared and 1 to other bits.

Note 3. When BRDYM = 0, clearing these interrupt status bits should be done before accessing the FIFO.

PIPEBRDY[9:0] bits (BRDY Interrupt Status for Each Pipe)

When the BRDY interrupt is detected for a pipe by this controller, the controller sets 1 in the corresponding PIPEBRDY[9:0] bit of the BRDYSTS register. Here, when 1 is written to the corresponding bit of BRDYENB register by using the software, the controller sets 1 to the BRDY bit of the INTSTS0 register.

The conditions for generating and clearing the BRDY interrupt depend on the settings of the SOFCFG.BRDYM bit and PIPECFG.BFRE bit for each pipe as described below.

- When BRDYM = 0 and BFRE = 0

With these settings, the BRDY interrupt indicates that the FIFO port is accessible.

On any of the following conditions, this controller generates the internal BRDY interrupt request trigger and sets 1 to the PIPEBRDY bit corresponding to the selected pipe.

1. For the pipe in the transmitting direction

- When the software has modified the DIR bit from 0 to 1
- When the controller ends the packet transmission of the selected pipe in the condition where writing is not possible from the CPU to the FIFO buffer that has been assigned to the selected pipe (when the BSTS bit is read as 0).
In continuous transmission/reception mode, the request trigger is generated when data of one plane of the FIFO buffer is complete.
- In double-buffer operation, one FIFO buffer being empty on completion of the writing of data to the other FIFO buffer.
The request trigger is not generated until completion of writing data to the currently-written FIFO buffer plane even if transmission to the other FIFO buffer is completed.
- When the hardware flushes the buffer of the pipe for isochronous transfers.
- The FIFO buffer making the transition from the write-disabled to the write-enabled state in response to writing of 1 to the ACLRM bit.
The request trigger is not generated for the DCP (that is, during data transmission for control transfers).

2. For the pipe in the receiving direction

- (a) When packet reception is completed successfully thus enabling the FIFO buffer to be read when read-access from the CPU to the FIFO buffer for the selected pipe is disabled (when the BSTS bit is read as 0).
The request trigger is not generated for the transaction in which DATA-PID disagreement occurs.
When a short packet is received, the request trigger is generated even if the FIFO buffer has available space.
When the transaction counter is used, the request trigger is generated on receiving the specified number of packets.
In this case, the request trigger is generated even if the FIFO buffer has available space.
- (b) In double-buffer operation, one FIFO buffer being ready for reading on completion of the reading of data from the other FIFO buffer.
The request trigger is not generated until completion of reading data from the currently-read FIFO buffer plane even if reception by the other FIFO buffer is completed.
The BRDY interrupt is not generated in the status stage of control transfers.
The PIPEBRDY interrupt state of the selected pipe can be cleared to 0 by writing 0 to the corresponding PIPEBRDY interrupt status bit in the BRDYSTS register. In this case, 1s should be written to the PIPEBRDY interrupt status bits for the other pipes.
Be sure to clear the BRDY state before accessing the FIFO buffer.

- When BRDYM = 0 and BFRE = 1

With these settings, this controller generates the BRDY interrupt on completion of the reading of all data for a single transfer using the pipe in the receiving direction, and sets 1 to the PIPEBRDY bit corresponding to the selected pipe. On any of the following conditions, this controller determines that the last data for a single transfer has been received.

1. When a short packet including a zero-length packet is received.
2. When the transaction counter register (TRNCNT bits) is used and the number of packets specified by the TRNCNT bits is completely received.

When the pertinent data is completely read out after any of the above determination conditions has been satisfied, this controller determines that all data for a single transfer has been completely read out.

When a zero-length packet is received when the FIFO buffer is empty, the controller determines that all data for a single transfer has been completely read out upon the FRDY and the DTLN bits of the FIFO port control register are set to 1 and 0, respectively.

In this case, write 1 to the BCLR bit of the corresponding FIFOCTR register by software to start the next transfer.

With these settings, this controller does not detect the BRDY interrupt for the pipe in the transmitting direction.

The PIPEBRDY interrupt state of the selected pipe can be cleared to 0 by writing 0 to the corresponding PIPEBRDY interrupt status bit. In this case, 1s should be written to the PIPEBRDY interrupt status bits for the other pipes.

In this mode, the BFRE bit setting should not be modified until all data for a single transfer has been processed.

When it is necessary to modify the BFRE bit before completion of processing, all the FIFO buffers for the selected pipe should be cleared using the ACLRM bit.

- When BRDYM = 1 and BFRE = 0

With these settings, the PIPEBRDY values are linked to the BSTS bit settings for each pipe. In other words, the BRDY interrupt status bits (PIPEBRDY) are set to 1 or 0 by this controller depending on the FIFO buffer state.

1. For the pipe in the transmitting direction

The setting of a BRDY interrupt status bit is 1 while the port register of the corresponding FIFO buffer is ready for written data and 0 when it is not able to accept written data.

However, the BRDY interrupt is not generated in response to setting of the bit to 1 if writing to the DCP in the transmitting direction can proceed.

2. For the pipe in the receiving direction

The setting of a BRDY interrupt status bit is 1 while the port register of the FIFO buffer is ready for reading and 0 when all data have been read.

When a zero-length packet is received when the FIFO buffer is empty, the corresponding bit is set to 1 and the BRDY interrupt continues to be effective until 1 is written to BCLR.

With this setting, the PIPEBRDY bit cannot be cleared to 0.

When BRDYM is set to 1, all the BFRE bits (for all pipes) should be cleared to 0.

When BRDYM is set to 1, the INTL bit should be set to 1 (level control).

31.2.7.4 NRDYSTS : NRDY Interrupt Status Register

Base address: USBF = 0x8020_1000

Offset address: 0x48

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	PIPENRDY[9:0]									
Value after reset:	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	PIPENRDY[9:0]	NRDY Interrupt Status for Each Pipe* ¹ Indicate the NRDY interrupt state for each pipe. When writing to these bits, only 0 can be written. 0: Interrupts not generated 1: Interrupts generated	R/W* ²
15:10	—	The read values are undefined. The write value should be 0.	R/W

Note: When a USB bus reset is detected, this register value is undefined.

Note 1. The bit number corresponds to the pipe number.

Note 2. To clear the state indicated by each bit of this register, write 0 only to the bit to be cleared and 1 to the other bits.

PIPENRDY[9:0] bits (NRDY Interrupt Status for Each Pipe)

On generating the internal NRDY interrupt request for the pipe whose PID bits are set to BUF, this controller sets the corresponding PIPENRDY bit in NRDYSTS in the register to 1. If the corresponding bit in the NRDYENB register is set to 1, this controller sets the NRDY bit in INTSTS0 in the register to 1.

The conditions on which this controller generates the internal NRDY interrupt request for a given pipe are described below.

However, the internal NRDY interrupt request is not generated during status stage execution of the control transfer.

1. When the pipe is in the transmitting direction

- (a) When an IN Token is received while there is no transmission data in the FIFO buffer and the corresponding PIPE PID bit is set to BUF (01):

In this case, this controller generates an NRDY interrupt request at the reception of the IN token, setting the PIPENRDY bit to 1.

For the pipe for the isochronous transfers in which an interrupt is generated, this controller transmits a zerolength packet, setting the OVRN bit to 1.

2. For the pipe in the receiving direction

- (a) When the PID bits for the corresponding pipe are set to BUF (01) and an OUT token is received while there is no open space in the FIFO buffer:

For the pipe for the isochronous transfers in which an interrupt is generated, this controller generates an NRDY interrupt request at the reception of the OUT token, setting the PIPENRDY bit to 1 and OVRN bit to 1.

For the pipe for the transfers other than isochronous transfers in which an interrupt is generated, this controller generates an NRDY interrupt request when an NAK handshake is transferred after the data following the OUT token was received, setting the PIPENRDY bit to 1.

However, during re-transmission (due to DATA-PID disagreement), the NRDY interrupt request is not generated. In addition, if an error occurs in the DATA packet, the NRDY interrupt request is not generated.

- (b) When the PID bits for the corresponding pipe are set to BUF (01) and a PING token is received while there is no open space in the FIFO buffer:

In this case, this controller generates an NRDY interrupt request at the reception of the PING token, setting the PIPENRDY bit to 1.

- (c) In an isochronous transfer pipe, when the PID bits are set to BUF (01) and data is not received successfully within the interval frame:

In this case, this controller generates an NRDY interrupt request at the reception of an SOF, and sets the PIPENRDY bit to 1.

31.2.7.5 BEMPSTS : BEMP Interrupt Status Register

Base address: USBF = 0x8020_1000

Offset address: 0x4A

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	PIPEBEMP[9:0]									
Value after reset:	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	PIPEBEMP[9:0]	BEMP Interrupt Status for Each Pipe* ¹ Indicate the BEMP interrupt state for each pipe. When writing to these bits, only 0 can be written. 0: Interrupts not generated 1: Interrupts generated	R/W* ²
15:10	—	The read values are undefined. The write value should be 0.	R/W

Note: When a USB bus reset is detected, this register value is undefined.

Note 1. The bit number corresponds to the pipe number.

Note 2. To clear the state indicated by each bit of this register, write 0 only to the bit to be cleared and 1 to the other bits.

PIPEBEMP[9:0] bits (BEMP Interrupt Status for Each Pipe)

On detecting the BEMP interrupt for the pipe whose PID bits are set to BUF by software, this controller sets the corresponding PIPEBEMP bit in BEMPSTS to 1. If the corresponding bit in BEMPENB is set to 1 by software, this controller sets the BEMP bit in INTSTS0 to 1.

The following describes the conditions on which this controller generates the internal BEMP interrupt request.

- For the pipe in the transmitting direction, when the FIFO buffer of the corresponding pipe is empty on completion of transmission (including zero-length packet transmission).
In single-buffer operation, the internal BEMP interrupt request is generated simultaneously with the BRDY interrupt for the pipe other than DCP.
However, the internal BEMP interrupt request is not generated on any of the following conditions.
 - When writing data to the FIFO buffer on the CPU (DMAC) side is started by software on completion of transmitting data of one plane in double-buffer operation.
 - When the buffer is cleared (emptied) by setting the ACLRM or BCLR bit to 1.
 - When IN transfer (zero-length packet transmission) is performed during the control transfer status stage.
- For the pipe in the receiving direction
When the successfully-received data packet size exceeds the specified maximum packet size.
In this case, this controller generates the BEMP interrupt request, setting the corresponding PIPEBEMP bit to 1, and discards the received data and modifies the setting of the PID bits of the corresponding pipe to STALL (11). However, the internal BEMP interrupt request is not generated on any of the following conditions.
 - When a CRC error or bit stuffing error is detected in the received data.
 - When a setup transaction is being performed.

Writing 0 to the PIPEBEMP bit clears the state; writing 1 to the PIPEBEMP bit has no effect.

31.2.8 Frame Number Registers

31.2.8.1 FRMNUM : Frame Number Register

Base address: USBF = 0x8020_1000

Offset address: 0x4C

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	OVRN	CRCE	—	—	—	FRNM[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
10:0	FRNM[10:0]	Frame Number Indicate the latest frame number.	R
13:11	—	These bits are read as 0. The write value should be 0.	R/W
14	CRCE	CRC Error Detection Status Indicates whether a CRC error has been detected in the pipe during isochronous transfer. When writing to this bit, only 0 can be written. 0: No error 1: An error occurred.	R/W
15	OVRN	Overflow/Underflow Detection Status*1 Indicates whether an overflow/underflow error has been detected in the pipe during isochronous transfer. When writing to this bit, only 0 can be written. 0: No error 1: An error occurred.	R/W

Note: When a USB bus reset is detected, this register value is undefined.

Note 1. The OVRN bit is for use in debugging. When designing a system, control the timing so that neither overflow nor underflow occurs.

FRNM[10:0] bits (Frame Number)

This controller sets these bits to indicate the latest frame number, which is updated every time an SOF packet is issued or received (every 1 ms).

When reading these bits by software, repeat reading until the same value is read twice.

CRCE bit (CRC Error Detection Status)

In an isochronous transfer pipe, this controller sets 1 to this bit when a CRC error or bit stuffing error has been detected.

This bit can be cleared by writing 0 to it by software. In this case, write 0x8000 when OVRN is not to be cleared at the same time.

This controller generates an internal NRDY interrupt request when it detects a CRC error. For details, see NRDY Interrupt Enable Bit for Each Pipe (PIPENRDYE).

OVRN bit (Overflow/Underflow Detection Status)

For the isochronous transfer pipe, this controller sets 1 to this bit when an overflow/underflow is detected.

When an overflow/underflow is detected, the controller issues an internal NRDY request. See NRDY Interrupt Enable Bit for Each Pipe (PIPENRDYE) for details.

This bit can be cleared writing 0 to it by software. In this case, write 0x4000 when CRCE is not to be cleared at the same time.

This controller sets this bit to 1 on any of the following conditions.

1. For the isochronous transfer pipe in the transmitting direction, the IN token is received before all the transmit data has been written to the FIFO buffer.
2. For the isochronous transfer pipe in the receiving direction, the OUT token is received when no FIFO buffer planes are empty.

31.2.8.2 UFRMNUM : Frame Number Register

Base address: USBF = 0x8020_1000

Offset address: 0x4E

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	UFRNM[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	UFRNM[2:0]	Microframe Number Indicate the μ frame number.	R
15:3	—	These bits are read as 0.	R

Note: When a USB bus reset is detected, this register value is undefined.

UFRNM[2:0] bits (Microframe Number)

This controller sets these bits to indicate the μ frame number during high-speed operation. During operation other than high-speed operation, this controller sets these bits to 0x00.

When reading these bits, repeat reading until the same value is read twice.

31.2.9 USB Address

31.2.9.1 USBADDR : USB Address Register

Base address: USBF = 0x8020_1000

Offset address: 0x50

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	USBADDR[6:0]						
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
6:0	USBADDR[6:0]	USB Address Indicate the USB address assigned by the host.	R
15:7	—	These bits are read as 0.	R

Note: When a USB bus reset is detected, The USBADDR[6:0] is set to 0000000b and other bits are undefined.

USBADDR[6:0] bits (USB Address)

These bits indicate the USB address assigned by the host when the SetAddress request is successfully processed.

If a USB bus reset is detected by the controller, 0x00 is set to this bit.

31.2.10 USB Request Registers

The USB request registers are used to store the setup requests for control transfers. The values of the USB request that have been received are stored.

31.2.10.1 USBREQ : USB Request Type Register

Base address: USBF = 0x8020_1000

Offset address: 0x54

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	BREQUEST[7:0]								BMREQUESTTYPE[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	BMREQUESTTYPE [7:0]	Request Type USB request bmRequestType value	R
15:8	BREQUEST[7:0]	Request USB request bRequest value	R

BMREQUESTTYPE[7:0] bits (Request Type)

These bits indicate the USB request data value received by this controller in the setup transaction. Writing to these bits by software has no effect.

BREQUEST[7:0] bits (Request)

These bits indicate the USB request data value received by this controller in the setup transaction. Writing to these bits by software has no effect.

31.2.10.2 USBVAL : USB Request Value Register

Base address: USBF = 0x8020_1000

Offset address: 0x56

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	WVALUE[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	WVALUE[15:0]	Value USB request wValue value	R

WVALUE[15:0] bits (Value)

These bits are used to read the value of the USB request wValue. Bits 7 to 0 are lower-order byte.

These bits indicate the USB request wValue value received by this controller in the setup transaction. Writing to these bits by software has no effect.

31.2.10.3 USBINDX : USB Request Index Register

Base address: USBF = 0x8020_1000

Offset address: 0x58

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	WINDEX[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	WINDEX[15:0]	Index USB request wIndex value	R

WINDEX[15:0] bits (Index)

These bits are used to read the value of the USB request wIndex. Bits 7 to 0 are lower-order byte.

These bits indicate the USB request wIndex value received by this controller in the setup transaction. Writing to these bits by software has no effect.

31.2.10.4 USBLENG : USB Request Length Register

Base address: USBF = 0x8020_1000

Offset address: 0x5A

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	WLENGTH[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	WLENGTH[15:0]	Length USB request wLength value	R

WLENGTH[15:0] bits (Length)

These bits are used to read the value of the USB request wLength. Bits 7 to 0 are lower-order byte.

These bits indicate the USB request wLength value received in setup transaction by the controller. Writing to these bits by software has no effect.

31.2.11 DCP Configuration**31.2.11.1 DCPCFG : DCP Configuration Register**

Base address: USBF = 0x8020_1000

Offset address: 0x5C

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	CNTM D	SHTN AK	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
6:0	—	These bits are read as 0. The write value should be 0.	R/W
7	SHTNAK	Disabling PIPE at the End of Transfer Specifies whether to change PID to NAK at the end of transfer when receiving direction is specified for the default control pipe. 0: The pipe continues at the end of transfer. 1: The pipe is disabled at the end of transfer.	R/W
8	CNTMD	Continuous Transfer Mode Specifies whether to enable the default control pipe communication in continuous transfer mode. 0: Continuous transfer mode disabled 1: Continuous transfer mode enabled	R/W
15:9	—	These bits are read as 0. The write value should be 0.	R/W

Note: When a USB bus reset is detected, this register value is undefined.

31.2.11.2 DCPMAXP : DCP Maximum Packet Size Register

Base address: USBF = 0x8020_1000

Offset address: 0x5E

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	MXPS[6:0]						
Value after reset:	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0

Bit	Symbol	Function	R/W
6:0	MXPS[6:0]	Maximum Packet Size Specify the maximum data payload (maximum packet size) for the DCP.	R/W
15:7	—	These bits are read as 0. The write value should be 0.	R/W

Note: When a USB bus reset is detected, this register value is undefined.

MXPS[6:0] bits (Maximum Packet Size)

Set the maximum data payload (maximum packet size) for the DCP in these bits. 0x40 (64 bytes) is the value after a reset.

The MXPS bits should be set to the value based on the USB specification.

The MXPS bits should be set while PID = NAK and before setting the CURPIPE bits.

Before modifying these bits after changing the setting of the PID bits for the selected pipe from BUF to NAK, check that PBUSY = 0. However, checking of the PBUSY bit is not required if the setting of the PID bits has been changed to select NAK response by the controller.

While MXPS is 0, do not write to the FIFO buffer or do not set PID to BUF.

31.2.11.3 DCPCTR : DCP Control Register

Base address: USBF = 0x8020_1000

Offset address: 0x60

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	BSTS	—	—	—	—	—	—	SQCL R	SQSE T	SQMO N	PBUS Y	—	—	CCPL	PID[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	PID[1:0]	Response PID These bits control the response type of this controller during control transfer. 0 0: NAK response 0 1: BUF response (depending on the buffer state) 1 0: STALL response 1 1: STALL response	R/W
2	CCPL	Control Transfer End Enable Setting this bit to 1 enables the status stage of the control transfer to be completed. 0: End of the control transfer disabled 1: End of the control transfer enabled	R/W
4:3	—	These bits are read as 0. The write value should be 0.	R/W
5	PBUSY	PIPE Busy Indicates whether the selected pipe is currently being used by the USB bus. 0: The pipe is not being used by the USB bus. 1: The pipe is being used by the USB bus.	R

Bit	Symbol	Function	R/W
6	SQMON	Sequence Toggle Bit Monitor Indicates the expected value of the sequence toggle bit for the next transaction during the DCP transfer. 0: DATA0 1: DATA1	R
7	SQSET	Toggle Bit Set Specifies DATA1 as the expected value of the sequence toggle bit for the next transaction during the DCP transfer. This bit is read as 0. When writing to this bit, only 1 can be written. 0: Writing has no effect 1: Specifies DATA1	R/W
8	SQCLR	Toggle Bit Clear Specifies DATA0 as the expected value of the sequence toggle bit for the next transaction during the DCP transfer. This bit is read as 0. When writing to this bit, only 1 can be written. 0: Writing has no effect 1: Specifies DATA0	R/W
10:9	—	These bits are read as 0. The write value should be 0.	R/W
11	—	This bit is read as 0. The write value should be 1.	R/W
12	—	The read value is undefined. The write value should be 0.	R/W
14:13	—	These bits are read as 0. The write value should be 1.	R/W
15	BSTS	Buffer Status Indicates whether access to the DCP FIFO buffer is enabled or disabled. 0: Buffer access disabled 1: Buffer access enabled	R

Note: When a USB bus reset is detected, The CCPL is set to 0 and PID[1:0] are set to 0 and other bits are undefined.

PID[1:0] bits (Response PID)

Change the setting of these bits from NAK to BUF by software during data stage or status stage of control transfers.

This controller changes the setting of these bits in the following conditions:

1. This controller changes the setting of the PID bits to select NAK response (00) on receiving the setup packet. Here, this controller sets VALID to 1. The setting of the PID bits cannot be changed until VALID is set to 0.
2. This controller sets the PID bits to select STALL response (11) on receiving the data of the size exceeding the maximum packet size when the PID bits have been set to select BUF response.
3. This controller sets the PID bits to select STALL response (1x) on detecting the control transfer sequence error.
4. This controller sets the PID bits to select NAK response on detecting the USB bus reset.

This controller does not reference to the setting of the PID bits while the SET_ADDRESS request is processed (auto processing).

CCPL bit (Control Transfer End Enable)

Setting this bit to 1 while the corresponding PID bits are set to BUF enables the stage of the control transfer to be completed.

Specifically, during control read transfer, this controller transmits the ACK handshake in response to the OUT transaction from the USB host, and outputs the zero-length packet in response to the IN transaction from the USB host during control writing or no-data control transfer. However, on detecting the SET_ADDRESS request, this controller operates in auto response mode from the setup stage up to the status stage completion irrespective of the setting of this bit.

This controller modifies this bit from 1 to 0 on receiving the new setup packet.

1 cannot be written by software to this bit while VALID is 1.

PBUSY bit (PIPE Busy)

This controller modifies this bit from 0 to 1 upon start of the USB transaction for the selected pipe, and modifies the bit from 1 to 0 upon completion of one transaction.

Reading this bit after the PID bits have been set to NAK allows checking that modification of the pipe settings is possible.

SQMON bit (Sequence Toggle Bit Monitor)

This bit indicates the expected value of the sequence toggle bit of the selected pipe.

This bit is toggled upon normal completion of the transaction. However, this bit is not toggled when a DATA-PID disagreement occurs during the receiving transfer.

This controller sets this bit to 1 (specifies DATA1 as the expected value) upon normal reception of the setup packet.

This controller does not reference to this bit during the IN/OUT transaction of the status stage, and does not allow this bit to toggle upon normal completion.

SQSET bit (Toggle Bit Set)

Setting this bit to 1 by software allows this controller to set DATA1 as the expected value of the sequence toggle bit of the selected pipe. This bit always indicates 0.

Do not set the SQCLR and SQSET bits to 1 simultaneously.

Set this bit to 1 while PID is NAK and before setting the CURPIPE bits.

Before setting this bit to 1 after changing the setting of the PID bits for the selected pipe from BUF to NAK, check that PBUSY is 0. However, checking of the PBUSY bit is not required if the setting of the PID bits has been changed to select NAK response by the controller.

SQCLR bit (Toggle Bit Clear)

Setting this bit to 1 by software allows this controller to set DATA0 as the expected value of the sequence toggle bit of the selected pipe. This bit always indicates 0.

Do not set the SQCLR and SQSET bits to 1 simultaneously.

Set this bit to 1 while PID is NAK and before setting the CURPIPE bits.

Before setting this bit to 1 after changing the setting of the PID bits for the selected pipe from BUF to NAK, check that PBUSY = 0. However, checking of the PBUSY bit is not required if the setting of the PID bits has been changed to select NAK response by the controller.

BSTS bit (Buffer Status)

This bit indicates whether access from the CPU to the FIFO buffer assigned to the DCP is possible.

The meaning of the BSTS bit depends on the ISEL bit setting as follows.

1. When ISEL = 0: indicates whether the received data can be read from the buffer.
2. When ISEL = 1: indicates whether the data for transmission can be written to the buffer.

31.2.12 Pipe Configuration Register

Pipes 1 to 9 should be set using the PIPESEL, PIPECFG, PIPEBUF, PIPEMAXP, PIPEPERI, PIPExCTR, PIPExTRE and PIPExTRN registers.

After selecting the pipe using the PIPESEL register, functions of the pipe should be set using the PIPECFG, PIPEBUF, PIPEMAXP and PIPEPERI registers. The PIPExCTR, PIPExTRE and PIPExTRN registers can be set regardless of the pipe selection in the PIPESEL register.

31.2.12.1 PIPESEL : Pipe Window Select Register

Base address: USBF = 0x8020_1000

Offset address: 0x64

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	PIPESEL[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	PIPESEL[3:0]	Pipe Window Select Specify the pipe corresponding to the PIPECFG, PIPEBUF, PIPEMAXP, PIPEPERI registers. 0x0: No pipe selected 0x1: Pipe 1 0x2: Pipe 2 0x3: Pipe 3 0x4: Pipe 4 0x5: Pipe 5 0x6: Pipe 6 0x7: Pipe 7 0x8: Pipe 8 0x9: Pipe 9 Others: Reserved	R/W
15:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: When a USB bus reset is detected, this register value is undefined.

PIPESEL[3:0] bit (Pipe Window Select)

When a value between 0001b and 1001b is set in these bits, the information and settings for the corresponding pipe can be read from the PIPECFG, PIPEBUF, PIPEMAXP, PIPEPERI registers. After a pipe is specified by setting these bits, the values set by software to the PIPECFG, PIPEBUF, PIPEMAXP, PIPEPERI registers are reflected in the corresponding pipe transfer type by this controller.

When 0000b is set in PIPESEL, 0 is read from all the bits in the PIPECFG, PIPEBUF, PIPEMAXP, and PIPEPERI registers. When 0000b is set in PIPESEL, writing to the bits in these registers has no effect.

31.2.12.2 PIPECFG : Pipe Configuration Register

Base address: USBF = 0x8020_1000

Offset address: 0x68

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	TYPE[1:0]	—	—	—	BFRE	DBLB	CNTMD	SHTNAK	—	—	DIR	EPNUM[3:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	EPNUM[3:0]	Endpoint Number Specify endpoint number for the selected pipe.	R/W
4	DIR	Transfer Direction Specifies pipe transfer direction for the selected pipe. 0: Receiving direction 1: Transmitting direction	R/W
6:5	—	These bits are read as 0. The write value should be 0.	R/W
7	SHTNAK	Pipe Disable at the End of Transfer Specifies whether to change PID to NAK at the end of transfer when the selected pipe is in the receiving direction. 0: Pipe continued at end of transfer 1: Pipe disabled at end of transfer	R/W
8	CNTMD	Continuous Transfer Mode Specifies whether to use the selected pipe in continuous transfer mode. 0: Non-continuous transfer mode 1: Continuous transfer mode	R/W
9	DBLB	Double Buffer Mode Selects either single- or double-buffer operation for the FIFO buffer used by the selected pipe. 0: Single-buffer operation 1: Double-buffer operation	R/W

Bit	Symbol	Function	R/W
10	BFRE	BRDY Interrupt Operation Specification Specifies the BRDY interrupt generation timing from this controller to the CPU with respect to the selected pipe. 0: BRDY interrupt notification upon transmitting or receiving data 1: BRDY interrupt notification upon reading data	R/W
13:11	—	These bits are read as 0. The write value should be 0.	R/W
15:14	TYPE[1:0]	Transfer Type Specify the transfer type for the pipe selected by the PIPESEL bits (selected pipe). 0 0: Pipe is not in use. 0 1: Bulk transfer 1 0: Interrupt transfer 1 1: Isochronous transfer	R/W

Note: When a USB bus reset is detected, this register value is undefined.

EPNUM[3:0] bits (Endpoint Number)

Specify the endpoint number for the selected pipe in these bits by software.

Setting 0000b indicates that the pipe is not being used.

These bits should be modified while PID is NAK, and the pipe is not selected by the CURPIPE bits.

Before modifying these bits after changing the setting of the PID bits for the selected pipe from BUF to NAK, check that PBUSY = 0. However, checking of the PBUSY bit is not required if the setting of the PID bits has been changed to select NAK response by the controller.

Do not make the settings such that the combination of the set values in the DIR and EPNUM bits should be the same for two or more pipes (EPNUM = 0000b (the selected pipe is not in use) can be set for all the pipes).

DIR bit (Transfer Direction)

When this bit is set to 0 by software, this controller uses the selected pipe in the receiving direction, and when this bit is set to 1, this controller uses the selected pipe in the transmitting direction.

This bit should be modified while PID is NAK and the pipe is not selected by the CURPIPE bits.

To modify this bit after completing USB communication using the selected pipe, write 1 and then 0 to ACLRM continuously to clear the FIFO buffer assigned to the selected pipe while the PID and CURPIPE bits are in the abovedescribed state.

Before modifying this bit after changing the setting of the PID bits for the selected pipe from BUF to NAK, check that PBUSY = 0. However, checking of the PBUSY bit is not required if the setting of the PID bits has been changed to select NAK response by the controller.

SHTNAK bit (Pipe Disable at the End of Transfer)

This bit is valid when pipes 1 to 5 are selected in the receiving direction.

When this bit is set to 1 for the selected pipe in the receiving direction, this controller modifies the PID bits for the selected pipe to NAK on determining the end of the transfer. This controller determines that the transfer has ended on any of the following conditions.

1. A short packet (including a zero-length packet) is successfully received.
2. The transaction counter is used and the number of packets specified for the counter is successfully received.

This bit should be modified while PID is NAK.

Before modifying this bit after changing the setting of the PID bits for the selected pipe from BUF to NAK, check that PBUSY = 0. However, checking of the PBUSY bit is not required if the setting of the PID bits has been changed to select NAK response by the controller.

This bit should be cleared to 0 for the pipe in the transmitting direction.

CNTMD bit (Continuous Transfer Mode)

This bit is valid when pipes 1 to 5 are selected and the transfer type of the selected pipe is set to bulk.

According to the setting value of this bit, this controller determines transmission/reception completion for the FIFO buffer assigned to the selected pipe. See [Table 31.10](#).

Table 31.10 Relationship between transfer mode settings by CNTMD bit and timings at which reading data or transmitting data from FIFO buffer is enabled

CNTMD bit setting	When reading data or transmitting data is enabled
0	In the receiving direction (DIR = 0), the FIFO buffer is ready for reading in the following case. This controller receives one packet. In the transmitting direction (DIR = 1), the FIFO buffer is ready for transmission in either of the following cases. 1. Data of the maximum packet size is written to the FIFO buffer by software (or DMAC). 2. Data of the short packet size (including 0 byte data) is written to the FIFO buffer and then 1 is written to BVAL by software (or DMAC).
1	In the receiving direction (DIR = 0), the FIFO buffer is ready for reading in any of the following cases. 1. The number of the data bytes received in the FIFO buffer assigned to the selected pipe is equal to the number of assigned data bytes ((BUFSIZE + 1) × 64). 2. This controller receives a short packet other than a zero-length packet. 3. This controller receives a zero-length packet when data is already stored in the FIFO buffer assigned to the selected pipe. 4. This controller receives the number of packets equal to the transaction counter value specified for the selected pipe by software. In the transmitting direction (DIR = 1), the FIFO buffer is ready for transmission in any of the following cases 1 to 3. 1. The number of the data bytes written to the FIFO buffer by software (or DMAC) is equal to the number of data bytes in a single FIFO buffer plane assigned to the selected pipe. 2. The number of data bytes less than the size of a single FIFO buffer plane (including 0 byte data) assigned to the selected pipe is written to the FIFO buffer and then 1 is written to BVAL by software (or DMAC). 3. The number of data bytes less than the size of a single FIFO buffer plane (including 0 byte data) assigned to the selected pipe is written to the FIFO buffer and then the DENDx_N signal is asserted at the same time as writing the last data by software (or DMAC).

This bit should be modified while PID is NAK and the pipe is not selected by the CURPIPE bits.

To modify this bit after completing USB communication using the selected pipe, write 1 and then 0 to ACLRM continuously to clear the FIFO buffer assigned to the selected pipe while the PID and CURPIPE bits are in the abovedescribed state.

Before modifying this bit after changing the setting of the PID bits for the selected pipe from BUF to NAK, check that PBUSY = 0. However, checking of the PBUSY bit is not required if the setting of the PID bits has been changed to select NAK response by the controller.

DBLB bit (Double Buffer Mode)

This bit is valid when pipes 1 to 5 are selected.

When this bit is set to 1, this controller assigns two planes of the FIFO buffer size specified by the BUFSIZE bits in PIPEBUF to the selected pipe.

Specifically, the following expression determines the FIFO buffer size assigned to the selected pipe by this controller.

$$(\text{BUFSIZE} + 1) \times 64 \times (\text{DBLB} + 1) [\text{bytes}]$$

This bit should be modified while PID is NAK and the pipe is not selected by the CURPIPE bits.

To modify this bit after completing USB communication using the selected pipe, write 1 and then 0 to ACLRM continuously by software to clear the FIFO buffer assigned to the selected pipe while the PID and CURPIPE bits are in the above-described state.

Before modifying this bit after changing the setting of the PID bits for the selected pipe from BUF to NAK, check that PBUSY = 0. However, checking of the PBUSY bit is not required if the setting of the PID bits has been changed to select NAK response by the controller.

BFRE bit (BRDY Interrupt Operation Specification)

This bit is valid when pipes 1 to 5 are selected.

When this bit is set to 1 and the selected pipe is in the receiving direction (the DIR bit is set to 0), this controller detects the transfer completion and generates the BRDY interrupt on having read the pertinent packet.

When the BRDY interrupt is generated with the above conditions, 1 must be written to BCLR. The FIFO buffer assigned to the selected pipe is not enabled for reception until 1 is written to BCLR.

When this bit is set to 1 and the selected pipe is in the transmitting direction (the DIR bit is set to 1), this controller does not generate the BRDY interrupt. For details, see the description of the PIPEBRDY interrupt register.

This bit should be modified while PID is NAK and the pipe is not selected by the CURPIPE bits.

To modify this bit after completing USB communication using the selected pipe, write 1 and then 0 to ACLRM continuously to clear the FIFO buffer assigned to the selected pipe while the PID and CURPIPE bits are in the abovedescribed state.

Before modifying this bit after changing the setting of the PID bits for the selected pipe from BUF to NAK, check that PBUSY = 0. However, checking of the PBUSY bit is not required if the setting of the PID bits has been changed to select NAK response by the controller.

TYPE[1:0] bits (Transfer Type)

Specify the USB transfer type for the pipe selected by the PIPESEL bits (selected pipe) in the TYPE bits.

Table 31.11 lists the selected pipes and transfer types that can be set in the TYPE bits.

Table 31.11 Selected pipes and transfer types that can be set in TYPE bits

Selected pipe	TYPE bits	USB transfer type
Pipe 1 or Pipe 2	01b or 11b	Bulk or isochronous transfer
Pipe 3 to Pipe 5	01b	Bulk transfer
Pipe 6 to Pipe 9	10b	Interrupt transfer

Before setting PID to BUF for the selected pipe (before starting USB communication using the selected pipe), be sure to set these bits to the value other than 00b.

These bits should be modified while the PID bits for the selected pipe are set to NAK. Before modifying these bits after changing the setting of the PID bits for the selected pipe from BUF to NAK, check that PBUSY is 0. However, checking of the PBUSY bit is not required if the setting of the PID bits has been changed to select NAK response by the controller.

31.2.12.3 PIPEBUF : Pipe Buffer Specification Register

Base address: USBF = 0x8020_1000

Offset address: 0x6A

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	BUFSIZE[4:0]						—	—	BUFNMB[7:0]						
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	BUFNMB[7:0]	Buffer number Specify the pipe FIFO buffer number for the selected pipe. (0x04 to 0x80)	R/W
9:8	—	These bits are read as 0. The write value should be 0.	R/W
14:10	BUFSIZE[4:0]	Buffer size Specify the FIFO buffer size for the pipe specified in PIPESEL bits (selected pipe). 0x00: 64 bytes 0x01: 128 bytes ⋮ 0x1F: 2 KB	R/W
15	—	This bit is read as 0. The write value should be 0.	R/W

Note: When a USB bus reset is detected, this register value is undefined.

Note: The bits in this register should be modified while PID is NAK and the pipe is not selected by the CURPIPE bits.

Note: Before modifying the bits of this register after changing the setting of the PID bits for the selected pipe from BUF to NAK, check that PBUSY = 0. However, checking of the PBUSY bit is not required if the setting of the PID bits has been changed to select NAK response by the controller.

BUFNMB[7:0] bits (Buffer number)

The first block number in the FIFO buffer to be allocated for the selected pipe should be set in these bits. The FIFO buffer blocks allocated for the selected pipe by this controller are determined as follows:

Block number: BUFNMB to block number of $\text{BUFNMB} + (\text{BUFSIZE} + 1) \times (\text{DBLB} + 1) - 1$

These bits must be set to a value that does not exceed the size of installed memory (0 [0x00] to 8192 [0x80] for 8 KB memory). Observe the following conditions:

- 0x00 is used exclusively for DCP.
- 0x04 is used exclusively for pipe 6. However, when pipe 6 is not used, 0x04 can be used for other pipes. When pipe 6 is selected, writing to these bits has no effect and 0x04 is automatically assigned by this controller.
- 0x05 is used exclusively for pipe 7. However, when pipe 7 is not used, 0x05 can be used for other pipes. When pipe 7 is selected, writing to these bits has no effect and 0x05 is automatically assigned by this controller.
- 0x06 is used exclusively for pipe 8. However, when pipe 8 is not used, 0x06 can be used for other pipes. When pipe 8 is selected, writing to these bits has no effect and 0x06 is automatically assigned by this controller.
- 0x07 is used exclusively for pipe 9. However, when pipe 9 is not used, 0x07 can be used for other pipes. When pipe 9 is selected, writing to these bits has no effect and 0x07 is automatically assigned by this controller.

BUFSIZE[4:0] bits (Buffer size)

Specify the size of the FIFO buffer for the selected pipe in these bits in terms of blocks, where one block comprises 64 bytes. When the DBLB bit is set to 1 by software, this controller assigns two planes of the FIFO buffer size specified by the BUFSIZE bits to the selected pipe. Specifically, the following expression determines the FIFO buffer size assigned to the selected pipe by this controller.

$(\text{BUFSIZE} + 1) \times 64 \times (\text{DBLB} + 1)$ [bytes]

Set the following value in these bits.

1. When pipes 1 to 5 are selected, any value from 0x00 to 0x1F can be set.
2. When pipes 6 to 9 are selected, only 0x0 should be set.

When used with CNTMD = 1, set an integral multiple of the maximum packet size to the BUFSIZE bits.

31.2.12.4 PIPEMAXP : Pipe Maximum Packet Size Register

Base address: USBF = 0x8020_1000

Offset address: 0x6C

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	MXPS[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
10:0	MXPS[10:0] ^{*1}	Maximum Packet Size Specify the maximum data payload (maximum packet size) of the selected pipe. Pipes 6 to 9 can be set to 0x01 to 0x40 bytes.	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W

Note: When a USB bus reset is detected, this register value is undefined.

Note 1. The value after a reset of the MXPS bit is 0x00 when no pipe is selected with the PIPESEL bits in PIPESEL and 0x40 when a pipe is selected with the PIPESEL bits in PIPESEL.

MXPS[10:0] bits (Maximum Packet Size)

Set the maximum data payload (maximum packet size) for the selected pipe in these bits.

For pipes 1 and 2, a value between 1 byte (0x1) to 1024 bytes (0x400) can be set.

For pipes 3 to 5, any value from 8 bytes (0x8), 16 bytes (0x10), 32 bytes (0x20), 64 bytes (0x40) and 512 bytes (0x200) can be set (bits [2:0] are not provided).

The value after a reset is 0x040 (64 bytes).

These bits should be set to the appropriate value for each transfer type based on the USB specification.

Set these bits to 1 while PID is NAK and before setting the CURPIPE bits.

Before modifying these bits after changing the setting of the PID bits for the selected pipe from BUF to NAK, check that PBUSY = 0. However, checking of the PBUSY bit is not required if the setting of the PID bits has been changed to select NAK response by the controller.

While MXPS is 0, do not write to the FIFO buffer or set PID to BUF.

31.2.12.5 PIPEPERI : Pipe Timing Control Register

Base address: USBF = 0x8020_1000

Offset address: 0x6E

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	IFIS	—	—	—	—	—	—	—	—	—	IITV[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	IITV[2:0]	Interval Error Detection Spacing Specify the interval error detection timing for the selected pipe in terms of frames, which is expressed as n-th power of 2.	R/W
11:3	—	These bits are read as 0. The write value should be 0.	R/W
12	IFIS	Isochronous IN Buffer Flush Specifies whether to flush the buffer when the pipe selected by the PIPESEL bits (selected pipe) is used for isochronous IN transfers. 0: The buffer is not flushed 1: The buffer is flushed	R/W
15:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: When a USB bus reset is detected, this register value is undefined.

IITV[2:0] bits (Interval Error Detection Spacing)

Specify the interval error detection timing in terms of frames, which is expressed as n-th power of 2, in these bits.

These bits should be modified while PID is NAK and before setting the CURPIPE bits.

Before modifying these bits after changing the setting of the PID bits for the selected pipe from BUF to NAK, check that PBUSY = 0. However, checking of the PBUSY bit is not required if the setting of the PID bits has been changed to select NAK response by the controller.

Before modifying these bits after USB communication has been completed with these bits set to a certain value, set PID to NAK and then set ACLRM to 1 to initialize the interval timer.

The IITV bits are invalid for pipes 3 to 5; set these bits to 0 for these pipes.

These bits can be set when the selected pipe transfer is for isochronous transfers.

When the selected pipe is for isochronous OUT transfers

This controller generates the NRDY interrupt when it fails to receive a data packet within the interval set for (μ) frames by the IITV bits.

This controller generates the NRDY interrupt when this controller fails to receive a data packet because of a CRC error or other errors contained in the packet, or because of the FIFO buffer being full because reading data from the FIFO buffer by software (DMAC) is slow.

This controller generates the NRDY interrupt on receiving an SOF packet. Even if the SOF packet is corrupted, the internal interpolation is used and allows the interrupt to be generated at the timing to receive the SOF packet. However, when the IITV bits are set to the value other than 0, this controller generates the NRDY interrupt on receiving an SOF packet for every interval after starting interval counting operation. When the PID bits are set to NAK by software after starting the

interval timer, this controller does not generate the NRDY interrupt on receiving an SOF packet. The interval counting starts at the different timing depending on the IITV bit setting as follows.

- When IITV = 0: The interval counting starts when the PID bits for the selected pipe are set to BUF. See [Figure 31.2](#).
- When IITV ≠ 0: The interval counting starts on completion of successful reception of the first data packet after the PID bits for the selected pipe have been modified to BUF. See [Figure 31.3](#)

(Micro) Frame	S O F		S O F		S O F	O U T	D A T A 0	S O F	O U T	D A T A 0
PID bit value	N A K		B U F		B U F	B U F	B U F	B U F	B U F	
Token issuance (0: Issued -: Not issued)	-		-		0	0				
Interval counting start					↑					

Figure 31.2 Relationship between (μ) frames and expected token reception when IITV = 0

(Micro) Frame	S O F		S O F		S O F	O U T	D A T A 0	S O F		S O F	O U T	D A T A 0	S O F		S O F	O U T	D A T A 0
PID bit value	N A K		B U F		B U F	B U F	B U F	B U F	B U F	B U F	B U F	B U F	B U F	B U F	B U F	B U F	B U F
Token reception expectation flag (0: Reception waited -: Reception not waited)	-		-		0	-	0	-	0	-	0	-	0	-	0	-	0
Interval counting start					↑												

Figure 31.3 Relationship between (μ) frames and expected token reception when IITV = 1

When the selected pipe is for isochronous IN transfers

The IFIS bit should be 1 for this use. When IFIS = 0, this controller transmits a data packet in response to the received IN token irrespective of the IITV bit setting.

When IFIS = 1, this controller clears the FIFO buffer when this controller fails to receive an IN token within the interval set for (μ) frames by the IITV bits in a state in which there is data to be transmitted in the FIFO buffer.

This controller also clears the FIFO buffer when this controller fails to receive an IN token successfully because of a bus error such as a CRC error contained in the token.

This controller clears the FIFO buffer on receiving an SOF packet. Even if the SOF packet is corrupted, the internal interpolation is used and allows the FIFO buffer to be cleared at the timing to receive the SOF packet. The interval counting starts at the different timing depending on the IITV bit setting (similar to the timing during OUT transfers).

The counting conditions for the interval counter are any of the following.

- When a hardware reset is applied to this controller (at this point, the value of the IITV bits is also cleared to 0).
- When the ACLRM bit is set to 1 by software.
- When the controller detects a USB bus reset.

IFIS bit (Isochronous IN Buffer Flush)

When the selected pipe is for isochronous IN transfers, this controller automatically clears the FIFO buffer when this controller fails to receive the IN token from the USB host within the interval set by the IITV bits in terms of (μ) frames.

In double-buffer operation (DBLB = 1), this controller only clears the data in the plane used earlier.

This controller clears the FIFO buffer on receiving the SOF packet immediately after the (μ) frame in which this controller has expected to receive the IN token. Even if the SOF packet is corrupted, this controller also clears the FIFO buffer at the right timing to receive the SOF packet by using the internal interpolation.

When the selected pipe is not for the isochronous transfer, set this bit to 0.

31.2.13 Pipe Control Registers

31.2.13.1 PIPEnCTR : PIPEn Control Register (n = 1 to 5)

Base address: USBF = 0x8020_1000

Offset address: 0x070 + 0x2 × (n - 1)

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	BSTS	INBUF M	—	—	—	ATRE PM	ACLR M	SQCL R	SQSE T	SQMO N	PBUS Y	—	—	—	PID[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	PID[1:0]	Response PID Specify the response type for the next transaction of the selected pipe. 0 0: NAK response 0 1: BUF response (depending on the buffer state) 1 0: STALL response 1 1: STALL response	R/W
4:2	—	These bits are read as 0. The write value should be 0.	R/W
5	PBUSY	PIPE Busy Indicates whether or not the selected pipe is being currently used for the USB bus. 0: The pipe is not being currently used for the USB bus. 1: The pipe is being currently used for the USB bus.	R
6	SQMON	Toggle Bit Confirm Sets the expected value of the sequence toggle bit in the next transaction of the selected pipe. 0: DATA0 1: DATA1	R
7	SQSET	Toggle Bit Set Specifies 1 when setting the expected value of the sequence toggle bit in the next transaction of the selected pipe, to DATA1. This bit is read as 0. When writing to this bit, only 1 can be written. 0: Writing has no effect. 1: Specifies DATA1	R/W
8	SQCLR	Toggle Bit Clear Specifies 1 when clearing the expected value of the sequence toggle bit in the next transaction of the selected pipe, to DATA0. This bit is read as 0. When writing to this bit, only 1 can be written. 0: Writing has no effect. 1: Specifies DATA0	R/W

Bit	Symbol	Function	R/W
9	ACLRM	Auto Buffer Clear Mode Enables or disables automatic buffer clear mode for the selected pipe. 0: Disabled 1: Enabled (all buffers are initialized)	R/W
10	ATREPM	Auto Response Mode Enables or disables auto response mode for the selected pipe. 0: Disabled 1: Enabled (a zero-length packet response while sending, NAK response and an NRDY interrupt is issued while receiving)	R/W
12:11	—	These bits are read as 0. The write value should be 0.	R/W
13	—	This bit is read as 0. The write value should be 1.	R/W
14	INBUFM	Transfer Buffer Monitor Indicates the selected FIFO buffer state when the selected pipe is in the transmitting direction. 0: FIFO buffer contains no transmittable data. 1: FIFO buffer contains transmittable data.	R
15	BSTS	Buffer Status Indicates the FIFO buffer state of the selected pipe. 0: Buffer access is disabled. 1: Buffer access is enabled.	R

Note: When a USB bus reset is detected, The PID[1:0] is set to 00b and other bits are undefined.

PID[1:0] bits (Response PID)

Set a response of the controller in each pipe in this bit by the software.

The default value of this bit is NAK. When executing a USB transfer through the selected pipe, change the setting of the PID bits to select BUF. The basic operations (operations when there is no error in the communication packet) of this controller for each value of the PID bit are given in [Table 31.12](#).

If the selected pipe is in USB communication, when the setting of the PID bits is changed from BUF to NAK by software, after writing NAK, to check whether the USB transfer of the selected pipe is actually shifted to NAK, check that PBUSY = 0. However, checking of the PBUSY bit is not required if the setting of the PID bits has been changed to select NAK response by the controller.

In following cases, the controller modifies the value of this bit:

1. When the selected pipe is for receiving and when the software has written 1 to the SHTNAK bit of the selected pipe, the controller sets PID = NAK upon identifying the transfer end.
2. For the selected pipe, when the data packet of payload exceeding the maximum packet size is received, the controller sets PID = STALL (11b).
3. When the USB bus reset is detected, the controller sets PID = NAK.

Write 10b to shift from PID = NAK (00b) to PID = STALL.

Write 11b to shift from BUF (01b) to STALL.

First write 10b and then write 00b to shift from STALL (11b) to NAK.

First, shift to NAK and then to BUF to shift from STALL to BUF.

Table 31.12 List of controller operations according to the PID bit (1 of 2)

PID bit value	Transfer type (Type bits value)	Transfer direction (DIR bit value)	Controller operations
00b (NAK)	Bulk (TYPE = 01b) or Interrupt (TYPE = 10b)	Not dependent on setup value	NAK response is sent to the token from the USB host.
		Reception (DIR = 0)	Does not respond to the token from the USB host.
	Isochronous (TYPE = 11b)	Transmission (DIR = 1)	A zero-length packet is sent to the token from the USB host.

Table 31.12 List of controller operations according to the PID bit (2 of 2)

PID bit value	Transfer type (Type bits value)	Transfer direction (DIR bit value)	Controller operations
01b (BUF)	Bulk (TYPE = 01b)	Reception (DIR = 0)	When an OUT token is sent from the USB host, if the FIFO buffer for the selected pipe is ready for reception, the data is received and an ACK or NYET response is returned. A NAK response is returned if not ready. When a PING token is sent from the USB host, if the FIFO buffer of the selected pipe is ready for reception, an ACK response is returned. A NAK response is returned if not ready.
	Interrupt (TYPE = 10b)	Reception (DIR = 0)	For the Out token from the USB host, if the FIFO buffer of the selected pipe is ready for reception, the data is received and an ACK response is sent. A NAK response is returned if not ready.
	Bulk (TYPE = 01b) or Interrupt (TYPE = 10b)	Transmission (DIR = 1)	If the corresponding FIFO buffer is ready for transmission, the data is sent for the token from the USB. A NAK response is returned if not ready.
	Isochronous (TYPE = 11b)	Reception (DIR = 0)	For the Out token from the USB host, if the FIFO buffer of the selected pipe is ready for reception, the data is received. The data is deleted if not ready.
		Transmission (DIR = 1)	If the corresponding FIFO buffer is ready for transmission, the data is sent for the token from the USB. A zero-length packet is sent if not ready.
10b (STALL) or 11b (STALL)	Bulk (TYPE = 01b) or Interrupt (TYPE = 10)	Not dependent on setup value	A STALL response is sent to the token from the USB host.
	Isochronous (TYPE = 11b)	Not dependent on setup value	Returns nothing in response to the token from the USB host.

PBUSY bit (PIPE Busy)

This controller modifies this bit from 0 to 1 upon start of the USB transaction for the selected pipe, and modifies the bit from 1 to 0 upon completion of one transaction.

Reading this bit after PID has been set to NAK allows checking that modification of the pipe settings is possible.

SQMON bit (Toggle Bit Confirm)

This bit indicates the expected value of the sequence toggle bit of the selected pipe.

When the selected pipe is not for the isochronous transfer, this bit is toggled upon normal completion of the transaction. However, this bit is not toggled when a DATA-PID disagreement occurs during the receiving transfer.

SQSET bit (Toggle Bit Set)

Setting this bit to 1 by software allows this controller to set DATA1 as the expected value of the sequence toggle bit of the selected pipe. This bit always indicates 0.

Set this bit to 1 while PID is NAK.

Before setting this bit to 1 after changing the setting of the PID bits for the selected pipe from BUF to NAK, check that PBUSY = 0. However, checking of the PBUSY bit is not required if the setting of the PID bits has been changed to select NAK response by the controller.

SQCLR bit (Toggle Bit Clear)

Setting this bit to 1 by software allows this controller to set DATA0 as the expected value of the sequence toggle bit of the selected pipe. This bit always indicates 0.

Set this bit to 1 while PID is NAK.

Before setting this bit to 1 after changing the setting of the PID bits for the selected pipe from BUF to NAK, check that PBUSY = 0. However, checking of the PBUSY bit is not required if the setting of the PID bits has been changed to select NAK response by the controller.

ACLRM bit (Auto Buffer Clear Mode)

To clear the contents in the FIFO buffer assigned to the selected pipe completely, write 1 and then 0 to this bit continuously.

Table 31.13 shows the contents cleared by writing 1 and 0 to this bit continuously.

Table 31.14 shows the cases in which clearing the contents is necessary.

Table 31.13 Contents cleared by this controller by setting ACLRM = 1

No.	Contents cleared by ACLRM bit manipulation
(1)	All the contents in the FIFO buffer assigned to the selected pipe (all the information in two FIFO buffer planes in double-buffer operation)
(2)	The interval count value when the selected pipe is for isochronous transfer

Table 31.14 Cases where setting ACLRM = 1 is required

No.	Case where clearing is required
(1)	All the contents in the FIFO buffer assigned to the selected pipe are to be cleared.
(2)	The interval counter is to be reset.
(3)	When the BFRE setting is modified
(4)	When the DBLB setting is modified
(5)	When the transaction count function is forcibly terminated

Set this bit to 1 while PID is NAK and before specifying the pipe in CURPIPE bits. Before setting this bit to 1 after changing the setting of the PID bits for the selected pipe from BUF to NAK, check that PBUSY = 0. However, checking of the PBUSY bit is not required if the setting of the PID bits has been changed to select NAK response by the controller.

ATREPM bit (Auto Response Mode)

When the selected pipe is for bulk transfer, this bit can be set to 1.

When this bit is set to 1, this controller responds to the token from the USB host as described below.

1. When the selected pipe is for bulk IN transfer (when TYPE = 01b and DIR = 1)
When ATREPM = 1 and PID = BUF, this controller transmits a zero-length packet in response to the IN token. This controller updates (toggles) the sequence toggle bit (DATAPID) each time this controller receives the ACK from the USB host (in a single transaction, IN token is received, zero-length packet is transmitted, and then ACK is received). In this case, this controller does not generate the BRDY or BEMP interrupt.
2. When the selected pipe is for bulk OUT transfer (when TYPE = 01b and DIR = 0)
When ATREPM = 1 and PID = BUF, this controller returns NAK in response to the OUT token (or PING token) and generates the NRDY interrupt.

This bit should be modified while PID is NAK. Before setting this bit to 1 after changing the setting of the PID bits for the selected pipe from BUF to NAK, check that PBUSY = 0. However, checking of the PBUSY bit is not required if the setting of the PID bits has been changed to select NAK response by the controller.

For USB communication in auto response mode, set this bit to 1 while the FIFO buffer is empty. Do not write to the FIFO buffer during USB communication in auto response mode.

When the selected pipe is for isochronous transfer, be sure to set this bit to 0.

INBUFM bit (Transfer Buffer Monitor)

When the selected pipe is in the transmitting direction (DIR = 1), this controller sets this bit to 1 when writing to at least one FIFO buffer plane is completed by software (or DMAC).

This controller sets this bit to 0 when this controller completes transmitting the data from the FIFO buffer plane to which all data has been written. In double-buffer operation (DBLB = 1), this controller sets this bit to 0 after this controller has completed transmitting the data from both FIFO buffer planes but before it has completed writing data to a single FIFO buffer plane.

This bit indicates the same value as the BSTS bit when the selected pipe is in the receiving direction (DIR = 0).

BSTS bit (Buffer Status)

This bit indicates whether access from the CPU to the FIFO buffer assigned to the selected pipe is enabled or disabled. The meaning of the BSTS bit depends on the setting of the DIR, BFRE and DCLRM bits as follows.

Table 31.15 BSTS bit operations

DIR bit value	BFRE bit value	DCLRM bit value	Meaning of BSTS bit
0	0	0	1: The received data can be read from the FIFO buffer. 0: The received data has been completely read from the FIFO buffer.
		1	Setting prohibited
	1	0	1: The received data can be read from the FIFO buffer. 0: The BCLR bit has been set to 1 after the received data has been completely read from the FIFO buffer.
		1	1: The received data can be read from the FIFO buffer. 0: The received data has been completely read from the FIFO buffer.
1	0	0	1: The transmit data can be written to the FIFO buffer. 0: The transmit data has been completely written to the FIFO buffer.
		1	Setting prohibited
	1	0	Setting prohibited
		1	Setting prohibited

31.2.13.2 PIPEnCTR : PIPEn Control Register (n = 6 to 9)

Base address: USBF = 0x8020_1000

Offset address: 0x07A + 0x2 × (n - 6)

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field:	BSTS	—	—	—	—	—	ACL R M	SQCL R	SQSE T	SQMO N	PBUS Y	—	—	—	PID[1:0]
------------	------	---	---	---	---	---	------------	-----------	-----------	-----------	-----------	---	---	---	----------

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
1:0	PID[1:0]	Response PID Specify the response type for the next transaction of the selected pipe. 0 0: NAK response 0 1: BUF response (depending on the buffer state) 1 0: STALL response 1 1: STALL response	R/W
4:2	—	These bits are read as 0. The write value should be 0.	R/W
5	PBUSY	PIPE Busy Indicates whether or not the selected pipe is being currently used for the USB bus. 0: The pipe is not being currently used for the USB bus. 1: The pipe is being currently used for the USB bus.	R
6	SQMON	Toggle Bit Confirm Sets the expected value of the sequence toggle bit in the next transaction of the selected pipe. 0: DATA0 1: DATA1	R
7	SQSET	Toggle Bit Set Specifies 1 while clearing the expected value of the sequence toggle bit in the next transaction of the selected pipe to DATA1. This bit is read as 0. When writing to this bit, write 1. 0: Writing has no effect. 1: Specifies DATA1	R/W

Bit	Symbol	Function	R/W
8	SQCLR	Toggle Bit Clear Specifies 1 while clearing the expected value of the sequence toggle bit in the next transaction of the selected pipe to DATA0. This bit is read as 0. When writing to this bit, write 1. 0: Writing has no effect. 1: Specifies DATA0	R/W
9	ACLRM	Auto Buffer Clear Mode Enables or disables automatic buffer clear mode for the selected pipe. 0: Disabled 1: Enabled (all buffers are initialized)	R/W
12:10	—	These bits are read as 0. The write value should be 0.	R/W
13	—	This bit is read as 0. The write value should be 1.	R/W
14	—	This bit is read as 0. The write value should be 0.	R/W
15	BSTS	Buffer Status Indicates the FIFO buffer state of the selected pipe. 0: Buffer access is disabled 1: Buffer access is enabled	R

Note: When a USB bus reset is detected, The PID[1:0] are set to 00b and other bits are undefined.

As for the bits listed below, see the description of each bit in [section 31.2.13.1. PIPEnCTR : PIPEn Control Register \(n = 1 to 5\)](#)

- Response PID Bits (PID[1:0])
- PIPE Busy Bit (PBUSY)
- Buffer Status Bit (BSTS)
- Sequence Toggle Bit Monitor Bit (SQMON)
- Sequence Toggle Bit Set Bit (SQSET)
- Sequence Toggle Bit Clear Bit (SQCLR)

ACLRM bit (Auto Buffer Clear Mode)

To clear all contents from the FIFO buffer allocated to the selected pipe, write 1 and 0 sequentially in the ACLRM bit.

[Table 31.16](#) shows the contents of data to be cleared by the controller when 1 and 0 are written to this bit sequentially.

The cases that require this processing are listed in [Table 31.17](#).

Table 31.16 Contents cleared by the controller when ACLRM = 1

No.	Contents cleared by ACLRM bit manipulation
(1)	Entire contents of the FIFO buffer allocated to the selected pipe

Table 31.17 Cases where setting ACLRM = 1 is required

No.	Cases where clearing is required
(1)	All the contents in the FIFO buffer assigned to the selected pipe are to be cleared.
(2)	The interval counter is to be reset.
(3)	The value of the BFRE bit is changed.
(4)	The transaction count function is forcibly terminated.

The ACLRM bit should be set while PID = NAK and before specifying the pipe in the CURPIPE bits.

Before modifying this bit after changing the setting of the PID bits for the selected pipe from BUF to NAK, check that PBUSY is 0. However, checking of the PBUSY bit is not required if the setting of the PID bits has been changed to select NAK response by the controller.

31.2.14 Transaction Counters

31.2.14.1 PIPEnTRE : PIPEn Transaction Counter Enable Register (n = 1 to 5)

Base address: USBF = 0x8020_1000

Offset address: 0x090 + 0x4 × (n - 1)

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	TRENB	TRCLR	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	—	These bits are read as 0. The write value should be 0.	R/W
8	TRCLR	Transaction Counter Clear The transaction counter can be cleared to 0 by writing 1 to this bit. This bit is read as 0. When writing to this bit, write 1. 0: Invalid 1: The current counter value is cleared	R/W
9	TRENB	Transaction Counter Enable Enable or disables the transaction counter 0: The transaction counter is disabled 1: The transaction counter is enabled	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: When a USB bus reset is detected, this register value is undefined.

Note: Modify each bit in this register while PID is NAK.

Before modifying each bit after changing the setting of the PID bits for the selected pipe from BUF to NAK, check that PBUSY = 0. However, checking of the PBUSY bit is not required if the setting of the PID bits has been changed to select NAK response by the controller.

TRCLR bit (Transaction Counter Clear)

When this bit is set to 1 by software, this controller clears the current counter value of the transaction counter corresponding to the selected pipe and then sets this bit to 0.

TRENB bit (Transaction Counter Enable)

For the pipe in the receiving direction, setting this bit is set to 1 by software after setting the total number of the packets to be received in the TRNCNT bits allows the following control on having received the number of packets equal to the set value in the TRNCNT bits.

1. In continuous transmission/reception mode (CNTMD = 1), this module switches the FIFO buffer to the CPU side even if the FIFO buffer is not full on completion of reception.
2. While SHTNAK is 1, this module changes the setting of the PID bits to NAK for the corresponding pipe on having received the number of packets equal to the set value in the TRNCNT bits.
3. While DENDE is 1 and PKTMD is 0, the DEND signal is asserted when the number of packets specified in the TRNCNT bits is received and the last data is to be read.
4. While BFRE is 1, this module asserts the BRDY interrupt on having received the number of packets equal to the set value in the TRNCNT bits and then reading out the last received data.

For the pipe in the transmitting direction, set this bit to 0.

When the transaction counter is not used, set this bit to 0.

When the transaction counter is used, set the TRNCNT bits before setting this bit to 1. Set this bit to 1 before receiving the first packet to be counted by the transaction counter.

31.2.14.2 PIPEnTRN : PIPEn Transaction Counter Register (n = 1 to 5)

Base address: USBF = 0x8020_1000

Offset address: 0x092 + 0x4 × (n - 1)

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	TRNCNT[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	TRNCNT[15:0]	Transaction Counter When written to: Specify the total number of reception packets (number of transactions) to be received by the selected pipe. When read from: Indicate the specified number of transactions if TRENb is 0. Indicate the number of currently counted transaction if TRENb is 1.	R/W

Note: When a USB bus reset is detected, this register value is undefined.

TRNCNT[15:0] bits (Transaction Counter)

For the pipe in the receiving direction, setting these bits to 1 by software after setting the total number of the packets to be received in the TRNCNT bits allows the control of the transaction counter clear bit (TRCLR).

When TRENb = 0, these bits indicate the number of transactions set by software.

When TRENb = 1, these bits indicate the number of transactions being currently counted.

This controller increments the value of these bits by one when all the following conditions are satisfied on receiving the packet.

- TRENb = 1
- (TRNCNT setting ≠ current counter value + 1) on receiving the packet
- The payload of the received packet agrees with the set value in the MXPS bits.

This module clears the value of these bits to 0 when any of the following conditions are satisfied.

When all the following conditions are satisfied:

- TRENb = 1
- (TRNCNT setting ≠ current counter value + 1) on receiving the packet
- The payload of the received packet agrees with the set value in the MXPS bits.

When both of the following conditions are satisfied:

- TRENb = 1
- A short packet is received.

When the following condition is satisfied:

- When the TRCLR bit is set to 1 by software

For the pipe in the transmitting direction, set these bits to 0. When the transaction counter is not used, set these bits to 0.

These bits should be modified while PID is NAK and TRENb is 0.

Before modifying these bits after changing the setting of the PID bits for the selected pipe from BUF to NAK, check that PBUSY = 0. However, checking of the PBUSY bit is not required if the setting of the PID bits has been changed to select NAK response by the controller.

To modify the value of these bits, set TRCLR to 1 before setting TRENb to 1.

31.2.14.3 LPSTS : Low Power Status Register

Base address: USBF = 0x8020_1000

Offset address: 0x102

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	SUSPM	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
13:0	—	These bits are read as 0. The write value should be 0.	R/W
14	SUSPM	UTMI SuspendM Control Controls the SuspendM signal to the UTMI. 0: UTMI suspend mode 1: UTMI normal mode	R/W
15	—	This bit is read as 0. The write value should be 0.	R/W

Note: When a USB bus reset is detected, this register value is undefined.

SUSPM bit (UTMI SuspendM Control)

The USBf module controls output of the clock signal from the PLL in the USB-PHY layer. Accordingly, supply of the clock signal to the USBf module is stopped while SUSPM = 0.

After setting this bit to 1, wait for at least 100 μ s to allow oscillation of the UTMI clock to become stable.

Writing to this controller is disabled when the SUSPM bit is set to 0 (the UTMI clock is stopped). Reading from the controller area is allowed. However, writing to the registers listed in [Table 31.18](#) is enabled even when the SUSPM bit is set to 0.

Table 31.18 List of registers that can be written by software when SUSPM = 0

Address	Register name
0x8020_1000	SYSCFG0
0x8020_1002	SYSCFG1
0x8020_1032	INTENB1 ^{*1}
0x8020_1102	LPSTS

Note 1. Bit 0 in the INTENB1 register can be written when SUSPM = 0.

However, the value written to the SYSCFG0 register while the UTMI clock is stopped (SUSPM = 0) is reflected after the UTMI clock oscillation is started (SUSPM = 1).

31.2.15 Next Register Set

31.2.15.1 N0SA_n, N1SA_n : Next Source Address Register n (n = 0, 1)

Base address: USBF = 0x8020_1000

Offset address: 0x400 + 0x40 $\times$ n (N0SA_n)
0x40C + 0x40 $\times$ n (N1SA_n)

Bit position:	31	0																																
Bit field:	SAWD[31:0]																																	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	SAWD[31:0]	Source Address or Write Data These bits set the start address of the DMA transfer source in normal mode. These bits set write data in write-only mode.	R/W

Note: When a USB bus reset is detected, this register value is undefined.

Note: The N0SA_n register is overwritten by descriptor read data during a transfer in link mode.

This register is used to set write data in write-only mode (when the WONLY field in the CHCFG_n register = 1). For details, see [section 31.14.1.3. Write-Only Mode](#).

31.2.15.2 N0DA_n, N1DA_n : Next Destination Address Register n (n = 0, 1)

Base address: USBF = 0x8020_1000

Offset address: 0x404 + 0x40 × n (N0DA_n)
0x410 + 0x40 × n (N1DA_n)

Bit position: 31 0

Bit field: DA[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	DA[31:0]	Destination Address These bits set the start address of the DMA transfer destination.	R/W

Note: When a USB bus reset is detected, this register value is undefined.

Note: The N0DA_n register is overwritten by descriptor read data during a transfer in link mode.

31.2.15.3 N0TB_n, N1TB_n : Next Transaction Byte Register n (n = 0, 1)

Base address: USBF = 0x8020_1000

Offset address: 0x408 + 0x40 × n (N0TB_n)
0x414 + 0x40 × n (N1TB_n)

Bit position: 31 0

Bit field: TB[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	TB[31:0]	Transaction Byte These bits set the total number of transaction bytes. Do not start a DMA transaction with these bits set to 0.	R/W

Note: When a USB bus reset is detected, this register value is undefined.

Note: The N0TB_n register is overwritten by descriptor read data during a transfer in link mode.

31.2.16 Current Register Set

The current register set indicates the DMA transfer source address, the DMA transfer destination address, and the total number of transfer bytes. The set value is loaded from the Next 0/1 register set (in register mode) or from the descriptor read data (in link mode) by hardware. This current register set cannot be written by software.

31.2.16.1 CRSA_n : Current Source Address Register n (n = 0, 1)

Base address: USBF = 0x8020_1000

Offset address: 0x418 + 0x40 × n

Bit position: 31

0

Bit field:

CRSA[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	CRSA[31:0]	Source Address These bits indicate the read address of the next DMA transaction. The CRSA[31:0] value is automatically incremented during a DMA transaction, but remains unchanged when the SAD bit in the CHCFG_n register is 1 and is undefined when the WONLY bit in the CHCFG_n register is 1. The initial value of these bits is loaded from the following register. In register mode: The transfer source address is loaded from the Next 0/1 register set. In link mode: The transfer source address is loaded from the descriptor. (Descriptor read data is input to the N0SA_n register by hardware, and is loaded to the CRSA_n register at the beginning of transfer.) The value of these bits is incremented upon completion of a read transfer. Read this register after the ongoing DMA stops (TACT = 0 in the CHSTAT_n register). (Handle the value of these bits during DMA operation as a reference value.)	R

Note: When a USB bus reset is detected, this register value is undefined.

This register indicates the DMA transfer source address of DMA channel n (n = 1, 0).

31.2.16.2 CRDA_n : Current Destination Address Register n (n = 0, 1)

Base address: USBF = 0x8020_1000

Offset address: 0x41C + 0x40 × n

Bit position: 31

0

Bit field:

CRDA[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	CRDA[31:0]	Destination Address These bits indicate the write address of the next DMA transaction. The CRDA[31:0] value is automatically incremented during a DMA transaction, but remains unchanged when the DAD bit in the CHCFG_n register is 1. The initial value of these bits is loaded from the following register. In register mode: The transfer destination address is loaded from the Next 0/1 register set. In link mode: The transfer destination address is loaded from the descriptor. (Descriptor read data is input to the N0DA_n register by hardware, and is loaded to the CRDA_n register at the beginning of transfer.) The value of these bits is incremented upon completion of a write transfer. Read this register after the ongoing DMA stops (TACT = 0 in the CHSTAT_n register). (Handle the value of these bits during DMA operation as a reference value.)	R

Note: When a USB bus reset is detected, this register value is undefined.

This register indicates the DMA transfer destination address of DMA channel n (n = 1, 0).

31.2.16.3 CRTB_n : Current Transaction Byte Register n (n = 0, 1)

Base address: USBF = 0x8020_1000

Offset address: 0x420 + 0x40 × n

Bit position: 31

0

Bit field:

CRTB[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	CRTB[31:0]	Transaction Byte These bits indicate the remaining number of transfer bytes of the DMA transaction in progress. The CRTB[31:0] value is automatically decremented during a DMA transaction. The initial value of these bits is loaded from the following register. In register mode: The number of transfer bytes is loaded from the Next 0/1 register set. In link mode: The number of transfer bytes is loaded from the descriptor. (Descriptor read data is input to the NOTB_n register by hardware, and is loaded to the CRTB_n register at the beginning of transfer.) The value of these bits is decremented upon completion of a write transfer. Read this register after the ongoing DMA stops (TACT = 0 in the CHSTAT_n register). (Handle the value of these bits during DMA operation as a reference value.)	R

Note: When a USB bus reset is detected, this register value is undefined.

This register indicates the total number of transfer bytes of DMA channel n (n = 1, 0). This register is cleared to 0 at the end of a transfer.

31.2.17 Channel Register Set

31.2.17.1 CHSTAT_n : Channel Status Register n (n = 0, 1)

Base address: USBF = 0x8020_1000

Offset address: 0x424 + 0x40 × n

Bit position:

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16

Bit field:

DNUM[7:0]

—

—

—

—

—

SWPR
Q

DMAR
QM

INTM

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit position:

15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field:

—

—

—

—

MODE

DER

DW

DL

SR

TC

END

ER

SUS

TACT

RQST

EN

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	EN	Enable Indicates that operation of DMA channel n is enabled or disabled. 0: Operation disabled 1: Operation enabled	R
1	RQST	Request Indicates that a transfer request has been received. 0: No DMA transfer request has been received. 1: A DMA transfer request has been received.	R
2	TACT	Transaction Active Indicates that the DMAC is operating to confirm that the channel is completely stopped. For details, see section 31.15.8. Transfer Status. 0: DMA of channel_n is inactive. 1: DMA of channel_n is in progress.	R

Bit	Symbol	Function	R/W
3	SUS	Suspend Indicates that the channel is suspended. For details, see section 31.15.8.2. Suspending a Transfer . 0: Channel_n is not suspended. 1: Channel_n is suspended.	R
4	ER	Error Indicates that an error response was received during a DMA transfer and a USB_FDMAERR interrupt was generated. 0: No error response was received. 1: An error response was received.	R
5	END	USB_FDMA _n Interrupted Indicates that a DMA transaction has been completed and an USB_FDMA _n interrupt was generated. 0: DMA transfer not completed 1: DMA transfer completed	R
6	TC	Terminal Count Indicates that a DMA transaction has been completed. 0: DMA transfer not completed 1: DMA transfer completed	R
7	SR	Selected Register Set Indicates the selected register set in register mode. 0: Next0 register set 1: Next1 register set	R
8	DL	Descriptor Load Indicates that the internal state is the descriptor read state. This bit remains 1 when a bus error is received during descriptor read. 0: Other than descriptor read 1: (When ER = 0) While the descriptor is read in link mode (When ER = 1) A bus error occurred while the descriptor was read in link mode.	R
9	DW	Descriptor WriteBack Indicates that the internal state is the descriptor write-back state. This bit remains 1 when a bus error is received during descriptor write-back. 0: Other than header write-back in link mode 1: (When ER in the CHSTAT_n register is 0) During header write-back in link mode (When ER in the CHSTAT_n register is 1) A bus error occurred during header write-back in link mode.	R
10	DER	Descriptor Error Indicates that the read descriptor is invalid (LV = 0). (Does not depend on the level of the DIM bit in the CHCFG_n register.) 0: No descriptor error occurred. 1: A descriptor error occurred.	R
11	MODE	DMA Mode Indicates the DMA mode that is specified by the DMS bit in the CHCFG_n register. 0: Register mode 1: Link mode	R
15:12	—	These bits are read as 0.	R
16	INTM	Interrupt Mask Indicates that the USB_FDMA _n interrupt is in the temporary mask status. 0: The temporary mask status is cleared. 1: In temporary mask status	R
17	DMARQM	DMAREQ Mask Indicates that a DMA transfer request from the USB control is in the temporary mask status. 0: The temporary mask status is cleared. 1: In temporary mask status	R

Bit	Symbol	Function	R/W
18	SWPRQ	Sweep Request Indicates the forced ejection request status. Indicates the software forced ejection request (activated by the SETSSWPRQ bit in the CHCTRL_n register). 0: The forced ejection request is not asserted. 1: The forced ejection request is asserted.	R
23:19	—	These bits are read as 0.	R
31:24	DNUM[7:0]	Data Number These bits indicate the amount of valid data in the buffer. Data is read from the transfer source, and then the amount of data that has not been written to the transfer destination is displayed (in bytes).	R

This register indicates the status of DMA channel n (n = 1, 0).

When the ER bit in the CHSTAT_n register is set to 1, handle the series of transfers as invalid.

To suspend the ongoing DMA transaction, mask or clear the transfer request or clear the EN bit in the CHSTAT_n register. For details on the procedure, see [section 31.15.8.3. Aborting a Transfer](#).

If a DMA transfer request from the USB control and a transfer request by software (by setting the STG bit in the CHCFG_n register to 1) are used together for the same channel, enabled activation sources cannot be identified. Use only a transfer request in the system.

To make a transfer request by software, make sure that the previously requested DMA transfer operation has been completed (by checking the current register or in other ways), and then set the next STG bit.

EN bit (Enable)

[Setting condition]

- 1 is written to SETEN in the CHCTRL_n register.

[Clearing conditions]

- 1 is written to SWRST in the CHCTRL_n register.
- 1 is written to CLREN in the CHCTRL_n register.
- When an error response is received during a transfer.
- When all DMA transactions are completed in register mode (REN = 0 in the CHCFG_n register)
- When DMA transfer of the last descriptor (LE = 1) is completed in link mode (write-back when WBD = 0)
- When descriptor read operation stops in link mode (LV = 0 and DRRP = 0 in the CHCFG_n register)

RQST bit (Request)

[Setting conditions]

- 1 is written to STG in the CHCTRL_n register.
- When a DMA transfer request is received from the USB control

[Clearing conditions]

- 1 is written to SWRST in the CHCTRL_n register.
- 1 is written to CLRRQ in the CHCTRL_n register.
- When a transfer is performed on the side specified by REQD in the CHCFG_n register in single transfer mode (TM = 0 in the CHCFG_n register)
- When all DMA transactions are completed in register mode (REN = 0 in the CHCFG_n register)
- When DMA transfer of the last descriptor (LE = 1) is completed in link mode
- When descriptor read operation stops in link mode (LV = 0 and DRRP = 0 in the CHCFG_n register)
- When a DMA transaction is completed in link mode with DEM in the CHCFG_n register cleared to 0
- When the master interface receives a bus error

TACT bit (Transaction Active)

[Setting condition]

- 1 is written to SETEN in the CHCTRL_n register (descriptor read start or waiting for a DMA request)

[Clearing condition]

- Idle internal state (EN in the CHSTAT_n register is cleared to 0 and all transfers are completed.)

SUS bit (Suspend)

[Setting condition]

- When the internal state becomes suspended state after 1 is written to SETSUS in the CHCTRL_n register while DMA transfer of channel n is in progress.

[Clearing conditions]

- 1 is written to CLRSUS in the CHCTRL_n register.
- 1 is written to CLREN in the CHCTRL_n register.
- Conditions for clearing EN in the CHSTAT_n register are met.

ER bit (Error)

[Setting condition]

- When an error response is received during a bus cycle.

[Clearing condition]

- 1 is written to SWRST in the CHCTRL_n register.

END bit (USB_FDMAn Interrupted)

[Setting conditions]

- Conditions for setting the TC bit are met when DEM in the CHCFG_n register is 0.
- When LV = 0 in the header, DRRP = 0 in the CHCFG_n register, and DIM = 0 during the descriptor read operation in link mode.

[Clearing conditions]

- 1 is written to CLREND in the CHCTRL_n register.
- 1 is written to SWRST in the CHCTRL_n register.

TC bit (Terminal Count)

[Setting conditions]

- When transfers for the total number of transfer bytes specified in the CRTB register are completed in register mode.
- When transfers for the total number of transfer bytes specified in the CRTB register are completed in link mode while WBD = 1 in the descriptor's header.
- When descriptor write-back is completed in link mode while WBD = 0 in the descriptor's header.

[Clearing conditions]

- 1 is written to CLRTC in the CHCTRL_n register.
- 1 is written to SWRST in the CHCTRL_n register.

SR bit (Selected Register Set)

[Setting condition]

- 1 is written to the RSEL bit in the CHCFG_n register to set it.

[Clearing condition]

- 0 is written to the RSEL bit in the CHCFG_n register to clear it.

DL bit (Descriptor Load)

[Setting condition]

- When descriptor read starts in link mode

[Clearing conditions]

- Descriptor read is completed with an OK response in link mode.
- 1 is written to SWRST in the CHCTRL_n register. If 1 is retained with an error response, this bit can be cleared to 0 only with the SWRST bit.

DW bit (Descriptor WriteBack)

[Setting condition]

- When header write-back starts in link mode.

[Clearing conditions]

- Header write-back is completed with an OK response in link mode.
- 1 is written to SWRST in the CHCTRL_n register. If 1 is retained with an error response, this bit can be cleared to 0 only with the SWRST bit.

DER bit (Descriptor Error)

[Setting condition]

- LV of the read descriptor is 0 when DRRP in the CHCFG_n register is 0 in link mode.

[Clearing conditions]

- 1 is written to the CLRDER bit in the CHCTRL_n register.
- 1 is written to the SWRST bit in the CHCTRL_n register.

INTM bit (Interrupt Mask)

[Setting condition]

- 1 is written to the SETINTM bit in the CHCTRL_n register.

[Clearing conditions]

- 1 is written to the CLRINTM bit in the CHCTRL_n register.
- 1 is written to the SWRST bit in the CHCTRL_n register.

DMARQM bit (DMAREQ Mask)

[Setting condition]

- 1 is written to the SETDMARQM bit in the CHCTRL_n register to set it.

[Clearing conditions]

- Header write-back in link mode is completed with an OK response.
- 1 is written to the CLRDARQM bit in the CHCTRL_n register.
- 1 is written to the SWRST bit in the CHCTRL_n register.

SWPRQ bit (Sweep Request)

[Setting condition]

- When SETSSWPRQ in the CHCTRL_n register is asserted

[Clearing conditions]

- The amount of data in the buffer becomes 0 because of forced ejection.

- 1 is written to the CLRHSWPRQM bit in the CHCTRL_n register.
- 1 is written to the SWRST bit in the CHCTRL_n register.

DNUM[7:0] bits (Data Number)

[Increment condition]

- When DMA read transfer is completed.

[Decrement condition]

- When DMA write transfer is completed.

[Clearing conditions]

- Conditions for clearing the EN bit are met.
- 1 is written to the SWRST bit in the CHCTRL_n register.

31.2.17.2 CHCTRL_n : Channel Control Register n (n = 0, 1)

Base address: USBF = 0x8020_1000

Offset address: 0x428 + 0x40 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	CLRD MARQ M	SETD MARQ M	CLRIN TM	SETIN TM
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	SETS SWPR Q	—	SETR EN	—	—	CLRS US	SETS US	CLRD ER	CLRT C	CLRE ND	CLRR Q	SWRS T	STG	CLRE N	SETE N
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SETEN	Set Enable Enables DMA transfer of DMA channel n. If this bit is set together with the SWRST bit, clearing by the SWRST bit takes precedence, and a transfer does not start. 0: No effect 1: DMA transfer is enabled. (The EN bit in the CHSTAT_n register is set)	W
1	CLREN	Clear Enable Clears the EN bit in the CHSTAT_n register. For details, see section 31.15.8.3. Aborting a Transfer . 0: No effect 1: DMA transfer is disabled. (The EN bit in the CHSTAT_n register is cleared)	W
2	STG	Software Trigger Activates the software (setting the internal transfer request). If this bit is set together with the SWRST bit, clearing by the SWRST bit takes precedence. 0: No effect 1: A transfer request is set by software. (The RQST bit in the CHSTAT_n register is set)	W
3	SWRST	Software Reset Clears each bit in the CHSTAT_n register. (For details on bits to be cleared, see descriptions of each bit.) Set this bit to 1 when the EN bit and the TACT bit are 0. 0: No effect 1: Clears each bit in the CHSTAT_n register	W

Bit	Symbol	Function	R/W
4	CLRRQ	Clear Request Clears the RQST bit in the CHSTAT_n register. 0: No effect 1: Clears the RQST bit in the CHSTAT_n register	W
5	CLREND	Clear End Clears the END bit in the CHSTAT_n register, and clears USB_FDMAn to low level. 0: No effect 1: Clears the END bit in the CHSTAT_n register	W
6	CLRTC	Clear TC Clears the TC bit in the CHSTAT_n register. 0: No effect 1: Clears the TC bit in the CHSTAT_n register	W
7	CLRDER	Clear DER Clears the DER bit in the CHSTAT_n register, and clears USB_FDMAn to low level. 0: No effect 1: Clears the DER bit in the CHSTAT_n register	W
8	SETSUS	Set Suspend Setting this bit to 1 when the EN bit in the CHSTAT_n register is 1 suspends the ongoing DMA transfer. 0: No effect 1: Suspends the ongoing DMA transfer	W
9	CLRSUS	Clear Suspend Setting this bit to 1 when the SUS bit in the CHSTAT_n register is 1 clears the suspended state. 0: No effect 1: Clears the suspended status of the ongoing DMA transfer	W
11:10	—	The write value should be 0.	W
12	SETREN	Set Register Set Enable Sets the REN bit in the CHCFG_n register. 0: No effect 1: Sets the REN bit in the CHCFG_n register to 1	W
13	—	The write value should be 0.	W
14	SETSSWPRQ	Set Software Sweep Request Forcibly ejects the data in the buffer to the transfer destination. See section 31.15.3. Forced Ejection Request . If the transfer destination asserts a hardware request (REQD = 1), forced ejection is not available (disabled by hardware). 0: No effect 1: Writes the buffer data (that has not been written to the transfer destination) to the transfer destination	W
15	—	The write value should be 0.	W
16	SETINTM	Set Interrupt Mask Temporarily masks USB_FDMAn and sets the INTM bit in the CHSTAT_n register to 1. 0: No effect 1: Masks USB_FDMAn	W
17	CLRINTM	Clear Interrupt Mask Clears the NT_DMA [n] pin output mask status, and clears the INTM bit in the CHSTAT_n register to 0. 0: No effect 1: Clears the mask status set by the SETINTM bit	W
18	SETDMARQM	SET DMAREQ Mask Temporarily masks a DMA transfer request from the USB control, and sets the DMARQM bit in the CHSTAT_n register to 1. 0: No effect 1: Masks a DMA transfer request from the USB control	W

Bit	Symbol	Function	R/W
19	CLRDMARQM	Clear DMAREQ Mask Clears the temporary mask status of the DMA transfer request from the USB control, and clears the DMARQM bit in the CHSTAT_n register to 0. 0: No effect 1: Clears the mask status set by the SETDMARQM bit	W
31:20	—	The write value should be 0.	W

The CHCTRL_n register controls DMA transfer of DMA channel n (n = 1, 0). This register is used to activate each function, and it does not retain written values.

31.2.17.3 CHCFG_n : Channel Configuration Register n (n = 0, 1)

Base address: USBF = 0x8020_1000

Offset address: 0x42C + 0x40 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	DMS	REN	RSW	RSEL	SBE	DIM	—	DEM	WONL Y	TM	DAD	SAD	DDS[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	SDS[3:0]				DRRP	AM[2:0]			—	LVL	HIEN	LOEN	REQD	—	—	SEL
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SEL	Terminal Select Selects the FIFO channel to be used on the USB control side. 0: D0FIFO is used 1: D1FIFO is used	R/W
2:1	—	These bits are read as 0. The write value should be 0.	R/W
3	REQD	Request Direction Specifies whether the USB control side is source or destination. 0: The USB control side is source (initial value) 1: The USB control side is destination	R/W
4	LOEN	Sets the transfer request signal between the USB control and the DMAC. Always set this bit to 0.	R/W
5	HIEN	Sets the transfer request signal between the USB control and the DMAC. Always set this bit to 1.	R/W
6	LVL	Sets the transfer request signal between the USB control and the DMAC. Always set this bit to 1.	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W
10:8	AM[2:0]	These bits set the transfer request signal between the USB control and the DMAC. Always set these bits to 000b.	R/W
11	DRRP	Descriptor Read Repeat Switches operation when LV in the header is 0 after reading the descriptor. (See Figure 31.22 .) 0: Sets the DER bit in the CHSTAT_n register to stop operation (initial value). 1: Reads the same descriptor continuously until LV becomes 1. When LV = 1, DMA transfer using the descriptor value starts. The descriptor read interval is controlled by the DSCITVL register.	R/W

Bit	Symbol	Function	R/W
15:12	SDS[3:0]	Source Data Size These bits set the DMA transfer size. When SDS[3] is set to 0, transfer mode is changed to normal mode. When SDS[3] is set to 1, transfer mode is changed to skip mode. Transfer size is set by SDS[2:0]. (For settable values, see Table 31.19.) 0 0 0: 8 bits (initial value) 0 0 1: 16 bits 0 1 0: 32 bits 0 1 1: 64 bits 1 0 0: 128 bits 1 0 1: 256 bits 1 1 0: 512 bits 1 1 1: 1024 bits	R/W
19:16	DDS[3:0]	Destination Data Size These bits set the DMA transfer size. When the USB control side is transfer destination, use the normal mode. When DDS[3] is set to 0, transfer mode is changed to normal mode. When SDS[3] is set to 1, transfer mode is changed to skip mode. Transfer size is set by DDS[2:0]. (For settable values, see Table 31.19.) 0 0 0: 8 bits (initial value) 0 0 1: 16 bits 0 1 0: 32 bits 0 1 1: 64 bits 1 0 0: 128 bits 1 0 1: 256 bits 1 1 0: 512 bits 1 1 1: 1024 bits	R/W
20	SAD	Source Address Direction Sets the count direction of the transfer source address of DMA channel n. When the USB control side is transfer source, set this bit to 1 (fixed). When using the skip mode on the transfer source side or when the transfer source is beat-unaligned, do not specify SAD = 1 (fixed). 0: Increment (initial value) 1: Fixed	R/W
21	DAD	Destination Address Direction Sets the count direction of the transfer destination address of DMA channel n. When the USB control side is transfer destination, set this bit to 1 (fixed). When using the skip mode on the transfer destination side or when the transfer destination is beat-unaligned, do not specify DAD = 1 (fixed). 0: Increment (initial value) 1: Fixed	R/W
22	TM	Sets the transfer request signal between the USB control and the DMAC. Always set this bit to 0.	R/W
23	WONLY	Write Only Mode Sets the write-only mode. For details, see section 31.14.1.3. Write-Only Mode. 0: Normal operation (initial value) 1: Write-only mode	R/W
24	DEM	USB_FDMAn Mask Masks USB_FDMAn interrupt detection. When this bit is 1 at USB_FDMAn timing, USB_FDMAn and END in the CHSTAT_n register are not asserted. In register mode, the DEM bit is automatically cleared to 0. In link mode, this bit is not cleared. 0: Does not mask USB_FDMAn interrupt detection (initial value) 1: Masks USB_FDMAn interrupt detection	R/W
25	—	This bit is read as 0. The write value should be 0.	R/W
26	DIM	Descriptor Interrupt Mask Sets masking for USB_FDMAn when LV = 0 after the descriptor's header is read. 0: Does not mask USB_FDMAn (initial value) 1: Masks USB_FDMAn	R/W

Bit	Symbol	Function	R/W
27	SBE	Sweep Buffer Enable Selects whether to sweep (by writing) the read data in the buffer before aborting the transfer when EN in the CHSTAT_n register is cleared to 0 during a DMA transaction. Sweep mode can be used only when REQD = 0. 0: Does not sweep buffer data and aborts the transfer (initial value) 1: Sweeps buffer data and aborts the transfer	R/W
28	RSEL	Register Set Select Selects the next register set to be executed next. This bit is valid only in register mode. When RSW = 1, this bit is automatically reversed upon completion of a DMA transaction. 0: Executes Next0 register set (initial value) 1: Executes Next1 register set	R/W
29	RSW	Register Select Switch The RSEL bit is automatically reversed upon completion of a DMA transaction. This bit is valid only in register mode. 0: Does not reverse the RSEL bit upon completion of a DMA transaction (initial value) 1: Reverses the RSEL bit upon completion of a DMA transaction	R/W
30	REN	Register Set Enable Performs DMA transaction of the next register set selected by the RSEL bit continuously after the current DMA transaction is completed. This bit is valid only in register mode. 0: Does not perform DMA transaction continuously 1: Performs DMA transaction continuously	R/W
31	DMS	DMA Mode Select Sets DMA mode. 0: Register mode (initial value) 1: Link mode	R/W

The CHCFG_n register controls DMA transfer of DMA channel n (n = 1, 0).

The settable size ranges of the SDS[2:0] and DDS[2:0] bits depend on the data bus width, number of implemented buffers, and alignment type of addresses to be accessed (beat-aligned or beat-unaligned). The following table shows the settable size ranges. These ranges are specified to prevent this controller from malfunctioning. Set appropriate values according to the module to be accessed.

Table 31.19 Settable size ranges of SDS[2:0] and DDS[2:0] bits

Data bus width [bit]	Number of buffers	Transfer address	Settable values of SDS[2:0] and DDS[2:0] bits
32	16	Beat-aligned	8 to 32 bits (000b to 010b) 128 to 512 bits (100b to 110b)
		Beat-unaligned	8 to 32 bits (000b to 010b) 128 to 256 bits (100b to 101b)
		Beat-unaligned	8 to 32 bits (000b to 010b)

If both transfer source and destination or either of these is beat-unaligned, set the SDS[2:0] and DDS[2:0] bits to values within the settable ranges for the beat-unaligned addresses from among the CHCFG_n register settings.

- Even when beat-aligned setting is made at the beginning of DMA transaction, if a skip transfer is used, beat-aligned setting may change to beat-unaligned setting during the DMA transaction. In this case, make beat-unaligned setting from the beginning.
- If the software cannot distinguish beat-aligned or beat-unaligned setting, use the beat-unaligned setting range.

DEM bit (USB_FDMA_n Mask)

[Clearing condition]

A DMA transaction is completed when DEM = 1.

RSEL bit (Register Set Select)

[Transition condition]

A DMA transaction is completed when RSW = 1.

31.2.18 Link Register Set

The link register set is used to set and display the descriptor address in link mode.

When the descriptor address is set in the NXLA_n register by software to activate the DMAC, the NXLA_n register value is loaded to the CRLA_n register by hardware and the descriptor is read. The DMAC starts a DMA transaction according to the read descriptor value. The NXLA_n register value is automatically updated by the next link address value in the read descriptor to be used as the descriptor address for the next DMA transaction.

31.2.18.1 NXLA_n : Next Link Address Register n (n = 0, 1)

Base address: USBF = 0x8020_1000

Offset address: 0x438 + 0x40 × n

Bit position: 31

0

Bit field:

NXLA[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	NXLA[31:0]	Next Link Address These bits set the address of the link destination. Because the two lower-order bits are fixed to 0, only word-aligned addresses can be set.	R/W

The NXLA_n register sets the link address of DMA channel n (n = 1, 0).

31.2.18.2 CRLA_n : Current Link Address Register n (n = 0, 1)

Base address: USBF = 0x8020_1000

Offset address: 0x43C + 0x40 × n

Bit position: 31

0

Bit field:

CRLA[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	CRLA[31:0]	Current Link Address These bits display the address of the descriptor that is being executed.	R

The CRLA_n register displays the link address of DMA channel n (n = 1, 0).

31.2.19 Skip Register Set

The skip register set is used for settings for skip (scatter/gather) transfer.

31.2.19.1 SCNT_n : Source Continuous Register n (n = 0, 1)

Base address: USBF = 0x8020_1000

Offset address: 0x600 + 0x20 × n

Bit position: 31

0

Bit field:

SCNT[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	SCNT[31:0]	Source Continuous These bits set the size (in bytes) of continuous access space for access to the transfer source address.	R/W

The SCNT_n register sets the space size for continuous access to the transfer source address of DMA channel n (n = 1, 0).

Use this register together with the SSKP_n register (see [Figure 31.4](#)).

When using this mode, set the SDS[3] bit in the CHCFG_n register to 1.

When performing a skip transfer on the transfer source side, do not set the SAD bit in the CHCFG_n register to 1 (fixed). In addition, do not perform a skip transfer when SCNT_n is set to 0x00000000.

31.2.19.2 SSKP_n : Source Skip Register n (n = 0, 1)

Base address: USBF = 0x8020_1000

Offset address: 0x604 + 0x20 × n

Bit position: 31

0

Bit field:

SSKP[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	SSKP[31:0]	Source Skip These bits set the skip amount (in bytes) during access to the source address.	R/W

The SSKP_n register sets the skip amount for skipping the next source address after accessing the data size set by the SCNT_n register during access to the source address of DMA channel n (n = 1, 0).

Use this register together with the SCNT_n register (see [Figure 31.4](#)).

When using this mode, set the SDS[3] bit in the CHCFG_n register to 1.

When performing a skip transfer on the transfer source side, do not set the SAD bit in the CHCFG_n register to 1 (fixed).

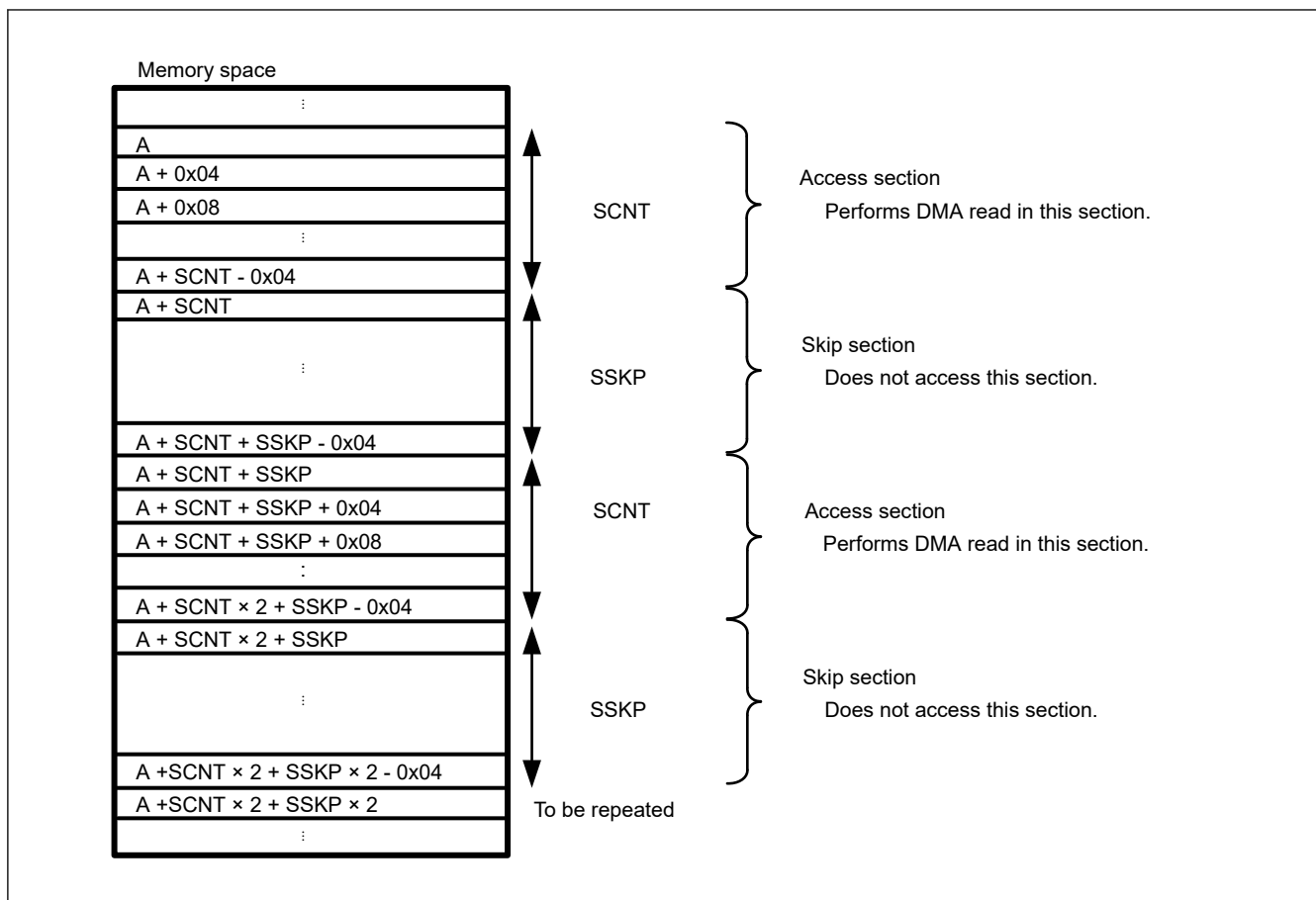


Figure 31.4 Relationship between SSKP and SCNT

The SCNT and SSKP_n values can be set regardless of the source address and the set value of the SDS field in the CHCFG_n register. The DMAC accesses a memory space with the size set in the SDS field, and acquires only valid data into the buffer. For details, see [section 31.16.1.1. Read Access](#).

31.2.19.3 DCNT_n : Destination Continuous Register n (n = 0, 1)

Base address: USBF = 0x8020_1000

Offset address: 0x608 + 0x20 × n

Bit position: 31

0

Bit field:

DCNT[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	DCNT[31:0]	Destination Continuous These bits set the space size (in bytes) for continuous access during an access to the destination address.	R/W

The DCNT_n register sets the space size for continuous access during an access to the destination address of DMA channel n (n = 1, 0).

Use this register together with the DSKP_n register (see [Figure 31.5](#)).

When using this mode, set the DDS[3] bit in the CHCFG_n register to 1.

When performing a skip transfer on the transfer destination side, do not set the DAD bit in the CHCFG_n register to 1 (fixed). In addition, do not perform a skip transfer when DCNT is set to 0x00000000.

31.2.19.4 DSKP_n : Destination Skip Register n (n = 0, 1)

Base address: USBF = 0x8020_1000

Offset address: 0x60C + 0x20 × n

Bit position: 31

0

Bit field:

DSKP[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	DSKP[31:0]	Destination Skip These bits set the skip amount (in bytes) during access to the destination address.	R/W

The DSKP_n register sets the skip amount for skipping the next destination address after accessing the data size set by the DCNT_n register during access to the destination address of DMA channel n (n = 1, 0).

Use this register together with the DCNT_n register (see [Figure 31.5](#)).

When using this mode, set the DDS[3] bit in the CHCFG_n register to 1.

When performing a skip transfer on the transfer destination side, do not set the DAD bit in the CHCFG_n register to 1 (fixed).

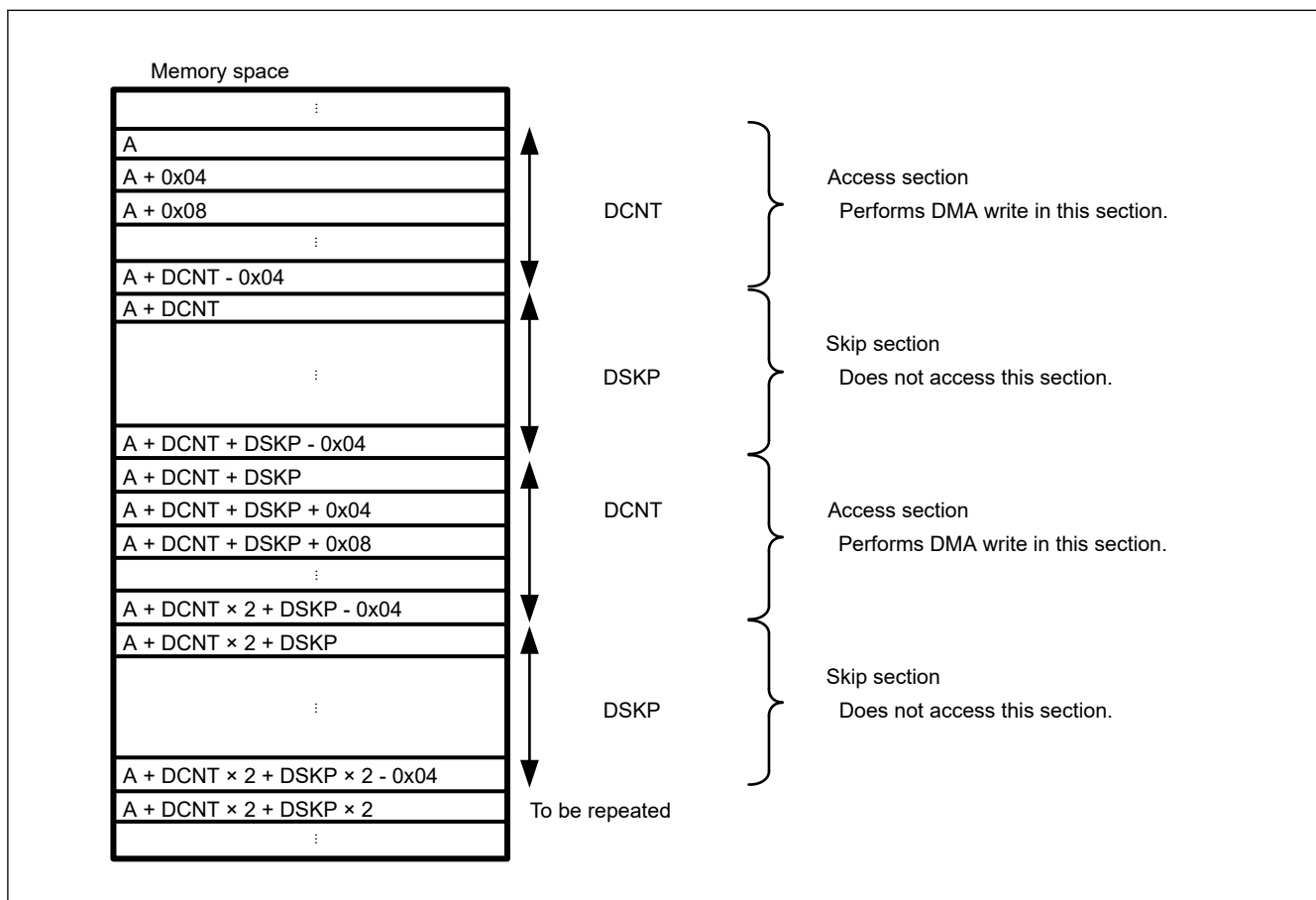


Figure 31.5 Relationship between DSKP and DCNT

The DCNT and DSKP_n values can be set regardless of the destination address and the set value of the DDS field in the CHCFG_n register. The DMAC performs write access only to the specified space with the size equal to or smaller than the value set in the DDS field. For details, see [section 31.16.1.2. Write Access](#).

31.2.20 DMA Register Set

The following registers are common to all channels.

31.2.20.1 DCTRL : DMA Control Register

Base address: USBF = 0x8020 1000

Offset address: 0x700

[illegible]

Bit	Symbol	Function	R/W
0	PR	Priority Sets transfer priority control mode. See section 31.15.2. DMA Channel Priority Control . 0: Fixed priority mode 1: Round-robin mode	R/W
15:1	—	These bits are read as 0. The write value should be 0.	R/W
19:16	LDPR[3:0]	Link Descriptor PROT These bits set the value to be output to the MHPROT[3:0] pins during descriptor read in link mode.	R/W
23:20	—	These bits are read as 0. The write value should be 0.	R/W
27:24	LWPR[3:0]	Link WriteBack PROT These bits set the value to be output to the MHPROT[3:0] pins during descriptor writeback in link mode.	R/W
31:28	—	These bits are read as 0. The write value should be 0.	R/W

The DCTRL register sets the transfer type for descriptor access and the arbitration between channels.

31.2.20.2 DSCITVL : Descriptor Interval Register

Base address: USBF = 0x8020 1000

Offset address: 0x704

[illegible]

Bit	Symbol	Function	R/W
7:0	—	These bits are read as 0. The write value should be 0.	R/W
15:8	DITVL[7:0]	Descriptor Interval These bits set the descriptor read interval. The descriptor is read again with an interval of (DITVL × 256) cycles.	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

The DSCITVL register sets the descriptor read interval.

By setting the DRRP bit in the CHCFG_n register to 1, the descriptor is continuously read until it reaches LV = 1. This register sets the read interval.

31.2.20.3 DSTAT_EN : DMA Status EN Register

Base address: USBF = 0x8020_1000

Offset address: 0x710

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	EN1	EN0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	EN0	Channel 0 EN Displays the EN bit status of DMA channel 0.	R
1	EN1	Channel 1 EN Displays the EN bit status of DMA channel 1.	R
31:2	—	These bits are read as 0.	R

The DSTAT_EN register displays the EN bit status of all channels.

Writing data to this register does not change the value of each bit.

31.2.20.4 DSTAT_ER : DMA Status ER Register

Base address: USBF = 0x8020_1000

Offset address: 0x714

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ER1	ER0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ER0	Channel 0 ER Displays the ER bit status of DMA channel 0.	R
1	ER1	Channel 1 ER Displays the ER bit status of DMA channel 1.	R
31:2	—	These bits are read as 0.	R

The DSTAT_ER register displays the ER bit status of all channels.

Writing data to this register does not change the value of each bit.

31.2.20.5 DSTAT_END : DMA Status END Register

Base address: USBF = 0x8020_1000

Offset address: 0x718

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	END1	END0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	END0	Channel 0 END Displays the END bit status of DMA channel 0.	R
1	END1	Channel 1 END Displays the END bit status of DMA channel 1.	R
31:2	—	These bits are read as 0.	R

The DSTAT_END register displays the END bit status of all channels.

Writing data to this register does not change the value of each bit.

31.2.20.6 DSTAT_TC : DMA Status TC Register

Base address: USBF = 0x8020_1000

Offset address: 0x71C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	TC1	TC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TC0	Channel 0 TC Displays the TC bit status of DMA channel 0.	R
1	TC1	Channel 1 TC Displays the TC bit status of DMA channel 1.	R
31:2	—	These bits are read as 0.	R

The DSTAT_TC register displays the TC bit status of all channels.

Writing data to this register does not change the value of each bit.

31.2.20.7 DSTAT_SUS : DMA Status SUS Register

Base address: USBF = 0x8020_1000

Offset address: 0x720

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SUS1	SUS0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SUS0	Channel 0 SUS Displays the SUS bit status of DMA channel 0.	R
1	SUS1	Channel 1 SUS Displays the SUS bit status of DMA channel 1.	R
31:2	—	These bits are read as 0.	R

The DSTAT_SUS register displays the SUS bit status of all channels.

Writing data to this register does not change the value of each bit.

31.3 Operation

31.3.1 System Controls and Oscillation Controls

This section describes the register operations required for initial settings of this module, and the registers necessary for power consumption control. The startup sequence is as follows.

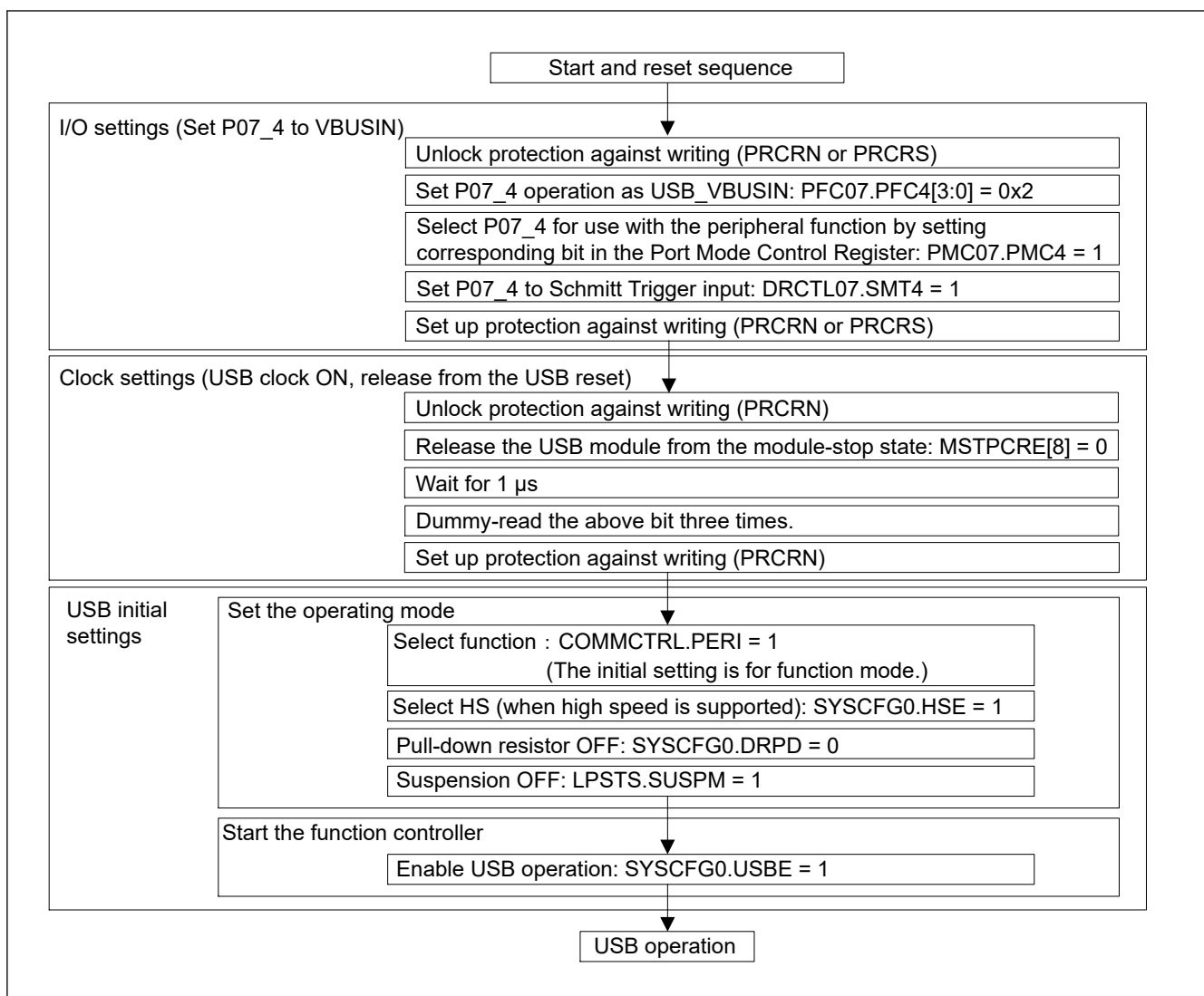


Figure 31.6 Startup sequence

31.3.2 Resets

Table 31.20 lists the types of reset for this controller. For the initial states of the registers following the reset operations, see .

Table 31.20 Types of reset

Name	Operation
Hardware reset	Low level input from the RES# pin
USB bus reset	Automatically detected by this module from the D+ and D- lines

31.3.3 USB Data Bus Resistor Controller

This controller controls the pull-up resistors for the D+ signal of the USB2.0-PHY port.

Make pull-up setting for the D+ signal using the DPRPU and DRPD bits of the SYSCFG0 register.

Confirm the connection to the USB Host before setting the DPRPU bit of the SYSCFG0 register to 1 to pull up D+.

This controller incorporates termination resistors for the D+ and D- signals (in high-speed operation mode) and output resistors (in full-speed operation mode). The switching of the internal resistors following the connection to the PC is automatically done by the controller during reset handshake, suspend, and resume processing.

If the DPRPU bit of the SYSCFG0 register is set to 0 during communication with the PC, the controller disables the pull-up resistors (or termination resistors) of the USB data lines, so that it can notify the host controller of the disconnection of the device.

31.3.4 Notes on Stopping Clocks

- The PHY clock can be stopped with the SUSPM register in suspend mode.
- If a clock is to be stopped while the controller is in USB suspended state, however, it is necessary to resume the supply of the clock during resume processing. The resumption of the PHY clock need be accomplished within 5.5 ms after a resume interrupt occurs.
- When the USB function is not in use, place the USB module in the module-stop state (MSTPCRE[8] = 1).

31.4 Interrupt Functions

31.4.1 Overview of Interrupt Functions

Table 31.21 lists the interrupt functions of this controller.

Table 31.21 List of interrupt functions

Bit	Interrupt name	Interrupt source	Related status
VBINT	VBUS interrupt	When a change in the state of the USB_VBUSIN input pin has been detected (low to high or high to low).	VBSTS
RESM	Resume interrupt	When a change in the state of the USB bus has been detected in the suspended state (J-state to K-state or J-state to SE0)	—
SOFR	Frame number update interrupt	If SOFRM = 0: When an SOF packet with a different frame number is received. If SOFRM = 1: When an SOF packet for a μ frame number of 0 cannot be received due to a corrupted packet.	—
DVST	Device state transition interrupt	When transition in device state is detected: A USB bus reset detected The suspended state detected Set Address request received Set Configuration request received	DVSQ
CTRT	Control transfer stage transition interrupt	When a stage transition is detected in control transfer: Setup stage completed Control write transfer status stage transition Control read transfer status stage transition Control transfer completed A control transfer sequence error occurred	CTSQ
BEMP	Buffer empty interrupt	When the buffer becomes empty after all data in the buffer memory has been transmitted. When a packet exceeding the maximum packet size is received.	PIPEBEMP
NRDY	Buffer not ready interrupt	When a token is received when PID = BUF and the buffer memory is not ready to transmit or receive data. When a CRC error or bit stuff error occurs while receiving data in isochronous transfer. When an interval error occurs while receiving data in isochronous transfer.	PIPENRDY
BRDY	Buffer ready interrupt	When the buffer is ready (readable or writable).	PIPEBRDY

Table 31.22 shows operations for a USBf interrupt output from this controller. In case more than one interrupt source is generated, the USBf interrupt output method can be set by using the INTL bit in the SOFCFG register. Set the USBf interrupt output operation according to the user system.

Table 31.22 Operations for USBf Interrupt (USB_FI)

INTL setting	USBf interrupt output pin operation	
	When one interrupt source is generated	When more than one interrupt source is generated
Edge sense (INTL = 0)	Holds the pin low until the interrupt source is cleared.	Negated (L pulse output) for 40 clock cycles at 60 MHz when one source is cleared.
Level sense (INTL = 1)	Holds the pin low until the interrupt source is cleared.	Holds the pin low until all sources are cleared.

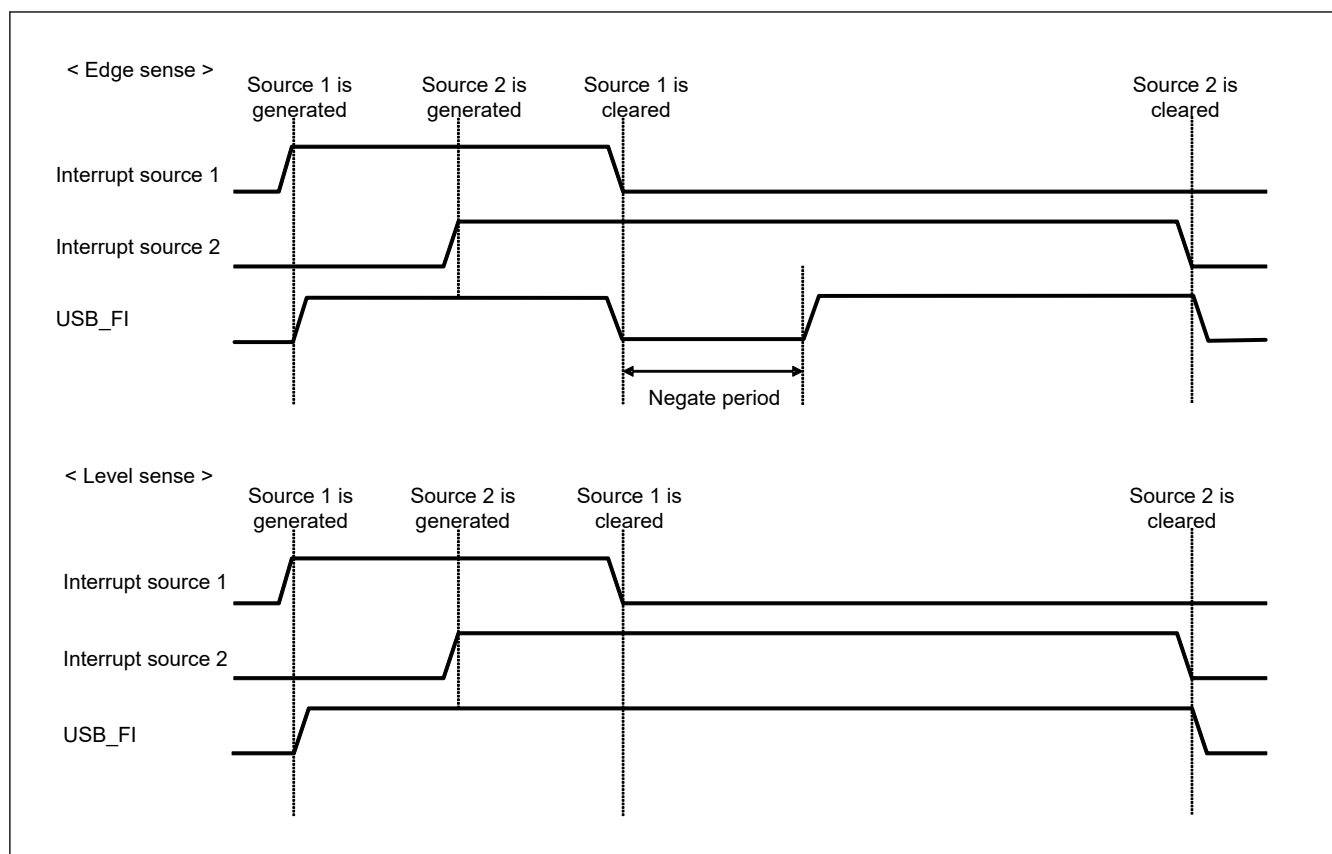


Figure 31.7 USBf interrupt output operation diagram

[Figure 31.8](#) shows the interrupt configurations for the controller.

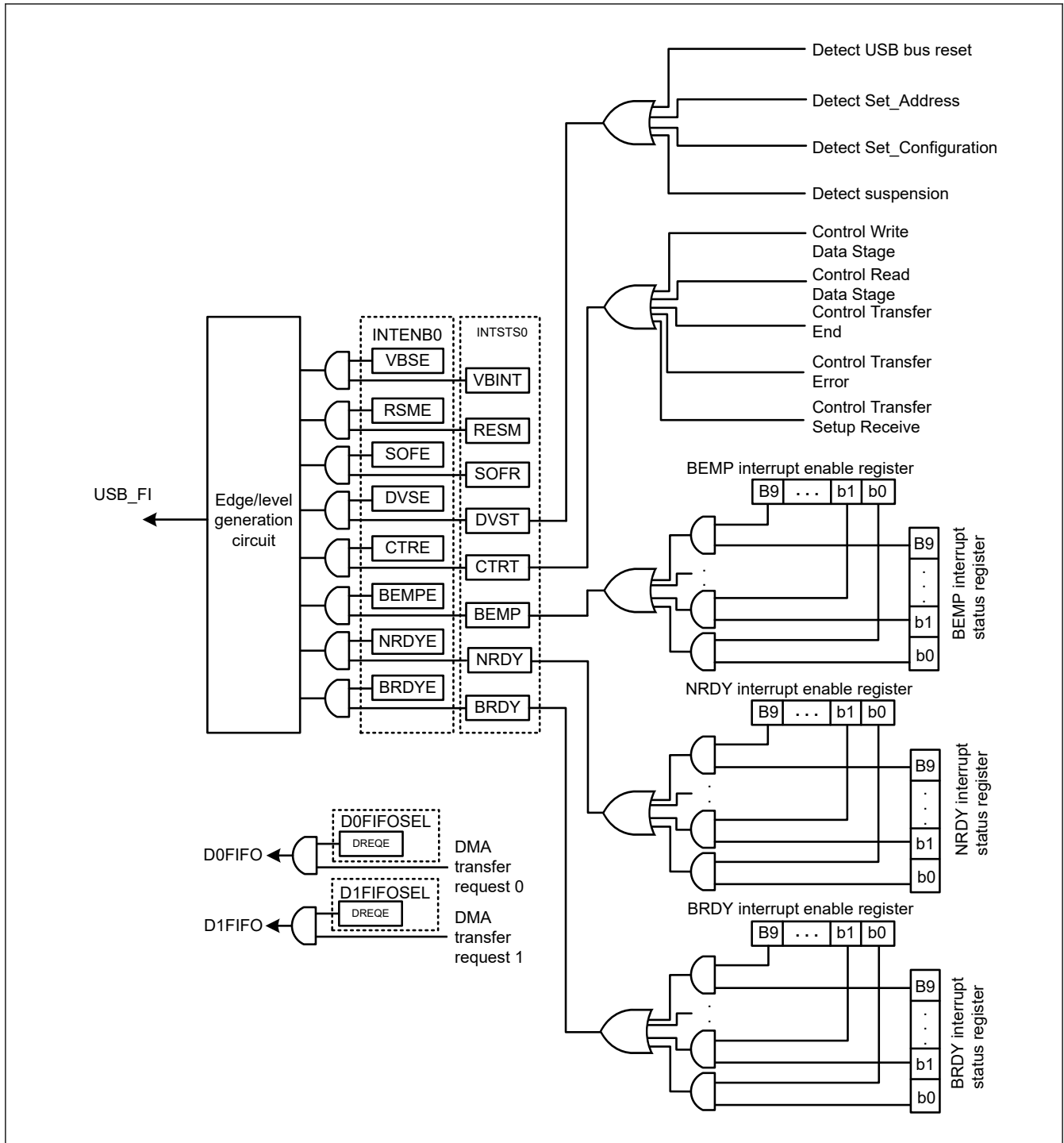


Figure 31.8 Interrupt configuration diagram

31.4.2 Device State Transition Interrupt

Figure 31.9 shows a diagram of how this module handles the device state transitions. The controller monitors the device states and generates the device state transition interrupts. However, recovery from the suspended state (resume signal detection) is detected by the resume interrupt. The device state transition interrupts can be enabled or disabled individually by setting the INTENB0 register. The device state after a transition can be confirmed using the DVSQ bits in the INTSTS0 register.

When making a transition to the default state, the device state transition interrupt is generated after the reset handshake protocol has been completed.

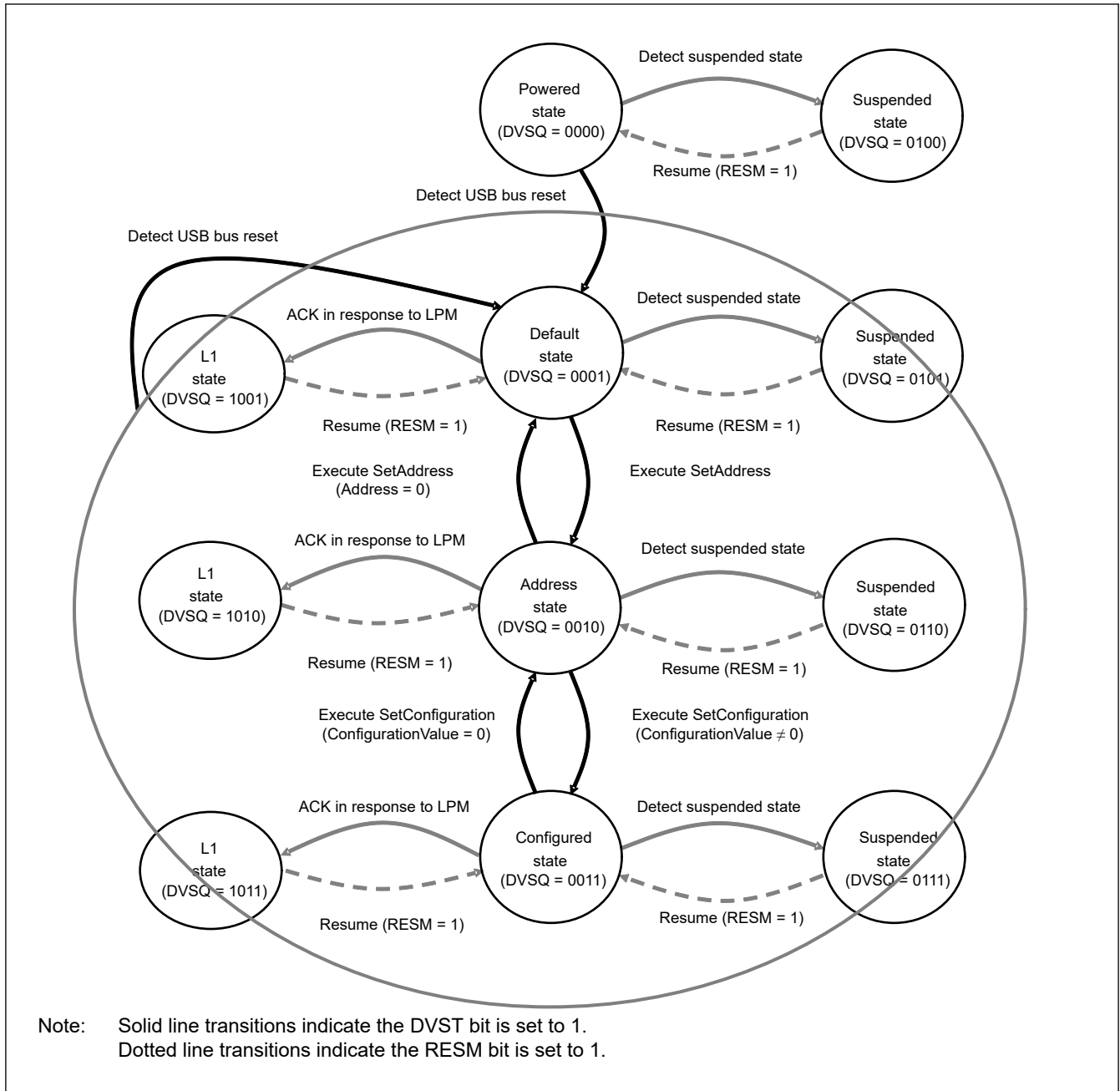


Figure 31.9 Device state transitions

31.4.3 Control Transfer Stage Transition Interrupt

Figure 31.10 shows a diagram of how this module handles the control transfer stage transition. The controller monitors the control transfer sequence and generates the control transfer stage transition interrupts. The control transfer stage transition interrupts can be enabled or disabled individually in the INTENB0 register. The control transfer stage after a transition can be confirmed in the CTSQ bits in the INTSTS0 register.

The control transfer sequence errors are described below. If an error occurs, the PID bits in DCPCTR are set to 1xb (STALL).

1. During control read transfers
 - (a) At the IN token of the data stage, an OUT or PING token is received when there have been no data transfers at all.
 - (b) An IN token is received at the status stage.
 - (c) A packet is received at the status stage for which the data packet is DATAPID = DATA0.
2. During control write transfers

- (a) At the OUT token of the data stage, an IN token is received when there have been no ACK response at all.
 - (b) A packet is received at the data stage for which the first data packet is DATAPID = DATA0.
 - (c) At the status stage, an OUT or PING token is received.
3. During control write no-data transfers
- (a) At the status stage, an OUT or PING token is received.

Note that in the control write transfer data stage, if the number of received data is more than the USB request wLength value, the control transfer sequence error cannot be recognized. Also, in the control read transfer status stage, when a packet other than a zero-length packet is received, an ACK response is returned and the transfer is successfully completed.

When a CTRT interrupt is generated due to a sequence error (SERR = 1), the value of CTSQ = 110b is retained until CTRT = 0 is written by the user system (clearing the interrupt state). Therefore, while CTSQ = 110b is retained, the CTRT interrupt for the completion of the setup stage is not generated, even when a new USB request is received. Events occurring after the setup stage are saved by the controller and the CTRT interrupt is generated after the interrupt state is cleared by software.

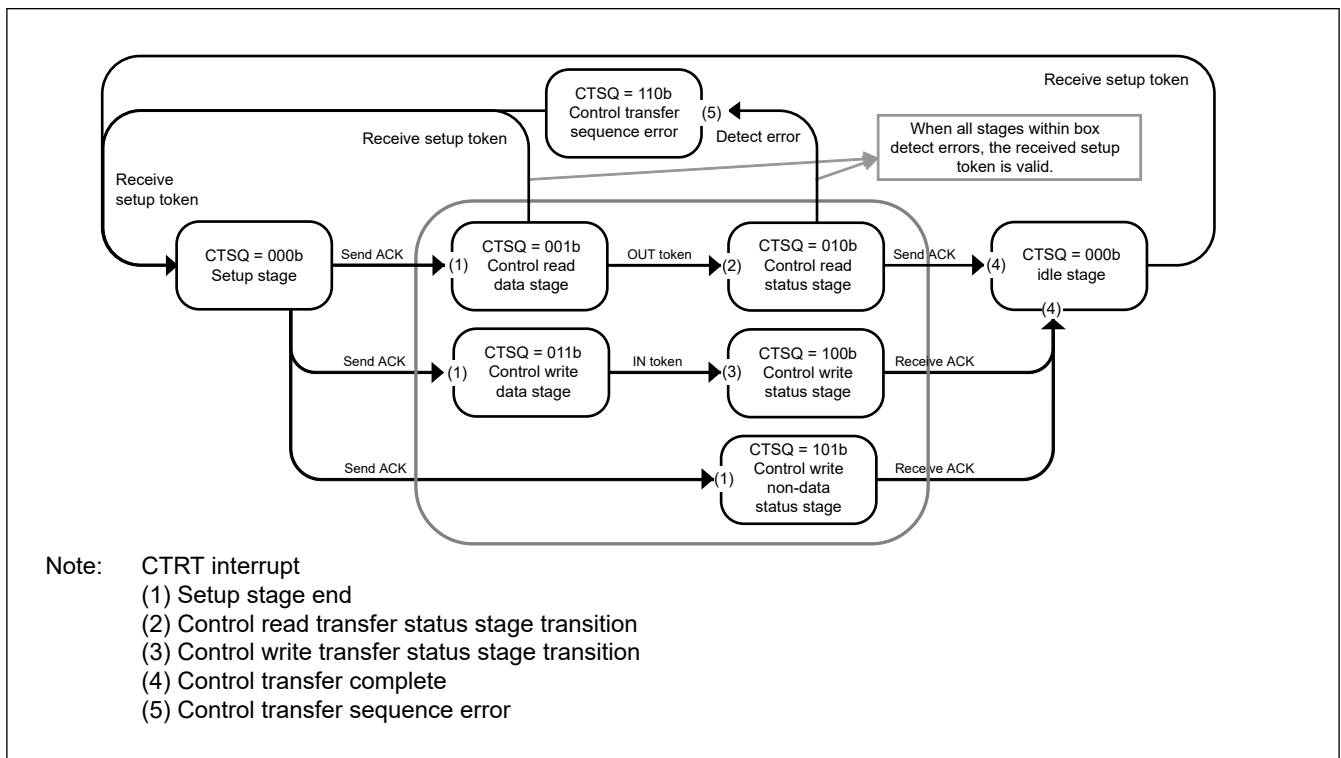


Figure 31.10 Control transfer stage transition

31.5 Pipe Control

Table 31.23 provides a list of pipe settings for the controller. In USB data transfers, data transmission is executed in logic pipes called endpoints. This controller comes with ten pipes for data transfer. Each pipe can be set to meet the requirements of the user system.

Table 31.23 Pipe settings

Register name	Bit name	Setting	Remarks
PIPECFG	TYPE	Specifies the transfer type	Can be set for pipes 1 to 9.
	BFRE	Selects BRDY interrupt mode	Can be set for pipes 1 to 5.
	DBLB	Selects double buffer configuration	Can be set for pipes 1 to 5.
	CNTMD	Selects continuous transfer or noncontinuous transfer	Can be set for pipes 1 and 2 only in bulk transfers. Can be set for pipes 3 to 5.
	DIR	Selects transfer direction	Set to IN or OUT
	EPNUM	Endpoint number	Can be set for pipes 1 to 9. Set to a value other than 0x0 when a pipe is in use.
	SHTNAK	Disables pipe when transfer is completed.	Can be set for pipes 1 and 2 only in bulk transfers. Can be set for pipes 3 to 5.
PIPEBUF	BUFSIZE	Buffer memory size	Cannot be set for DCP (fixed to 256 bytes). Up to 2 KB can be set for pipes 1 to 5. Cannot be set for pipes 6 to 9 (fixed to 64 bytes).
	BUFNMB	Buffer memory number	Cannot be set for DCP (fixed at areas 0x00 to 0x03). Can be set for pipes 1 to 5 (specifiable in area ranging from 0x08 to 0x80). Cannot be set for pipes 6 to 9 (fixed at areas ranging from 0x04 to 0x07).
DCPMAXP PIPEMAXP	MXPS	Maximum packet size	Setting conforming to the USB standard
PIPEPERI	IFIS	Buffer flush	Can be set for pipes 1 and 2 only in isochronous transfers. Cannot be set for pipes 3 to 5. Cannot be set for pipes 6 to 9.
	IITV	Interval counter	Can be set for pipes 1 and 2 only in isochronous transfers. Cannot be set for pipes 3 to 5. Cannot be set for pipes 6 to 9.
DCPCTR PIPEXCTR	BSTS	Buffer status	DCP state switched between receive and transmission buffer by ISEL bit
	INBUFM	IN buffer monitor	Available only for pipes 3 to 5.
	ATREPM	Auto response mode	Can be set for pipes 1 to 5.
	ACLRM	Auto buffer clear	Can be set for pipes 1 to 9.
	SQCLR	Sequence clear	Clears data toggle bit.
	SQSET	Sequence set	Sets data toggle bit.
	SQMON	Sequence check	Monitors data toggle bit.
	PBUSY	Pipe busy check	—
	PID	Response PID	—
PIPEXTRE	TRENB	Transaction count enable	Can be set for pipes 1 to 5.
	TRCLR	Current transaction counter clear	Can be set for pipes 1 to 5.
PIPEXTRN	TRNCNT	Transaction counter	Can be set for pipes 1 to 5.

31.5.1 Maximum Packet Size Setting

The MXPS bits in DCPMAXP and PIPEMAXP are used to specify the maximum packet size for each pipe. DCP and PIPE1 to PIPE5 can be set to any of the maximum packet sizes defined by the USB Specification. For PIPE6 to PIPE9, 64 bytes are the upper limit of the maximum packet size. The maximum packet size should be set before beginning the transfer (PID = BUF).

DCP: 64 should be set when using high-speed operation.

DCP: Select and set 8, 16, 32, or 64 when using full-speed operation.

Pipes 1 to 5: 512 should be set when using high-speed bulk transfer.

Pipes 1 to 5: Select and set 8, 16, 32, or 64 when using full-speed bulk transfer.

Pipes 1 to 2: Set a value between 1 and 1024 when using high-speed isochronous transfer.

Pipes 1 to 2: Set a value between 1 and 1023 when using full-speed isochronous transfer.

Pipes 6 to 9: Set a value between 1 and 64.

The high bandwidth transfers used with interrupt transfers and isochronous transfers are not supported.

31.5.2 Response PID

Set the response PID for each pipe with the PID bits of the DCPCTR and PIPExCTR registers.

1. Response PID setting

The response PID specifies the response to a transaction from the Host.

- (a) NAK setting: Always sends a NAK response when a transaction is issued.
- (b) BUF setting: Responds to the transaction in accordance with the buffer memory state.
- (c) STALL setting: Always sends a STALL response when a transaction is issued.

Regardless of the value set in the PID bit, an ACK is always sent as a response to a setup transaction and the USB request is stored in corresponding registers.

Based on the results of the transaction, the controller may trigger the PID bits to be written.

The controller will trigger a write event to the PID bit in the following cases.

2. Hardware setting of response PID

- (a) NAK setting:
 - (i) When SETUP token is received normally (only DCP)
 - (ii) In bulk transfers when the SHTNAK bit of the PIPECFG register is set to 1 and short packet is received.
 - (iii) In bulk transfers when SHTNAK bit is set to 1 and the transaction counter is completed.
- (b) BUF setting: The BUF cannot be written by the controller.
- (c) STALL setting:
 - (i) When a maximum packet size over error is detected in the received data packet
 - (ii) When a control transfer sequence error is detected

31.5.3 Pipe Control Register Switching Procedures

The following bits in the pipe control registers can be re-written only when USB transmission is disabled (PID = NAK).

Figure 31.11 shows the procedure for switching the pipe control register from the USB transmission enabled (PID = BUF) state.

Registers for which settings are prohibited when the USB transmission is enabled (PID = BUF):

- All bits of the DCPMAXP register
- Bits SQCLR and SQSET of the DCPCTR register
- All bits of the PIPECFG, PIPEBUF, PIPEMAXP and PIPEPERI registers
- ATREPM, ACLRM, SQCLR, and SQSET bits of the PIPExCTR register
- All bits of the PIPEXTRE and PIPExTRN registers

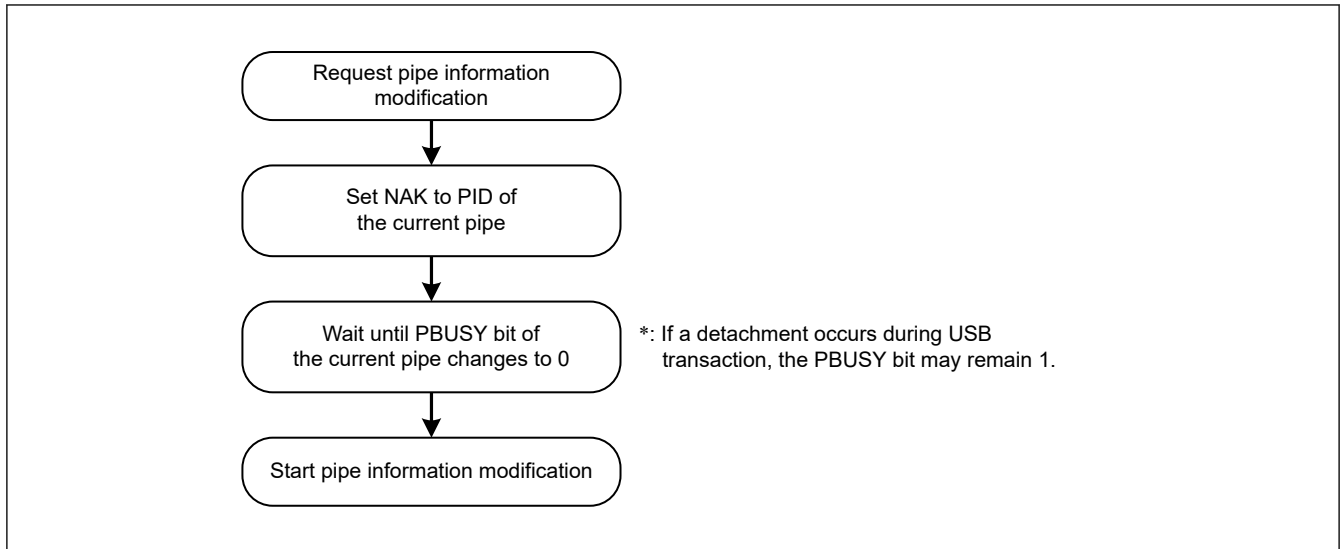


Figure 31.11 Procedure for changing pipe information from USB transmission enabled (PID = BUF) state

In addition, the following bits of the pipe control registers can only be re-written with pipe information that is not set in the CURPIPE bits of CPU/DMA0/DMA1-FIFO ports.

- Register for which settings are prohibited during setting of the CURPIPE bits in the FIFO port register:
 - All bits of the DCPMAXP register
 - All bits of the PIPECFG, PIPEBUF, PIPEMAXP and PIPEPERI registers
 - Bit ACLRM of the PIPExCTR register

When modifying information of a pipe, specify other pipe number in the CURPIPE bits. Also, after setting the DCP pipe information, clear the buffer using the BCLR bit.

31.5.4 Data PID Sequence Bit

When a normal data transfer occurs in the control transfer data stage, bulk transfer or interrupt transfer, the controller automatically toggles the data PID sequence bit. The next data PID sequence bit for data transfer can be confirmed in the SQMON bit in the DCPCTR or PIPExCTR registers. The sequence bit is switched in the ACK handshake receive timing when data is sent or in the ACK handshake send timing when data is received. The data PID sequence bit can also be modified for the SQCLR and SQSET bits of the DCPCTR and PIPExCTR registers.

In control transfers, the controller automatically sets the sequence bit for stage transitions. DATA1 is established at the end of the setup stage. In the status stage, the controller does not reference the sequence bit and returns a response with PID = DATA1. Therefore, the bit does not need to be set with software. Note that the data PID sequence bit must be set with software when a ClearFeature request is received.

Finally, the sequence bit cannot be manipulated through the SQSET bit for the isochronous transfer setup pipe.

31.6 FIFO Buffers

This section describes the operation of the FIFO buffers in this controller.

31.6.1 FIFO Buffer Allocation

Figure 31.12 shows an example of memory map for the FIFO buffers of this controller. The FIFO buffer area is shared by the CPU controlling the user system and this controller. The access right of the FIFO buffers may be given to the user system (CPU side) or to this controller (SIE side).

An independent FIFO buffer area is allocated for each pipe. The memory area is made up of memory blocks of 64 bytes and defined by the starting block number (1 block is 64 bytes long) and the number of blocks (specified by the BUFNMB and BUFSIZE bits of the PIPEBUF register). If the CNTMD bit of the PIPECFG register is set to continuous transfer mode, the value specified in the BUFSIZE bits must be an integral multiple of the maximum packet size. If the double buffer configuration is selected through the DBLB bit of the PIPECFG register, two planes of memory area the size of which is specified by the BUFSIZE bits of the PIPEBUF register are allocated to a single pipe.

Two FIFO ports are used to access an FIFO buffer (data read/write). The pipe to be assigned to an FIFO port is designated by specifying the pipe number in the CURPIPE bits of the CFIFOSEL or DxFIFOSEL register.

The FIFO buffer state of each pipe can be confirmed by using the BSTS and INBUFM bits of the DCPCTR and PIPExCTR registers. The access right of a FIFO port can be confirmed by using the FRDY bit of the CFIFOCTR or DxFIFOCTR register.

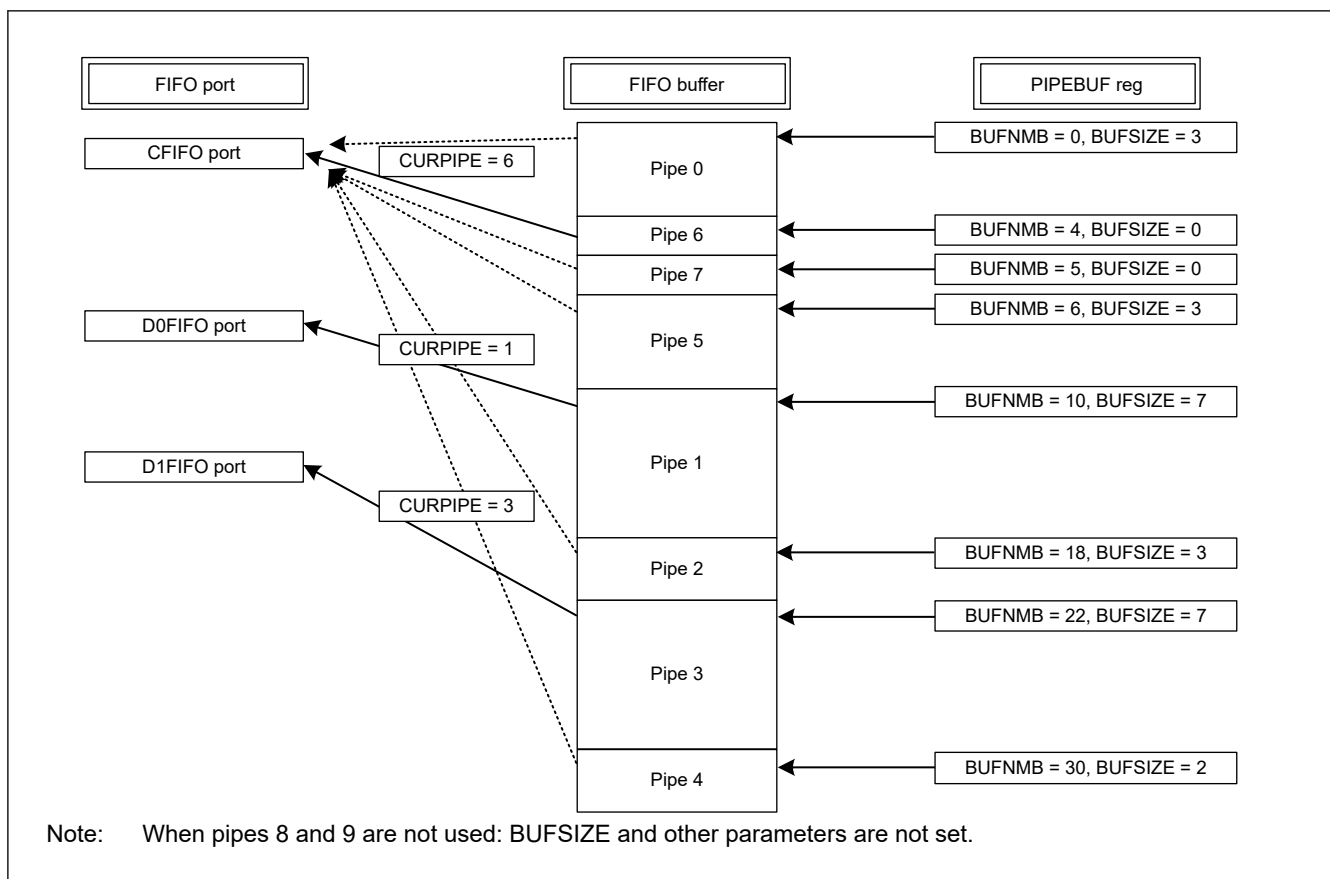


Figure 31.12 Example of FIFO buffer memory map

31.6.2 Clearing FIFO Buffers

Table 31.24 shows a list of modes of clearing the FIFO buffers by this controller. The FIFO buffers can be cleared by the 3 bits that are listed in the table.

Table 31.24 List of FIFO buffer clearing modes

Bit name	BCLR	DCLRM	ACLRM
Register	CFIFOCTR register DxFIFOCTR register	DxFIFOSEL register	PIPExCTR register
Function	The FIFO buffer on the CPU side is cleared.	The FIFO buffer is automatically cleared after the data is read from the designated pipe.	The buffer is automatically cleared to discard all the received packets.
Clearing method	Write 1 to clear.	1: Mode valid 0: Mode invalid	1: Mode valid 0: Mode invalid

31.7 FIFO Port Function

This section describes FIFO port functions. Table 31.25 lists settings of FIFO port functions of this controller. If writing data is continued until the buffer becomes full (up to the maximum packet size in discontinuous transfer mode) during the data write access, the FIFO port automatically enters the state where data can be transmitted to the USB bus. To make data with a size less than the buffer full (the maximum packet size in discontinuous transfer mode) transmittable, write completion must

be set by the BVAL bit in the CFIFOCTR or DxFIFOCTR register. Furthermore, to transmit a zero-length packet, clear the buffer by the BCLR bit in the CFIFOCTR or DxFIFOCTR register and set write completion by the BVAL bit.

When all data has been read during the read access, the FIFO port automatically enters the state where a new packet can be received. However, when a zero-length packet is received (DTLN = 0), the data cannot be read. In that case, clear the buffer by the BCLR bit in the CFIFOCTR or DxFIFOCTR register. The receive data length is checked by the DTLN bits in the CFIFOCTR or DxFIFOCTR register.

Table 31.25 FIFO Port function settings

Register name	Bit name	Function	Notes
C/DxFIFOSEL	RCNT	DTLN read mode selection	
	REW	Buffer memory rewind (re-read, re-write)	
	DCLRM	Automatic clearing of buffer memory after specified pipe received data is read	DxFIFO only
	DREQE	DREQ signal assertion	DxFIFO only
	MBW	FIFO port access bit width	
	BIGEND	FIFO port endian selection	
	ISEL	FIFO port access direction	DCP only
	CURPIPE	Current pipe selection	
C/DxFIFOCTR	BVAL	Buffer memory write completion	
	BCLR	Clearing of CPU-side buffer memory	
	FRDY	Monitoring of FIFO port ready	
	DTLN	Confirmation of received data length	

31.7.1 FIFO Port Selection

Table 31.26 shows the list of pipes that can be selected in each FIFO port. The pipes to be accessed are selected with the CURPIPE bits of the CFIFOSEL or DxFIFOSEL register. After selecting the pipes, confirm that the value of the CURPIPE bits written was read correctly (if the previous pipe number is read out, this indicates the controller is still changing the pipe), then confirm that FRDY = 1 and access the FIFO port. Figure 31.13 shows the procedure for switching pipes for access to the FIFO port.

Also, select the bus width for the FIFO port access with the MBW bit. The buffer memory access direction is determined by the ISEL bit for DCP, and the DIR bit of the PIPEXCFG register for all other pipes.

Table 31.26 FIFO port access by pipe

Pipe	Access method	Usable ports
DCP	CPU access	CFIFO port register
Pipes 1 to 9	CPU access	CFIFO port register
	DMA access	DxFIFO port register

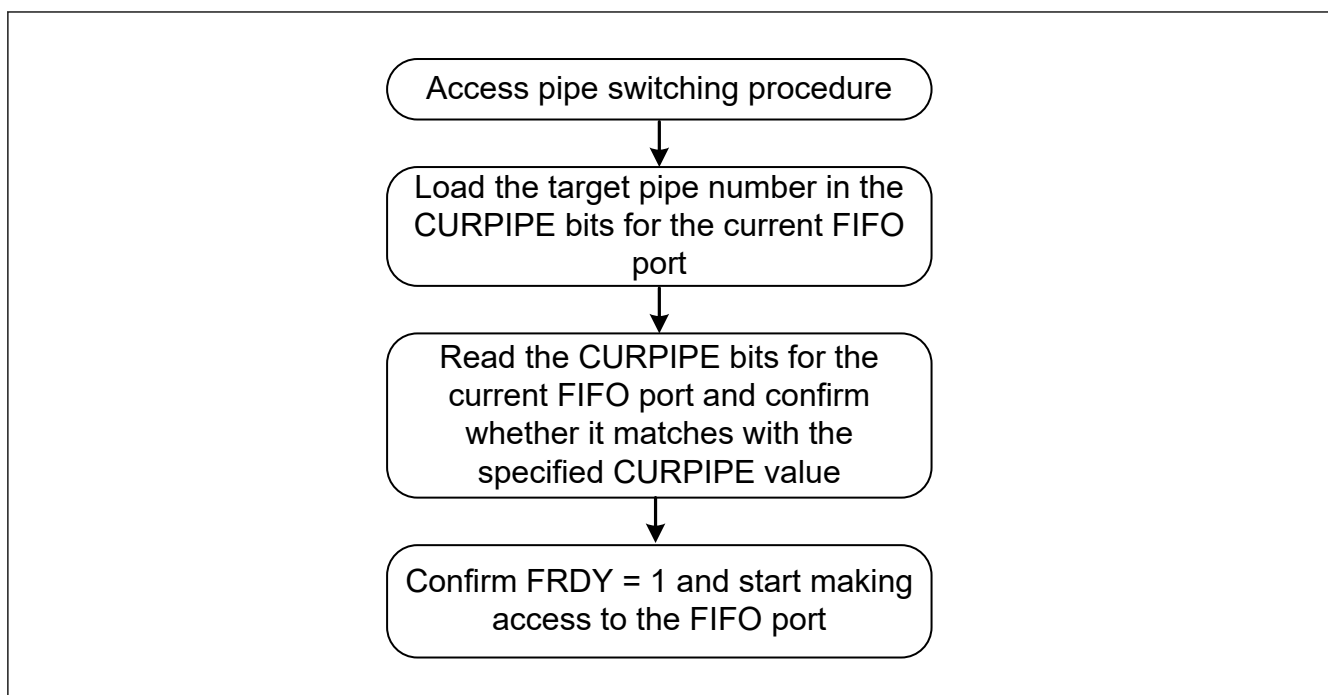


Figure 31.13 Pipe switching procedure for FIFO port access

31.7.2 DxFIFO Automatic Clear Mode (DxFIFO Port Read Direction)

When a data read event of the controller buffer memory is completed with setting the DCLRM bit of the DxFIFOSEL register to 1, the buffer memory of the corresponding pipe is automatically cleared.

Table 31.27 shows the correspondence between packet reception and buffer memory clearing by software in each setting.

As indicated in Table 31.27, the buffer clear conditions differ according to the set value of the BFRE bit, even for states in which clear is normally required, using the DCLRM bit eliminates the need for clearing of the buffer by software, enabling DMA transfers without the use of software.

Note that this function only supports the buffer memory read direction setting.

Table 31.27 Correspondence of packet reception and buffer memory clearing by software

Buffer state when packet is received	DCLRM = 0		DCLRM = 1	
	BFRE = 0	BFRE = 1	BFRE = 0	BFRE = 1
Buffer full	Clearing not required	Clearing not required	Clearing not required	Clearing not required
Zero-length packet received	Clearing required	Clearing required	Clearing not required	Clearing not required
Normal short packet received	Clearing not required	Clearing required	Clearing not required	Clearing not required
Transaction count end	Clearing not required	Clearing required	Clearing not required	Clearing not required

31.7.3 BRDY Interrupt Timing Selection

The BFRE bit of the PIPECFG register can be set so that the BRDY interrupt is not generated when a data packet of maximum packet size is received.

When using a DMA transfer, this function enables an interrupt to be generated only when the last data is received. The last data indicates either a short packet reception or the transaction count end. By setting BFRE = 1, the BRDY interrupt will be generated after the received data is read. By reading the DTLN bit of the DnFIFOCTR register, the receive data length of last data packet received just before the BRDY interrupt was generated can be confirmed.

Table 31.28 shows the timing of the BRDY interrupt.

Table 31.28 BRDY interrupt generation timing

Buffer state when packet is received	BFRE = 0	BFRE = 1
Buffer full (normal packet received)	When packet is received	No interrupt generated
Zero-length packet received	When packet is received	When packet is received
Normal short packet received	When packet is received	When read event of data received from buffer memory is completed
Transaction count end	When packet is received	When read event of data received from buffer memory is completed

The BFRE bit function is only valid in reading direction of the buffer memory. When in writing direction, fix the BFRE bit to 0.

31.8 Control Transfer (DCP)

Data transfers of the data stage of control transfers are done using the default control pipe (DCP). The DCP buffer memory is a 64 byte single buffer, and is a fixed area that is shared for both control reading and control writing. The buffer memory can be accessed only through the CFIFO port.

31.8.1 Setup Stage

The controller always responds with an ACK when it receives a normal setup packet. The controller operations in the setup stage are as follows.

- When a new setup packet is received, the controller sets the following bits.
 - Sets the VALID bit of the INTSTS0 register to 1.
 - Sets the PID bits of the DCPCTR register to NAK.
 - Sets the CCPL bit of the DCPCTR register to 0.
- When a data packet is received following the setup packet, the USB request parameters are stored in the following registers: USBREQ, USBVAL, USBINDX and USBLENG.

Always set VALID = 0 before the response process to a control transfer. While VALID = 1, PID = BUF will not be set and the data stage cannot be completed.

The function of the VALID bit allows the controller to temporarily stop a request in-process when it receives a new USB request during a control transfer, and respond to the newest request.

In addition, the controller automatically judges the direction bit (bmRequestType bit 8) and the request data length (wLength) of the received USB request and determines whether it is a control read transfer, control write transfer or control write no-data transfer, and then handles the stage transition. If the sequence is incorrect, a sequence error for the control transfer stage transition interrupt is generated and is notified to the software. For more information concerning the controller stage management, see [Figure 31.10](#).

31.8.2 Data Stage

Use the DCP for data transfers in response to receiving a USB request. Before accessing the DCP buffer memory, set the access direction in the ISEL bit of the CFIFOSEL register.

The transaction is executed by setting the PID bits of the DCPCTR register to BUF.

Data transfer completion is detected by the BRDY and BEMP interrupts. Use the BRDY interrupt for control write transfers and the BEMP interrupt for control read transfers.

For control write transfers in high-speed operation, a NYET handshake is sent in accordance with the buffer memory state.

31.8.3 Status Stage

When the setting of the PID bits of the DCPCTR register is BUF, set the CCPL bit to 1 to complete the control transfer.

After the above settings, the controller automatically executes the status stage in accordance with the data transfer direction fixed in the setup stage. The detailed process is as follows.

- Control read transfers:

The controller receives a zero-length packet from the USB host and sends an ACK response.

2. Control write transfers and no-data control transfers:

The controller sends a zero-length packet and receives an ACK response from the USB host controller.

31.8.4 Control Transfer Automatic Response

The controller automatically sends a response to a normal SET_ADDRESS request. If one of the following errors occurs, a response must be sent by software.

1. bmRequestType $\neq$ 0x00
2. wIndex $\neq$ 0x00
3. wLength $\neq$ 0x00
4. wValue $>$ 0x7F
5. DVSQ = 011b (configured)

All requests other than the SET_ADDRESS request must be responded to by software.

31.9 Bulk Transfer (Pipes 1 to 5)

The user can select the buffer memory usage method (single/double buffer, continuous/non-continuous transfer mode) for bulk transfer. The maximum size that can be set for the buffer memory is 2 KB. The controller manages the buffer memory state and automatically responds to PING packets and NYET handshakes.

31.9.1 NYET Handshake Control

Table 31.29 shows the list of responses to a token received in a bulk or control transfer. When an OUT token is received in a bulk or control transfer and there is only enough open space for one packet in the buffer memory, the controller sends a NYET response. However, when a short packet is received, the controller sends an ACK response instead of a NYET response, even under these conditions.

Table 31.29 List of responses to received tokens

PID bits setting	Buffer memory state	Received token	Response	Notes
NAK/STALL	—	SETUP	ACK	—
	—	IN/OUT/PING	NAK/STALL	—
BUF	—	SETUP	ACK	—
	RCV-BRDY*1	OUT/PING	ACK	When OUT token is received, data packet is received.
	RCV-BRDY*2	OUT	NYET	Data packet is received.
	RCV-BRDY*2	OUT (Short)	ACK	Data packet is received.
	RCV-BRDY*2	PING	ACK	
	RCV-NRDY*3	OUT/PING	NAK	
	TRN-BRDY*4	IN	DATA0/1	Data packet is transmitted.
	TRN-NRDY*5	IN	NAK	

Note 1. RCV-BRDY: Buffer memory has enough space for 2 packets or more when an OUT or PING token is received.

Note 2. RCV-BRDY: Buffer memory has only enough space for one packet when an OUT token is received.

Note 3. RCV-NRDY: Buffer memory has not enough space when a PING token is received.

Note 4. TRN-BRDY: Buffer memory has data for transmission when an IN token is received.

Note 5. TRN-NRDY: Buffer memory does not have data for transmission when an IN token is received.

31.10 Interrupt Transfer (Pipes 6 to 9)

This controller executes an interrupt transfer in accordance with the period managed by the host controller. The controller ignores (no response) PING packets in interrupt transfers. In addition, the controller does not send a NYET handshake, but responds with ACK, NAK or STALL.

The controller does not support high-bandwidth interrupt transfers.

31.11 Isochronous Transfer (Pipes 1 and 2)

The controller provides the following functions for isochronous transfers.

1. Isochronous transfer error information notification
2. Interval counter (IITV bit setting)
3. Isochronous IN transfer data setup control (IDLY function)
4. Isochronous IN transfer buffer flush function (IFIS bit setting)
5. SOF pulse output function

The controller does not support high-bandwidth isochronous transfers.

31.11.1 Isochronous Transfer Error Detection

The controller manages isochronous transfer errors by software and therefore has the following error information detection functions. [Table 31.30](#) and [Table 31.31](#) describe the procedure in which errors are confirmed and the interrupts that are generated.

1. PID error
When PID of the received packet is corrupted
2. CRC error and bit stuffing error
When an error occurs in CRC of the received packet or when the bit stuffing is corrupted
3. Maximum packet size over
This indicates the data size of the received packet is larger than the value set for the maximum packet size.
4. Overrun and underrun
When there is no data in the buffer memory when an IN token is received in an IN-direction (send) transfer When there is no empty space in the buffer memory when an OUT token is received in an OUT-direction (receive) transfer
5. Interval error
The following will generate interval errors.
 - (a) When an IN token could not be received in the interval frame of an isochronous IN transfer
 - (b) When an OUT token could not be received in the interval frame of an isochronous OUT transfer

Table 31.30 Errors detected in token reception/transmission

Detection priority	Error type	Generated interrupts and states at time of error detection
1	PID error	No interrupt generated (ignored as corrupted packet)
2	CRC error or bit stuffing error	No interrupt generated (ignored as corrupted packet)
3	Overrun or underrun errors	NRDY interrupt is generated and OVRN bit is set. A zero-length packet is sent in response to an IN token. A data packet is not received in response to an OUT token.
4	Interval error	NRDY interrupt generated

Table 31.31 Errors detected in data packet reception

Detection priority	Error type	Generated interrupts and states
1	PID error	No interrupt generated (ignored as corrupted packet)
2	CRC error or bit stuffing error	NRDY interrupt is generated and CRCE bit is set.
3	Maximum packet size over error	BEMP interrupt is generated and PID is set to STALL.

31.11.2 DATA-PID

This controller does not support high-bandwidth transfers. The following occurs in response to a received PID.

1. IN direction:
 - (a) DATA0: Transmitted as PID of the data packet
 - (b) DATA1: Not transmitted
 - (c) DATA2: Not transmitted
 - (d) mDATA: Not transmitted
2. OUT direction (in full-speed operation):
 - (a) DATA0: Received successfully as PID of the data packet
 - (b) DATA1: Received successfully as PID of the data packet
 - (c) DATA2: The packet is ignored.
 - (d) mDATA: The packet is ignored.
3. OUT direction (in high-speed operation):
 - (a) DATA0: Received successfully as PID of the data packet
 - (b) DATA1: Received successfully as PID of the data packet
 - (c) DATA2: Received successfully as PID of the data packet
 - (d) mDATA: Received successfully as PID of the data packet

31.11.3 Interval Counter

31.11.3.1 Overview of Operation

The isochronous transfer interval can be set in the IITV bit of the PIPEPERI register. [Table 31.32](#) shows the functions of the interval counter.

Table 31.32 Interval counter functions

Transfer direction	Function	Detection conditions
IN	Transfer buffer flush function	Cannot successfully receive IN token in interval frame during isochronous IN transfer.
OUT	Notifies that a token not being received	Cannot successfully receive OUT token in interval frame during isochronous OUT transfer.

Since the interval counting is performed upon reception of SOF or by the complemented SOF, the isochronism can be maintained even if the SOF is damaged. Frame intervals are set as 2^{IITV} (μ) frames.

31.11.3.2 Interval Counter Initialization

The controller initializes the interval counter under the following conditions.

1. Hardware reset
Initializes the IITV bit.
2. Clearing of the buffer memory by the ACLRM bit
This initializes the counter but not the IITV bit.
3. USB bus reset

After the interval counter is initialized and a packet is successfully transferred, the interval count starts under the following conditions.

1. SOF is received after data is sent in response to an IN token when PID = BUF.
2. SOF is received after data is received in response to an OUT token when PID = BUF.

Note that the interval counter is not initialized in the following conditions.

1. When the PID is set to NAK or STALL
The interval timer is not stopped at this time. The transaction will be attempted at the next interval.

2. USB bus reset or USB suspend

The IITV bit is not initialized at this time. When the SOF is received, the count starts from the value before the reception.

31.11.4 Isochronous Transfer Transmission Data Setup

In the isochronous data transmission by this controller, after data is written to the buffer memory, the data packet can be sent out in the next frame after the SOF packet is detected. This function, called the isochronous transfer transmission data setup, allows specification of the frame that started transmission.

When the buffer memory is used for double buffering and writing to both buffers has been completed, only transfer from the first buffer to have received data can proceed. Therefore, even when several IN tokens are received in the same frame, only one packet of data is sent by the buffer memory.

When an IN token is received, if the buffer memory is ready for transmission, the data is transferred and a normal response is returned. However, if the buffer memory is not ready for transmission, a zero-length packet is sent and an underrun error occurs.

Figure 31.14 shows an example of transmission using the isochronous transfer transmission data setup function with this controller when IITV = 0 (for each frame) is set.

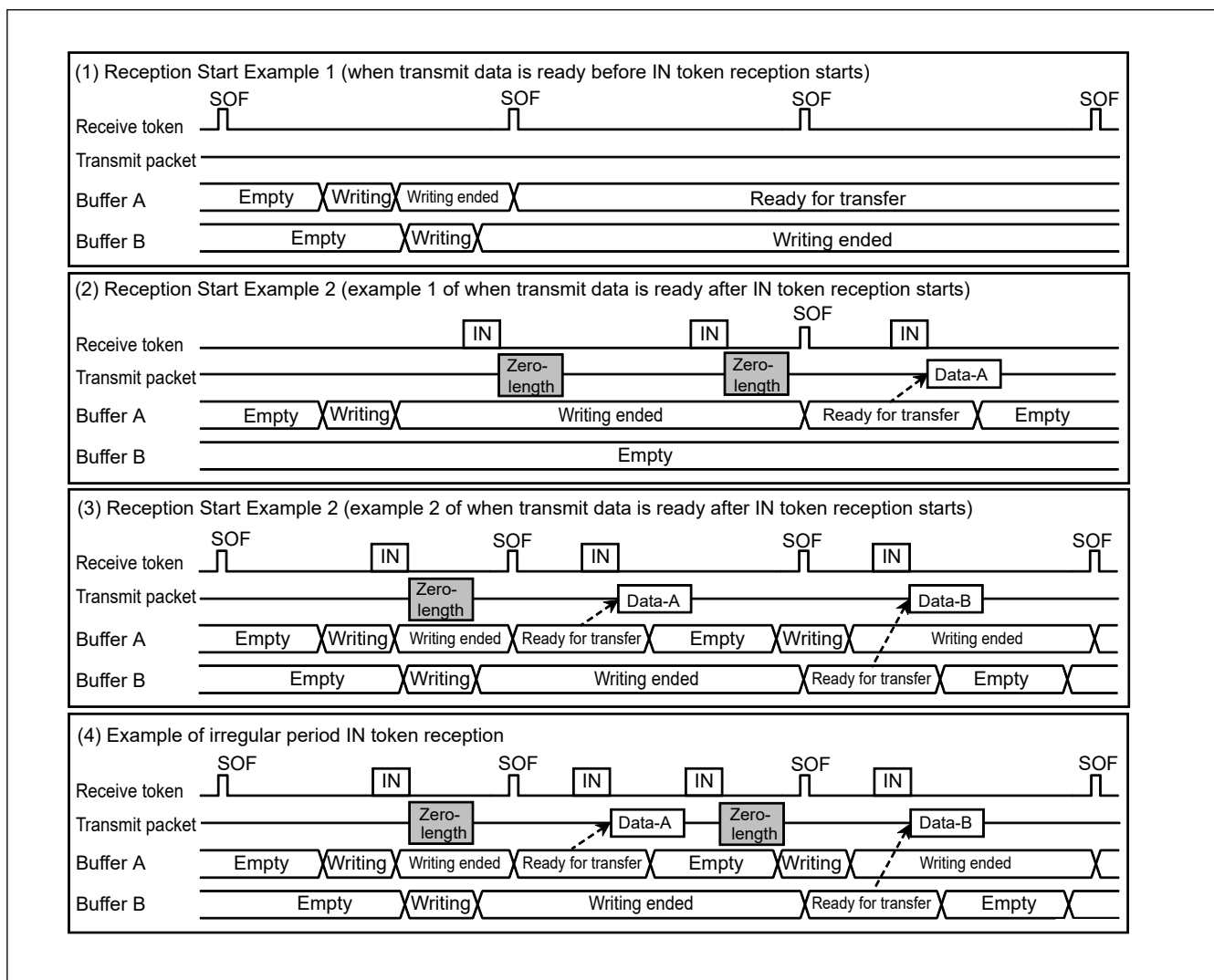


Figure 31.14 Example of data setup function operation

31.11.5 Isochronous Transfer Transmission Buffer Flush

When an SOF packet or a μ SOF packet of the next frame is received without receiving an IN token in the interval frame during isochronous data transmission, this controller operates as if a corrupted IN token was received, and clears the buffer which is ready for transmission, making that buffer ready for writing.

If a double buffer is being used and writing to both buffers has been completed, data are considered to have been sent from the buffer memory that was cleared in the same interval frame, and transmission is enabled for the buffer memory that is not discarded with SOF or μ SOF packets reception.

The timing at which the buffer flush function is activated varies depending on the setting of the IITV bits.

1. When $IITV = 0$
The buffer flush operation proceeds from the first frame after the pipe becomes valid.
2. When $IITV \neq 0$
The buffer flush operation proceeds after the first successful transaction.

Figure 31.15 shows an operation example of the buffer flush function of this controller. When an unanticipated token is received prior to the interval frame, this controller sends the written data or a zero-length packet as an underrun error according to the data setup state.

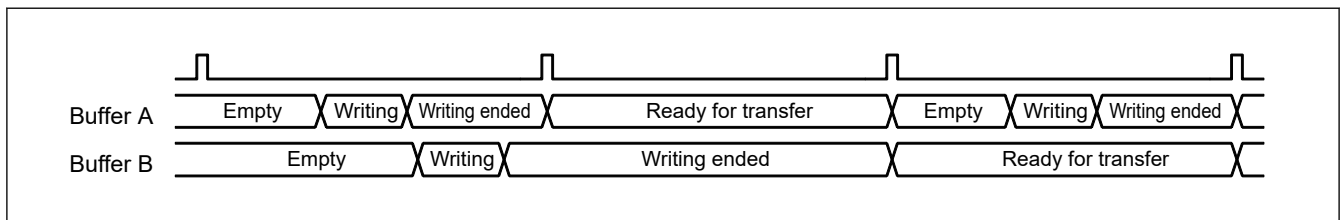


Figure 31.15 Buffer flush function operation example

Figure 31.16 shows an example of an interval error generated in the controller. There are five types of interval errors, as listed below. Timing 1 in the figure shows when the interval error occurs and how the buffer flush function operates.

When an interval error occurs during an IN transfer, the buffer flush function goes into operation; during an OUT transfer, the NRDY interrupt is generated.

Use the OVRN bit to determine whether an error is an NRDY interrupt, such as a receive packet error, or an overrun error.

Responses to the tokens in the shaded boxes are executed in accordance to the buffer memory state.

1. IN direction:
 - (a) If the buffer is ready for transfer, data is transferred as a normal response.
 - (b) If the buffer is not ready for transfer, a zero-length packet is sent and an underrun error occurs.
2. OUT direction:
 - (a) If the buffer is ready for reception, data is received as a normal response.
 - (b) If the buffer is not ready for reception, data is discarded and an overrun error occurs.

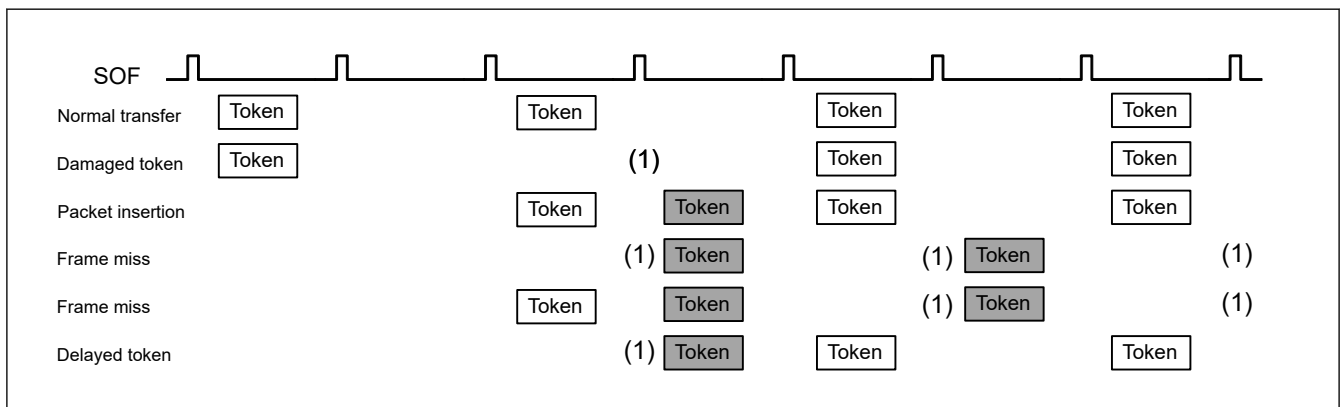


Figure 31.16 Example of interval error when IITV = 1

31.12 SOF Interpolation

If an SOF packet could not be received at intervals of 1 ms (in full-speed operation) or 125 μ s (in high-speed operation) because of corruption or missing, this controller interpolates the SOF. The SOF interpolation operation begins when both the USBE and SUSPM bits have been set to 1 and an SOF packet is received. The interpolation is initialized under the following conditions.

1. Hardware reset
2. USB bus reset
3. Suspended state detected

The SOF interpolation operates according to the following specifications.

1. The frame interval (125 μ s or 1 ms) is based on the results of the reset handshake protocol.
2. The interpolation does not operate until the SOF packet is received.
3. After a first SOF packet is received, interpolation of another SOF packet proceeds after a 125- μ s or 1-ms interval counted in cycles of the internal clock running at 48 MHz clock has elapsed.
4. Interpolation is performed in the previous reception intervals after the 2nd and subsequent SOF packets are received.
5. Interpolation is not performed in the suspended state or during reception of a USB bus reset.
When the controller goes to the suspended state in high-speed operation, interpolation continues for 3 ms from the last packet.

The SOF interpolation works for the following.

1. Updating of the frame number or micro-frame number
2. SOFR interrupt and μ SOF lock
3. SOF pulse output
4. Isochronous transfer interval count

When an SOF packet is lost during full-speed operation, the FRNM bit of the FRMNUM register is not updated.

When a μ SOF packet is lost during high-speed operation, the UFRNM bit of the UFRMNUM register is updated.

However, when a μ SOF packet when μ FRNM = 000b is lost, the FRNM bit is not updated. At this time, even if μ SOF packets when μ FRNM $\neq$ 000b are received successfully, the FRNM bit is not updated.

31.13 Link Power Management Processing

The Link Power Management standard redefines the existing suspended state as L2 state, and further defines L1 state that enables transitions and recovery with lower latency than the L2 (suspended) state.

The following table shows comparison between L2 (suspended) and L1.

Table 31.33 Comparison between L2 (suspended) and L1

Item	L1	L2 (suspended)
Transition	LPM transaction	3 ms idle
Recovery due to host	(Host) The host can specify the minimum drive period (75 μ s to 1.175 ms). (Device) 10 μ s K drive	(Host) Min 20 ms K drive (Device) 10 ms K drive
Recovery due to device	(Device) 50 μ s K drive (Host) 60 μ s to 990 μ s K drive (Device) 10 μ s K drive	(Device) 1 ms to 15 ms K drive (Host) Min 20 ms K drive (Device) 10 ms K drive
Signaling	Low and full speed idle	Low and full speed idle

The following describes the transition (to L1 state) processing and recovery processing.

31.13.1 Descriptor

When the GetDescriptor command is received, the own descriptor must be sent back.

It is necessary to change the contents of the descriptor to be returned depending on whether to respond to the transition to L1 and recovery processing because of the LPM transaction, as shown in the table below.

Table 31.34 Relationship between response to LPM and descriptor

Response to LPM	bcdUSB	Provision of USB2.0Ex.Des	LPM of USB2.0Ex.Desc	Response when LPM is received	Remarks
Does not respond	0x0200	Not provided	—	Not Respond	Standard operation when the descriptor does not respond to LPM
	0x0201	Provided	LPM = 0	STALL	Clear declaration that the descriptor does not respond to LPM. In this case, no response must be avoided and STALL must be sent in response.
Responds	0x0201	Provided	LPM = 1	ACK or NYET	Standard operation when the descriptor responds to LPM

Whether to respond to the transition to L1 and recovery processing is declared with the LPM bit in the USB2.0 extension descriptor. To provide the USB2.0 extension descriptor, a value of 0x0201 or larger must be set for the bcdUSB field of the device descriptor.

If the descriptor does not respond to LPM, the USB2.0 extension descriptor is not provided and the bcdUSB field is set to 0x0200. In this case, even if LPM token is received, ignore it.

If the descriptor does not respond to LPM, it is also possible to set the bcdUSB field to 0x0201 and clear the LPM bit in the USB2.0 extension descriptor to 0 (does not respond). In this case, however, LPM cannot be ignored and STALL must be sent in response. When the descriptor responds to LPM, set the bcdUSB field to 0x0201 and set the LPM bit in the USB2.0 extension descriptor to 1 (responds). This enables acknowledgment of NYET or ACK response to the LPM token.

31.13.2 Basic Processing

The following processing is required.

1. Send no response, ACK, NYET, or STALL response to the LPM token received from the host in accordance with the own state.
2. If re-transmission of LPM token is not detected for 8 μ s after an ACK response was sent, a transition to the L1 state is made.
3. Detect the K drive of the host and perform the processing for recovery to the idle state.
4. Perform the processing for recovery to the idle state by remote wakeup.

For step 1, hardware receives the LPM token, it sends the response. Hardware performs processing in step 2 in addition to re-transmission control and transition to the L1 state.

A transition to the L1 state can be checked by a DVST interrupt.

For step 3, a RESM interrupt is generated by detecting the K drive of the host in the L1 state.

For step 4, setting the WKUP bit to 1 by software gives an instruction to start remote wakeup to the hardware. In the specifications, software clears this bit at the time of recovery from the L2 state, but hardware clears this bit at the time of recovery from the L1 state.

31.13.3 HIRD Value Negotiation

The HIRD value included in the LPM token indicates the host K period for recovery from the L1 state. It is supposed that the desired HIRD value may vary depending on applicable applications. It is supposed that a small HIRD value is desired for applications focusing on improvement in transfer efficiency, but a large HIRD value is desired for applications focusing on low power consumption.

According to the L1NEGOMD and HIRDTHR bits in the L1CTRL register, an ACK response is returned if the received HIRD value is within the desired range. Otherwise, a NYET response is returned to request the host to correct the HIRD value.

Note: Using this HIRD value negotiation function requires that the negotiation processing be also supported on the host side.

31.14 DMA Mode

31.14.1 Register Mode/Link Mode

With the DMS bit in the CHCFG_n register, DMA mode can be switched between register mode and link mode.

Table 31.35 DMA mode setting

DMS(CHCFG)	Mode	Description
0	Register Mode	Performs a DMA transfer with the values set for the next register set.
1	Link Mode	Accesses the descriptor area and performs a DMA transfer with the value set for the descriptor. Repeats descriptor read and DMA transfer unless you set the descriptor or use the control register to stop them.

31.14.1.1 Register Mode

In register mode, you can perform a DMA transfer using the value set in the internal register. You can set two sets (Next0 register set and Next1 register set) of transfer source addresses, the transfer destination addresses, and the numbers of transfer bytes.

You can select the next register set to perform a transfer, or use two next register sets for continuous transfer.

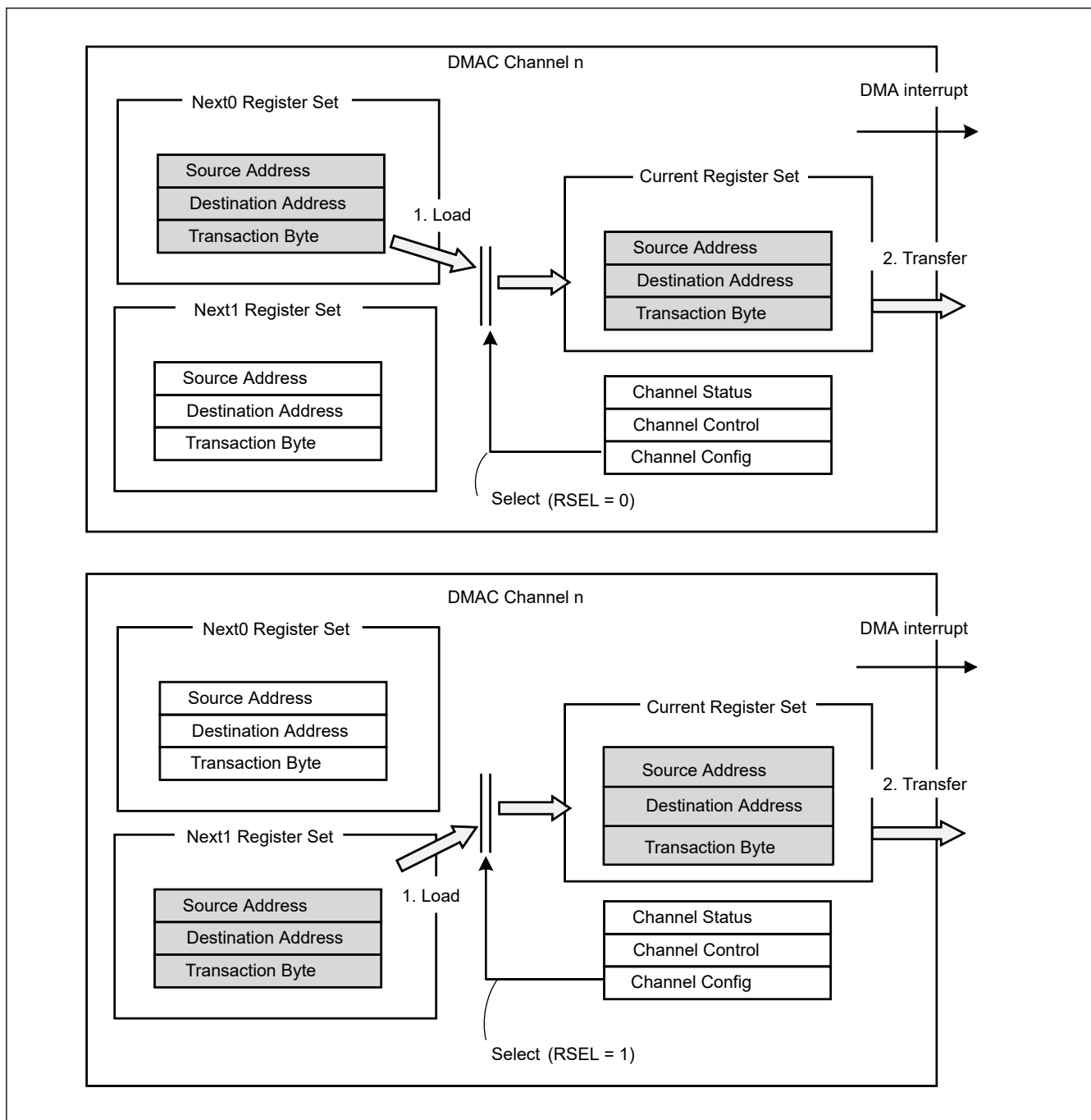


Figure 31.17 Overview of register normal mode

The above figure illustrates operation when Next0 register set is executed (above) and operation when Next1 register set is executed (below).

(1) Operation Flow in Register Mode

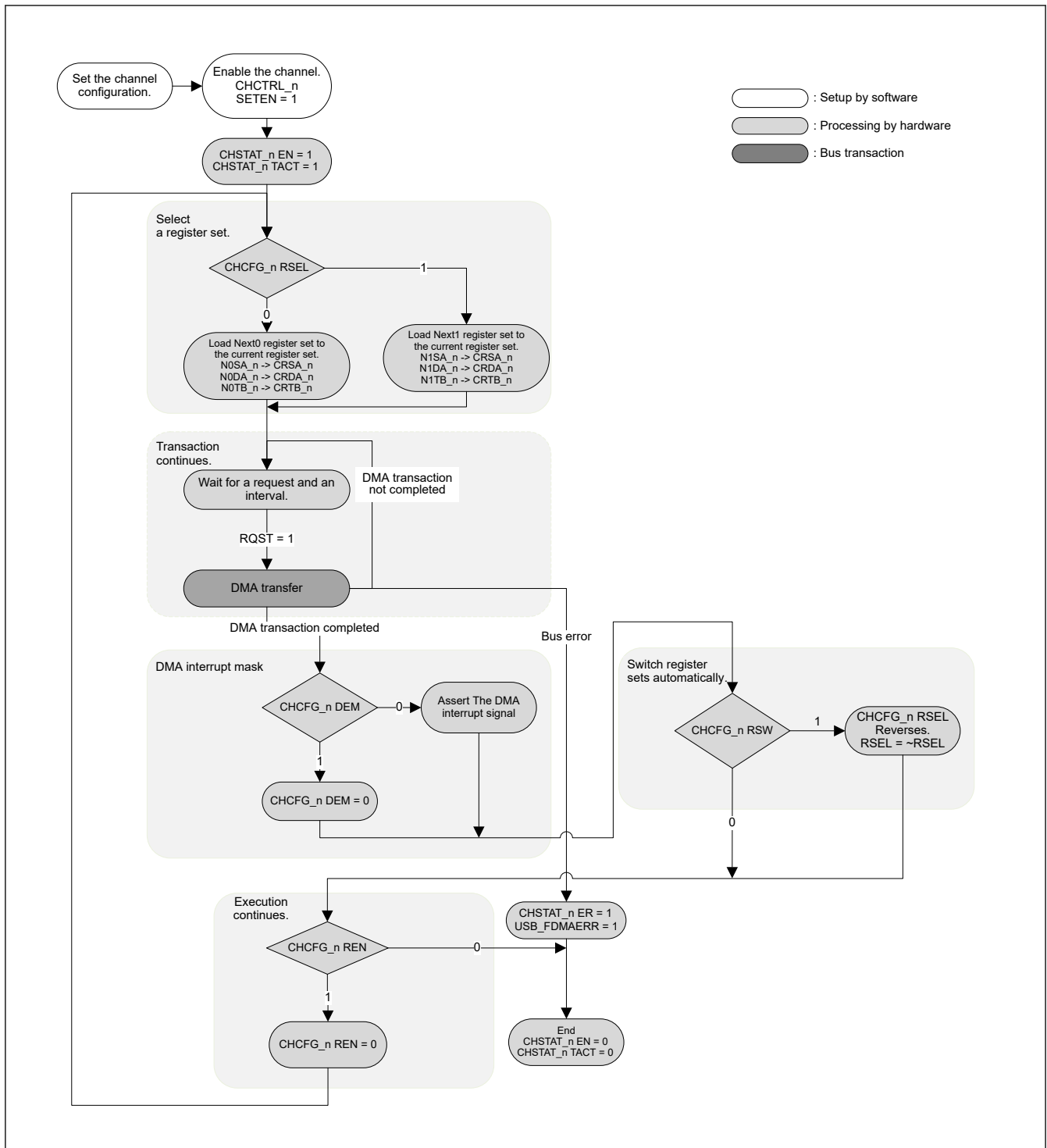


Figure 31.18 Register mode flow

<Description of register mode flow>

- Channel setting (set channel configuration)
Set Next0 or Next1 register set (the transfer destination address, the transfer source address, and the total number of transfer bytes). Also set the USB-controlled FIFO channels and the amount of transfer used for channel register set. (See [section 31.15. DMA Transfer.](#))
- Selecting register set

When 1 is written to the SETEN bit in the CHCTRL_n register, the EN bit and the TACT bit in the CHSTAT_n register are set to 1, and the set value of the next register set selected by the RSEL bit in the CHCFG_n register is loaded to the current register set.

3. DMA transaction (processing transaction)

A DMA transaction is performed according to the set value. For details on transfer, see [section 31.15. DMA Transfer](#).

4. Making USB_FDMan (USB_FDMan mask)

The USB_FDMan is masked according to the set DEM bit value in the CHCFG_n register. When DEM = 1, the USB_FDMan is masked. Immediately after that, the DEM bit is automatically cleared to 0.

5. Switching register sets automatically

Switching to the other next register set is determined according to the set RSW bit value in the CHCFG_n register.

6. Continuous execution

Whether to perform DMA transfers continuously is determined according to the set REN bit value in the CHCFG_n register. When REN = 0, the EN bit and the TACT bit in the CHSTAT_n register are cleared to 0, and the DMAC stops operation. When REN = 1, DMA transfers are continued. Immediately after that, the REN bit is automatically cleared to 0.

(2) Setting Register Mode

Register mode settings

Select the register set to execute.

Table 31.36 Register mode settings

DMS (CHCFG_n)	RSEL (CHCFG_n)	Description
0	0	Executes Next0 register set.
	1	Executes Next1 register set.

USB_FDMan mask settings

The USB_FDMan can be masked.

Table 31.37 USB_FDMan mask settings

DEM (CHCFG_n)	Description
0	When a DMA transaction is completed, the USB_FDMan is asserted.
1	Even if a DMA transaction is completed, the USB_FDMan is not asserted. Upon completion of a DMA transaction, the DEM bit is cleared to 0 by hardware.

Settings for executing register sets automatically

After a DMA transaction is completed, another DMA transaction can be performed successively.

Table 31.38 Settings for automatic execution of register sets

REN (CHCFG_n)	Description	Remarks
0	When a DMA transaction of the register set that is set in the RSEL bit is completed, the EN bit is cleared to 0 and the DMA operation ends.	Set this value if you want to perform a DMA transaction once.
1	After a DMA transaction is completed, a DMA transfer is performed for the register set that was selected to be performed successively. When the successive transfer is performed, the REN bit is cleared to 0.	Set this value if you want to perform DMA transactions for register sets successively.

Settings for switching register sets automatically

After a DMA transaction is completed, the register set to be used for the next DMA transaction can be switched.

Table 31.39 Settings for automatic switching of register sets

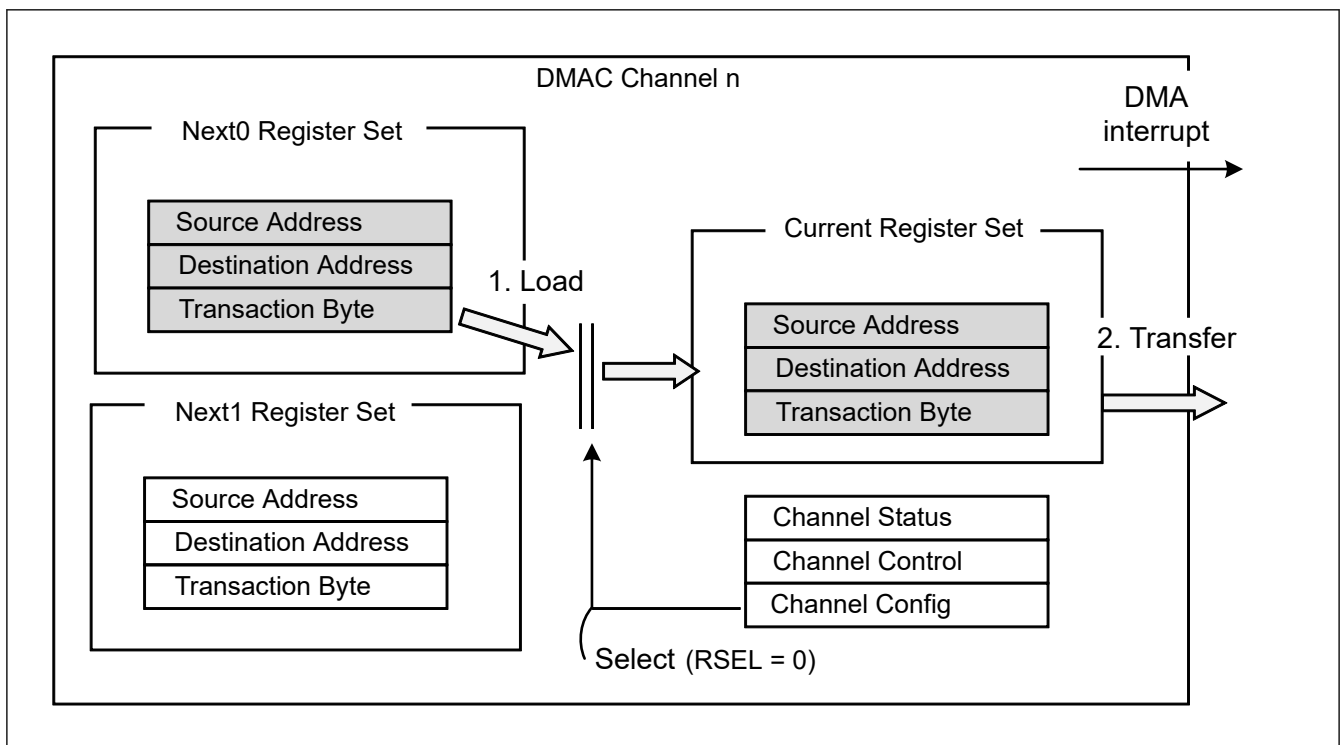
RSW (CHCFG_n)	Description	Remarks
0	After a DMA transaction is completed, register sets are not switched.	Set this value when you want to use only one register set.
1	When a DMA transaction is completed while REN = 1, the RSEL bit is reversed automatically and the other register set is selected.	Set this value when you want to switch register sets.

(3) Register Mode Setting Examples

Using the Next0 register set only

Table 31.40 Register mode setting example 1

DMS (CHCFG_n)	RSEL (CHCFG_n)	DEM (CHCFG_n)	RSW (CHCFG_n)	REN (CHCFG_n)
0 (Register mode)	0 (Next0)	0 (Do not mask)	0 (Do not switch)	0 (Continuous execution not in progress)

**Figure 31.19 Register mode setting example 1**

1. Writing 1 to the SETEN bit in the CHCTRL_n register sets the EN bit in the CHSTAT_n register to 1, and loads Next0 register set to the current register set.
2. According to the values set for the current register set and the channel register set, a DMA transaction is performed.
3. Because the DEM bit in the CHCFG_n register is 0, USB_FDMA_n is asserted after a DMA transaction is completed.
4. Because the REN bit in the CHCFG_n register is 0, the EN bit in the CHSTAT_n register is cleared to 0, and the operation ends.

Using two register sets successively

Table 31.41 Register mode setting example 2

DMS (CHCFG_n)	RSEL (CHCFG_n)	DEM (CHCFG_n)	RSW (CHCFG_n)	REN (CHCFG_n)
0 (Register mode)	0 (Next0)	1 (Mask)	1 (Switch)	1 (Continuous execution in progress)

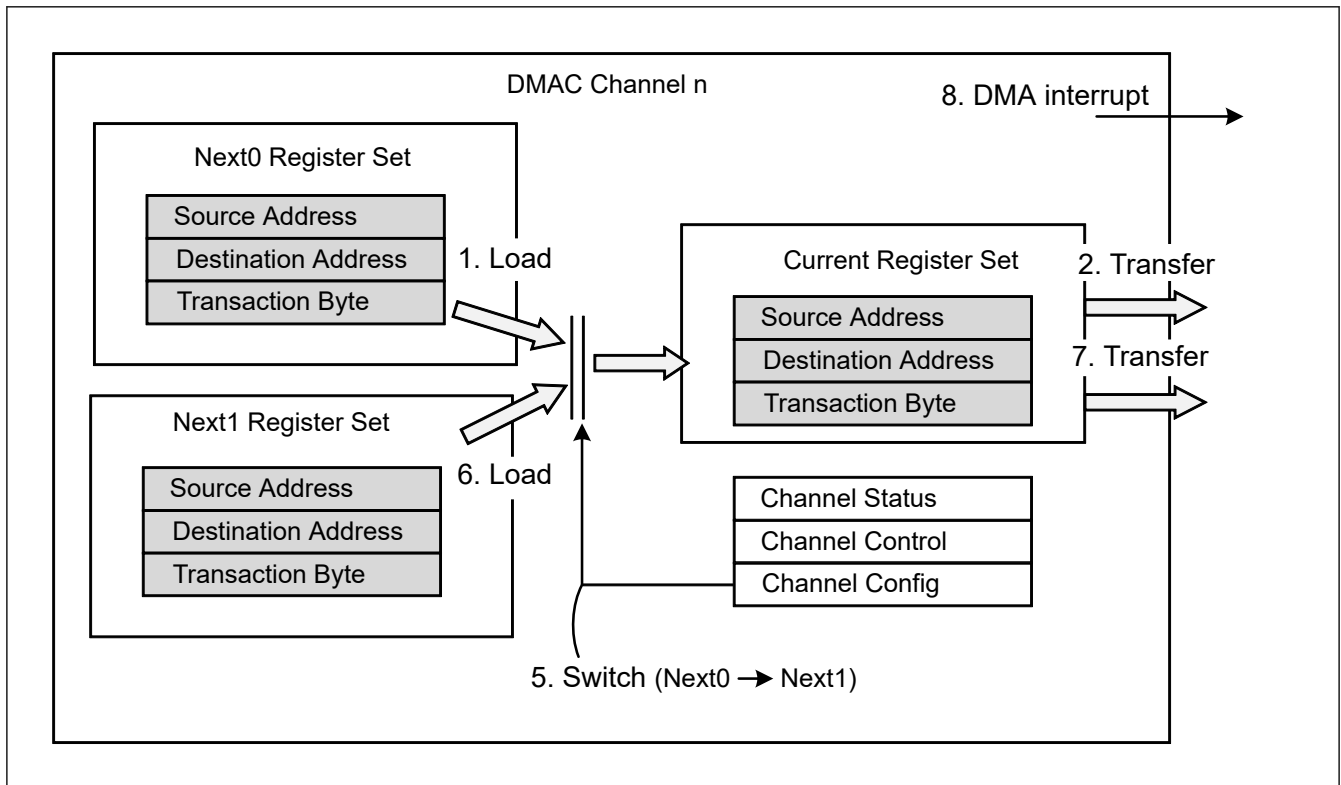


Figure 31.20 Register mode setting example 2

- Writing 1 to the SETEN bit in the CHCTRL_n register sets the EN bit in the CHSTAT_n register to 1, and loads Next0 register set to the current register set.
- According to the values of the current register set and the channel register set, a DMA transaction is performed.
- Because the DEM bit in the CHCFG_n register is 1, USB_FDMA_n is not asserted after a DMA transaction is completed, and the DEM bit is automatically cleared to 0.
- Because the REN bit in the CHCFG_n register is 1, DMA transactions are performed successively, and the REN bit is automatically cleared to 0.
- Because the RSW bit in the CHCFG_n register is 1, the register set to be executed next is switched (RSEL = 0 → 1).
- Loads Next1 register set to the current register set.
- According to the values of the current register set and the channel register set, DMA transactions are performed.
- Because the DEM bit in the CHCFG_n register is 0, USB_FDMA_n is asserted after a DMA transaction is completed.
- Because the REN bit in the CHCFG_n register is 0, the EN bit in the CHSTAT_n register is cleared to 0, and the operation ends.

31.14.1.2 Link Mode

In link mode, a DMA transaction is performed by reading a descriptor in an external memory area as a setting value. The DMAC incorporates the next link address (NXLA_n) register and the current link address (CRLA_n) register for each channel. Each of them is used to set the address of the descriptor to be executed next and to display the descriptor address of the current DMA transaction respectively.

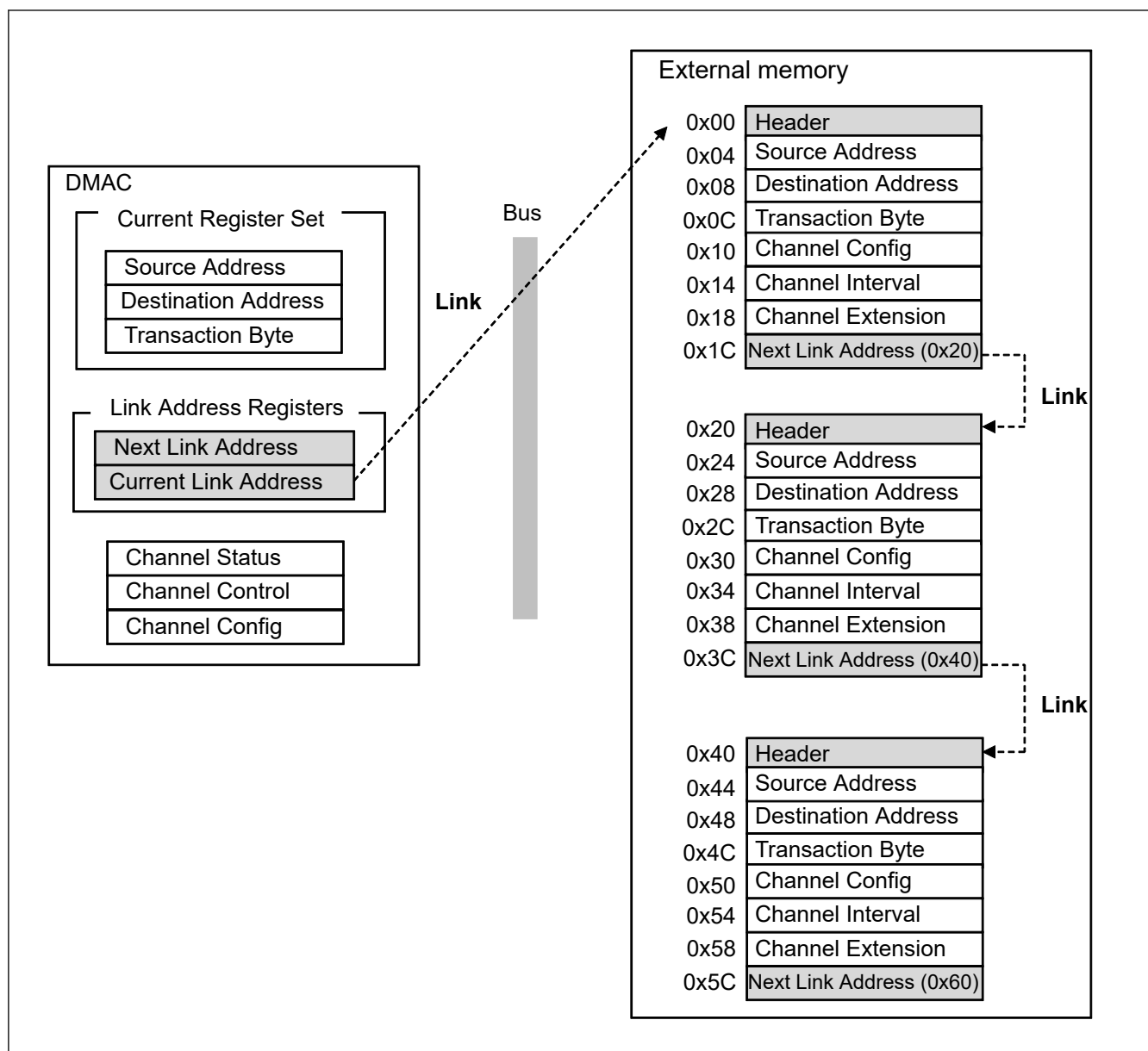


Figure 31.21 Overview of link mode

(1) Link Mode Operation Flow

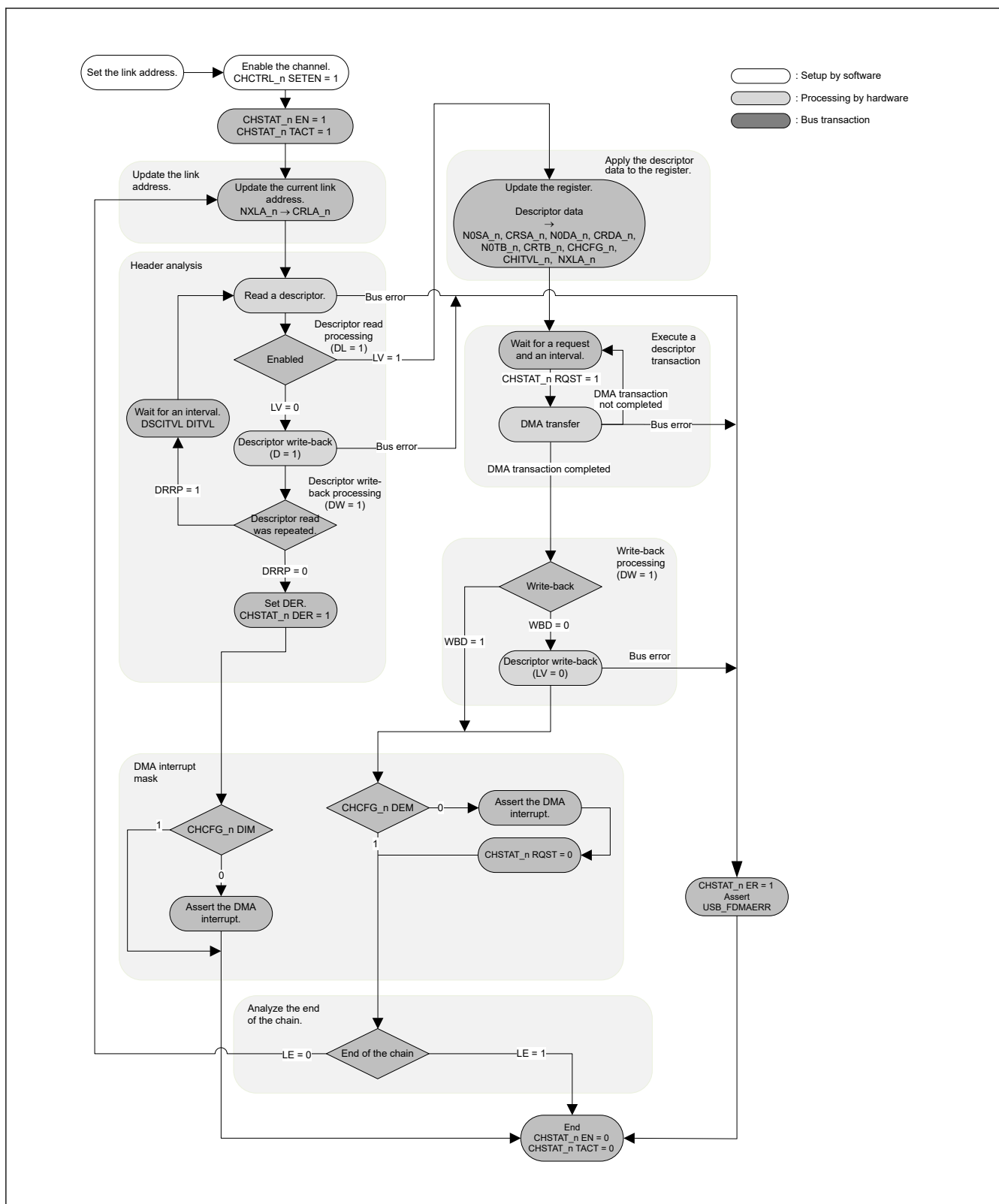


Figure 31.22 Link mode operation flow

Description of Link Mode Operation Flow

1. Channel setting

Set the start address of the link destination in the NXLA_n register.

2. Updating the link address
Writing 1 to the SETEN bit in the CHCTRL_n register sets the EN bit and the TACT bit in the CHSTAT_n register to 1, and loads the link address set for the NXLA_n register to the CRLA_n register.
3. Descriptor read and header judgment
The DMAC starts reading a descriptor and checks the header contents. When LV = 0, 1 is written back to the D bit in the header. After that, when the DRRP bit in the CHCFG_n register = 1, the same descriptor is read again after the number of cycles set in the DSCITVL register elapsed. When DRRP = 0, DER in the CHSTAT_n register is set to 1. This indicates the end state (EN = 0 and TACT = 0 in the CHSTAT_n register). At that time, if the DIM bit in the CHCFG_n register is 0, USB_FDMAn is asserted.
4. Descriptor setting
When LV = 1, the read descriptor data is loaded to the current register set and the channel register set. In addition, the next link destination is loaded to the NXLA_n register.
5. DMA transaction
According to the set values, a DMA transaction is performed. For details on transfers, see [section 31.15. DMA Transfer](#).
6. Writing back of header
When WBD in the header = 0, the DMAC writes LV = 0 back to the header area.
7. USB_FDMAn mask
When the DEM bit in the CHCFG_n register is 0, USB_FDMAn is asserted.
8. Link end judgment
When LE in the header = 1, the EN bit and the TACT bit in the CHSTAT_n register are cleared to 0, and the DMAC stops operation. When LE = 0, the current register set is updated and the next descriptor read is started.

(2) Register Settings

Link mode settings

When using link mode, set the DMS bit in the CHCFG_n register to 1.

Table 31.42 Link mode settings

DMS (CHCFG_n)	Description
1	Operates in link mode. This bit cannot be modified by using a descriptor.

LINK address settings

As registers that indicate the link destination, the next link address (NXLA_n) register and the current link address (CRLA_n) register are available.

To start link mode, set the link destination in the NXLA_n register.

The NXLA_n register is updated to the next link after a descriptor is read. In addition, the CRLA_n register indicates the link address of the currently executed descriptor.

Table 31.43 Link address register set

Register	Description
Next link address register (NXLA_n)	Sets and displays the next link destination. Before starting link mode, set the address of the link destination in this register.
Current link address register (CRLA_n)	Displays the currently executed link destination. This register is read-only.

(3) Descriptor Settings

The DMAC supports multiple descriptor formats. To switch formats, use the DSCFM field of bits [31:28] in the first word (header) of a descriptor.

The following table shows the relationship between the DSCFM bit value and descriptor formats.

Table 31.44 Descriptor format

DSCFM	Descriptor size	Next link address	Channel extension	Channel interval	Channel config	Transaction size	Destination address	Source address	Header
0x3	Four words	Y	— (Reload)	— (Reload)	— (Reload)	— (Header)	Y	Y	Y (with STS)
0x1	Eight words	Y	Y	Y	Y	Y	Y	Y	Y (without STS)
Other than above	If DSCFM is set to a value other than 1 and 3, operation is not guaranteed. Do not set a value other than 0x1 and 0x3 for DSCFM.								

Table 31.45 Description of availability

Field	Availability	Description	Remarks
Header	Y(with STS)	Indicates that the STS field of [15:0] in the header is enabled. The value set in the STS field is used as the total number of transfer bytes (transaction size).	—
	Y(without STS)	The STS field of [15:0] in the header is disabled. The transaction size of the descriptor is used as the total number of transfer bytes.	—
Source address	Y	Specifies the source address.	—
Destination address	Y	Specifies the destination address.	—
Transaction size	Y	Specifies the transaction size.	—
	— (Header)	Omits the transaction size. The value set in the STS field in the header is used as the total number of transfer bytes (transaction size).	Because the STS field is 16 bits, up to 65,535 bytes can be set.
Channel config Channel interval Channel extension	Y	Specifies the channel config, channel interval, and channel extension.	—
	— (Reload)	Omits the channel config, channel interval, and channel extension. Inherits the previous setting values (of the CHCFG_n, CHITVL_n, and CHEXT_n registers at that time).	—
Next link address	Y	Specifies the next descriptor address (next link address) to be read after a DMA transfer of this descriptor.	—

The DMAC interprets data obtained through descriptor read in order. If a value less than eight words is specified in the DSCFM field, place descriptor data marked with Y in [Table 31.44](#) in memory.

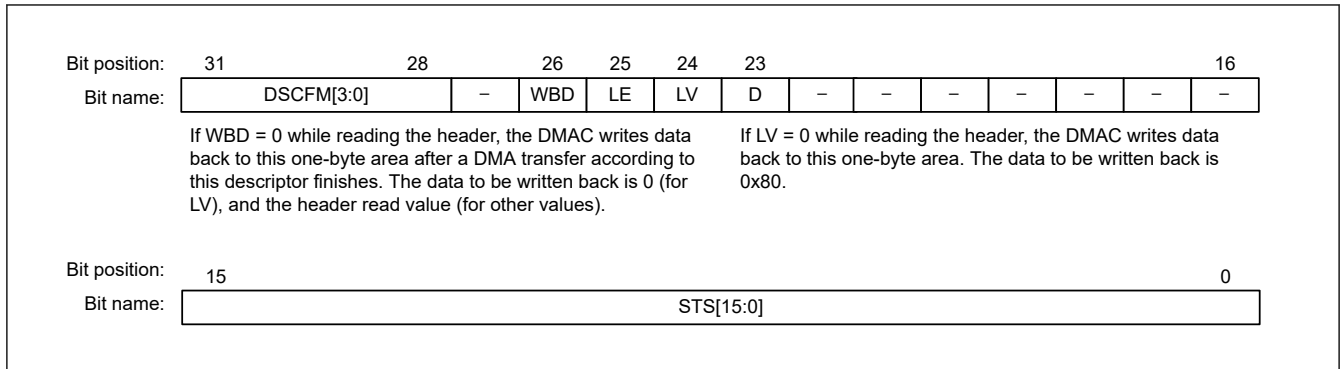
Table 31.46 Descriptor placement example

DSCFM	Address (link address + n)							
	+ 0x1C	+ 0x18	+ 0x14	+ 0x10	+ 0x0C	+ 0x08	+ 0x04	+ 0x00
0x3	—	—	—	—	Next link address	Destination address	Source address	Header
0x1	Next link address	Extension	Interval	Config	Transaction byte	Destination address	Source address	Header

Header

The header indicates descriptor states as shown below.

This area is read by the DMAC before starting a DMA transfer in link mode. Furthermore, the transfer status is written back by the DMAC after the DMA transfer.

**Figure 31.23 Header area****Table 31.47 Header area**

Bit position	Bit name	Description
15:0	STS	Short Transaction Size When DSCFM = 0x3, these bits set the transaction size in bytes. Up to 65535 bytes can be set. When DSCFM = 0x3, do not set the STS bits to 0. If 0 is set, the operation is not guaranteed.
22:16	—	A reserved area. Set 0.
23	D	Descriptor Error Indicates an access error of a descriptor. When LV = 0 while a descriptor is being read, the DMAC writes 1 back to this bit. 0: No descriptor error 1: LV = 0 during descriptor read
24	LV	Link Valid Indicates that this descriptor is enabled. When WBD = 0, the DMAC writes 0 after the DMA transaction written in the descriptor is completed. Set 1 when setting header. 0: The descriptor is disabled. 1: The descriptor is enabled.
25	LE	Link End Indicates that the link ends with the DMA transaction of this descriptor. To indicate the end of the link, set this bit to 1. 0: The link continues. 1: The link ends.
26	WBD	Write Back Disable Masks write-back execution of the LV bit. When this bit is 1, the DMAC does not perform writeback operation. 0: The LV bit is written back to 0. 1: The LV bit is not written back.
27	—	A reserved area. Set 0.
31:28	DSCFM	Descriptor Format Specifies the descriptor format (descriptor length and combination). For details, see Table 31.44 .

If descriptors are added sequentially while the DMAC is operating, an access of the CPU to set the LV bit to 1 and an access of the DMAC to write 1 back to the D bit may conflict. Because of this, the data that was written first is overwritten by the data that was written later.

To avoid this problem, the byte lane of the D bit and the byte lane of the LV bit are placed differently to separate write areas. The DMAC uses the byte write method to write back the D bit. Therefore, the CPU must set LV = 1 by using the byte write method.

Setting descriptors other than header

Data in descriptors except header has the same specifications as an on-chip register. For details on the on-chip register specifications, see [section 31.2. Register Descriptions](#).

Setting AHB during descriptor access

The MHPROT pin output during the descriptor access can be set in the LWPR and LDPR fields in the DCTRL register. Set it according to the access destination for which descriptor is placed.

Descriptor areas and DMA transfer areas

Figure 31.24 provides an overview of descriptor areas and DMA transfer areas which the DMAC accesses.

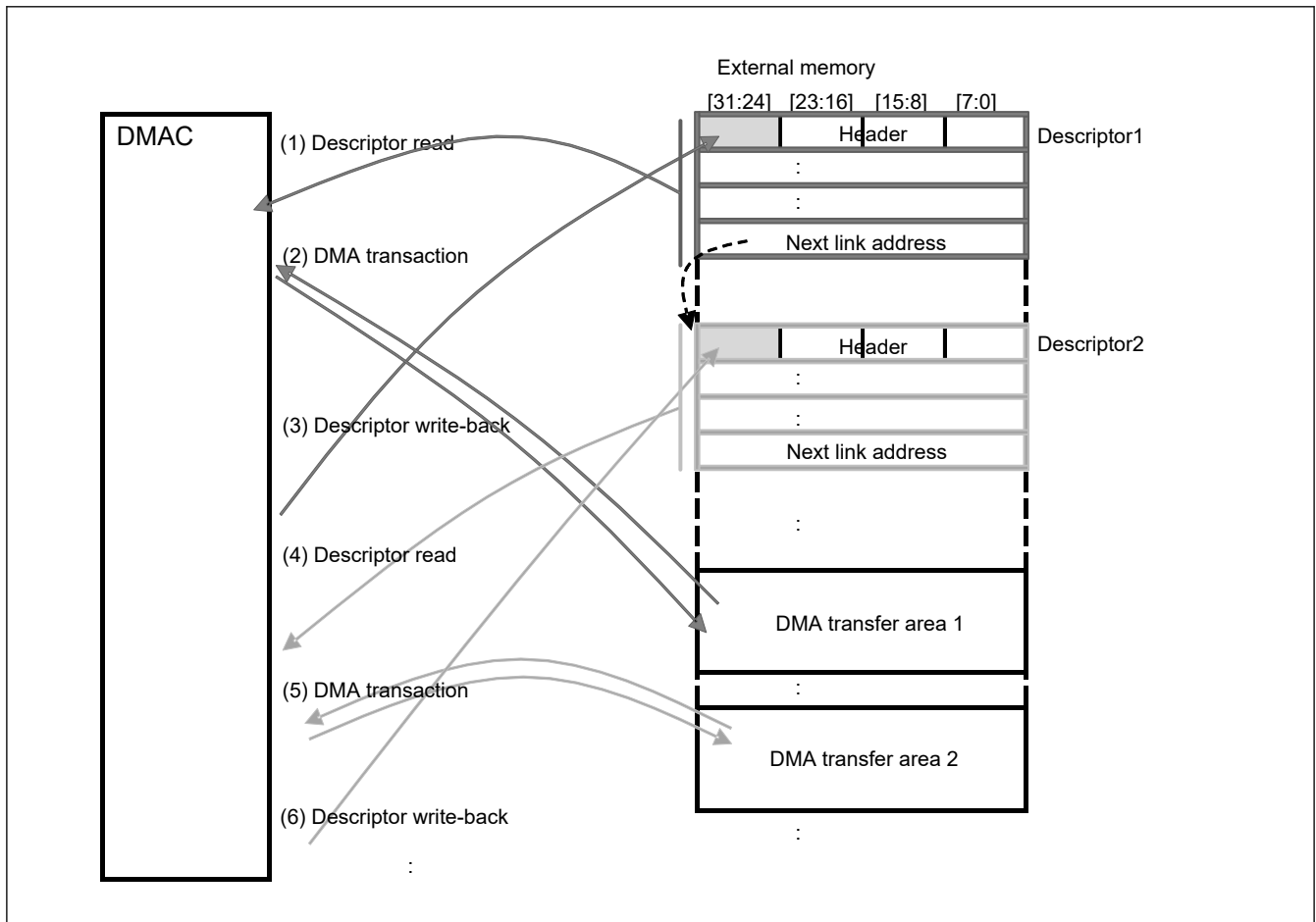


Figure 31.24 Overview of descriptor areas and DMA transfer areas

1. **Descriptor read**
Loads the value set for the on-chip NXLA_n register to the CRLA_n register, and then reads the descriptor from the external memory space (descriptor 1) indicated by the CRLA_n register.
2. **DMA transaction**
When the LV bit in the header of the descriptor is 1, performs a DMA transfer according to the descriptor information.
3. **Descriptor write-back**
If the WBD bit in the header is 0 after a DMA transfer for the set number of bytes, data (LV bit = 0 and the data that was read in step 1 in other fields) is written back to header[31:24] of descriptor 1 with a byte size.
4. **Descriptor read**
If the LE bit in the header of the descriptor that was read in step 1 is 0, reads the next descriptor from the address (descriptor 2) indicated by the next link address in the descriptor.
5. **DMA transaction**
When the LV bit in the header of the descriptor is 1, performs a DMA transfer according to the descriptor information.
6. **Descriptor write-back**
If the WBD bit in the header is 0 after a DMA transfer for the set number of bytes, data (LV bit = 0 and the data that was read in step 4 in other fields) is written back to header[31:24] of descriptor 2 with a byte size.

Hereafter, repeats steps 4 through 6.

When LE in the header is 1 and WBD = 0, a DMA transfer is performed with the descriptor settings, and 0 is written back to the LV bit in the header, and then operation ends.

When LE in the header is 1 and WBD = 1, a DMA transfer is performed with the descriptor settings, and then operation ends (without write-back).

When the LV bit in the header is 0, 1 is written back to the D bit in the header. After that, if the DRRP bit in the CHCFG_n register is 1, the descriptor is read again after the interval specified in the DITVL field in the DSCITVL_n register. When DRRP = 0, the operation stops.

Notes on descriptors

- In link mode, settings can be changed by reading a descriptor, but the timing for changing settings and for a hardware request cannot be synchronized. For this reason, when using a hardware request, set the AM, LVL, HIEN, LOEN, and SEL bits in the CHCFG_n register to 1 before setting the SETEN bit in the CHCTRL_n register. Besides, do not change these bits in a descriptor.
- In a descriptor, the settings of the DMS field in the CHCFG_n register cannot be changed (always link mode). In addition, settings of the REN, RSW, and RSEL fields in the CHCFG_n register can be changed in the descriptor, but that does not affect the operation.
- The DMAC references the DSCFM field and the LV bit in the header to determine if the descriptor is enabled or disabled. Therefore, initialize (DSCFM = 0x1 or 0x3 and LV = 1) the memory area equivalent to the DSCFM field and the LV bit of the descriptor before the DMAC accesses the memory area.
- If you want to set the next descriptor in the memory while the DMAC is operating, write 1 to the LV bit after setting the descriptors after header (source address, destination address, ... next link address). This is to avoid DMA transfer using descriptor values (source address, destination address, and so on) before the setting if descriptor settings by the CPU and descriptor read of the DMAC conflict and descriptor read by the DMAC interrupts descriptor settings by the CPU.
- If you want to keep the information written back to the D bit in the header, perform a byte access to write 1 to the LV bit in the header.

(4) LINK Configuration Example

In link mode, descriptors can be configured as shown below.

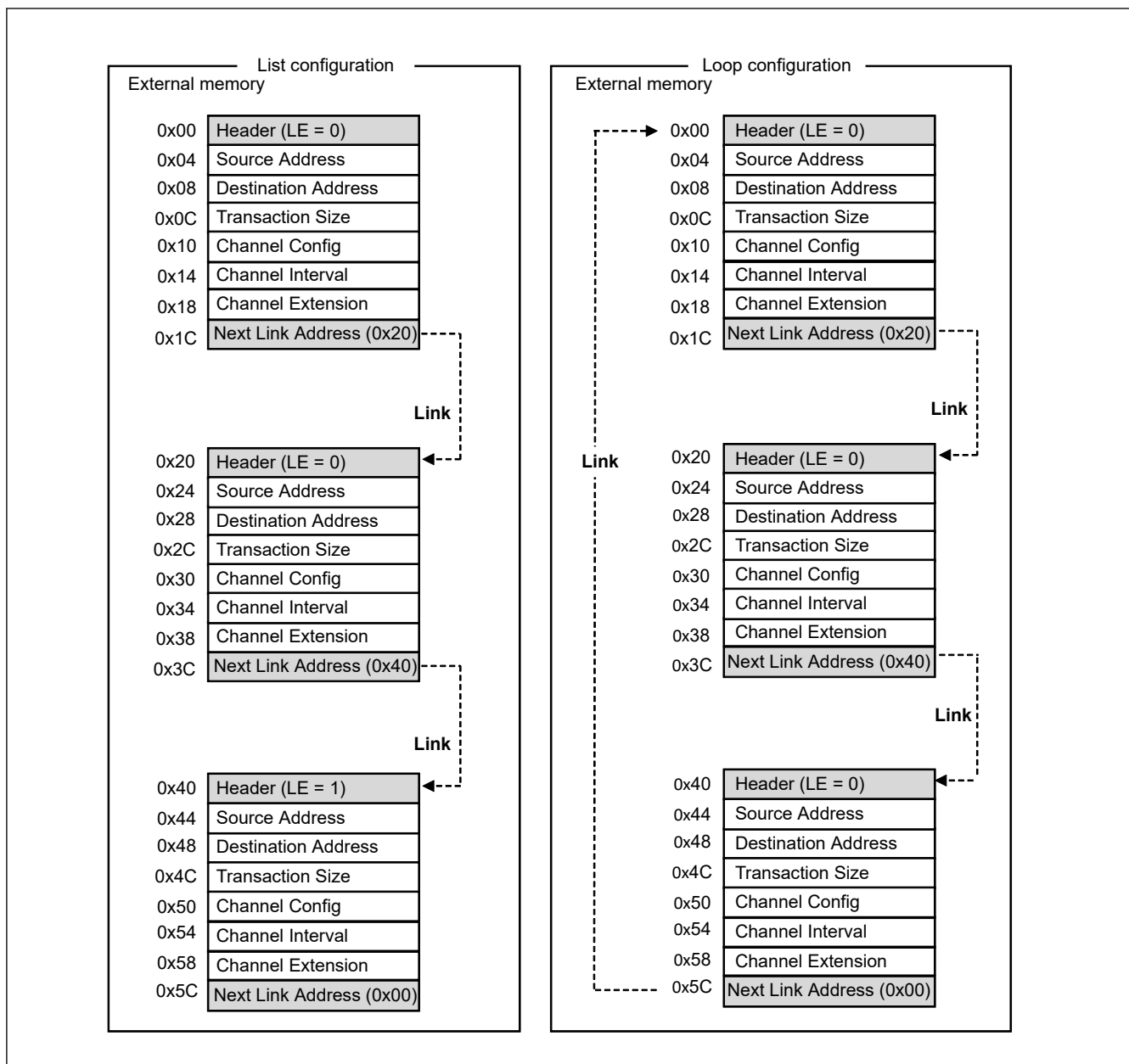


Figure 31.25 Link mode configuration example

List configuration

Writing 1 to the LE bit in the header of the last descriptor terminates the link.

Loop configuration

Setting the link destination of the last descriptor to the address of the previous descriptor configures descriptors in a loop. To terminate the loop, modify the LE bit in the header to 1 before the DMAC reads a descriptor, or follow the procedure to suspend a transfer.

31.14.1.3 Write-Only Mode

Setting the WONLY bit in the CHCFG_n register to 1 enables write-only mode.

Table 31.48 Write-only mode settings (1 of 2)

WONLY(CHCFG)	Mode	Description
0	Normal mode	Performs DMA transfers with the values set for the next register set.

Table 31.48 Write-only mode settings (2 of 2)

WONLY(CHCFG)	Mode	Description
1	Write-only mode	Performs DMA write transfers only without performing DMA read transfers.

In write-only mode, read operation for DMA transfers is not performed (descriptor read is performed in the same way as normal mode). In register mode, use the values set for the NxSA_n register (x = 0 when RSEL = 0, and x = 1 when RSEL = 1) as write data. In link mode, use the value in the SA field of a descriptor as write data.

Use this mode for initializing a memory area.

31.15 DMA Transfer

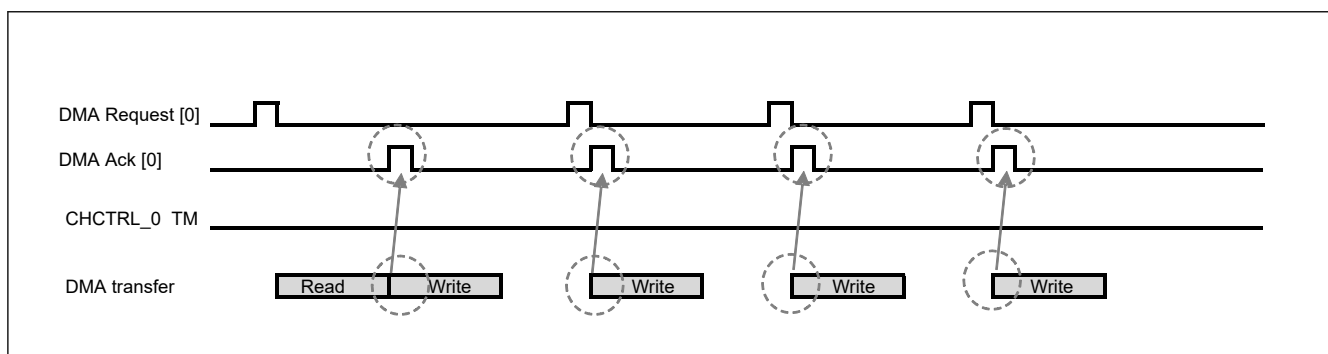
This section describes basic operation of DMA transfer.

31.15.1 Transfer Mode

Only single transfer mode is supported.

When a DMA transfer request from the USB control is accepted, a DMA transfer is performed on the side (transfer source or transfer destination) indicated by the REQD bit in the CHCFG_n register to assert the internal DMA acknowledge signal from the internal USB control to the DMAC control. Each time a transfer request is accepted, transfer proceeds. This operation is repeated until it reaches the transfer size loaded to the CRTB_n register. (Arbitration between channels is performed for each DMA transfer.)

The DMA acknowledge timing differs depending on the setting for the REQD bit in the CHCFG_n register or settings for the transfer size (DDS[2:0] and SDS[2:0] in the CHCFG_n register). For details, see [section 31.15.7. Differences in Operation According to the Transfer Data Size](#).

**Figure 31.26 Single transfer mode (REQD = 1, SDS > DDS)**

31.15.2 DMA Channel Priority Control

As an arbitration method between channels, fixed priority mode and round-robin mode are supported. To select a mode, use the PR bit in the DCTRL register. When the PR bit is 0, fixed priority mode is selected. When the PR bit is 1, round-robin mode is selected.

Table 31.49 Priority control settings

Mode	PR (DCTRL)	Description	Application
Fixed priority	0	Controls requests in fixed priority (high: CH0 > CH1: low) mode.	Use this mode if channels have priority.
Round-robin	1	Controls requests in round-robin mode.	Use this mode if you want to execute requests equally.

31.15.2.1 Fixed Priority Mode

In fixed priority mode, the priority among channels is fixed as shown below.

[High] CH0 > CH1 [Low]

If DMA transfer requests are generated in multiple channels simultaneously, a DMA transfer request with a smaller channel number takes precedence. The following figure provides an example where another DMA transfer request with higher priority is generated during a DMA transfer in fixed priority mode.

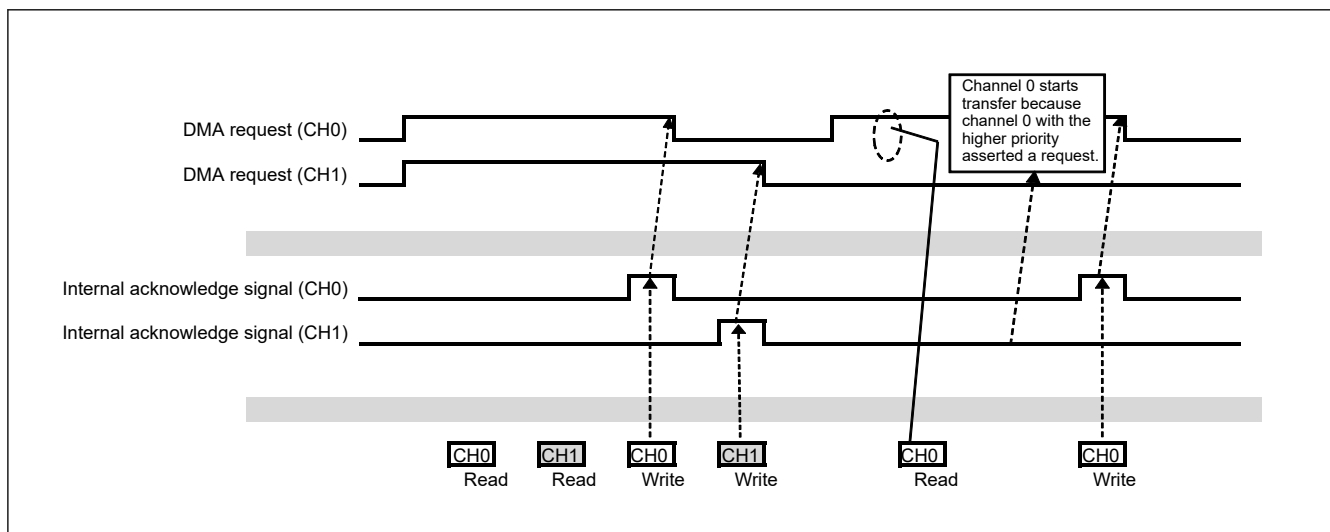


Figure 31.27 Fixed priority mode (four channels, REQD = 1)

Note: Channel 0 is handled with the highest priority, but the transfer with the next highest priority is performed to increase the bus usage rate while the transfer of channel 0 switches.

31.15.2.2 Round-Robin Mode

In round-robin mode, priority is changed each time a transfer of each channel is accepted so that the lowest priority is given to the channel in which the last transfer is performed.

The priority immediately after a reset is the same as the one in fixed priority mode as shown below.

[High] CH0 > CH1 [Low]

In this state, if no transfer request is generated in DMA channel 0, but a request is generated in DMA channel 1, a transfer of DMA channel 1 is performed. The priority after the transfer is shown below.

[High] CH1 > CH0 [Low]

The following figure provides an example of a DMA transfer in round-robin mode.

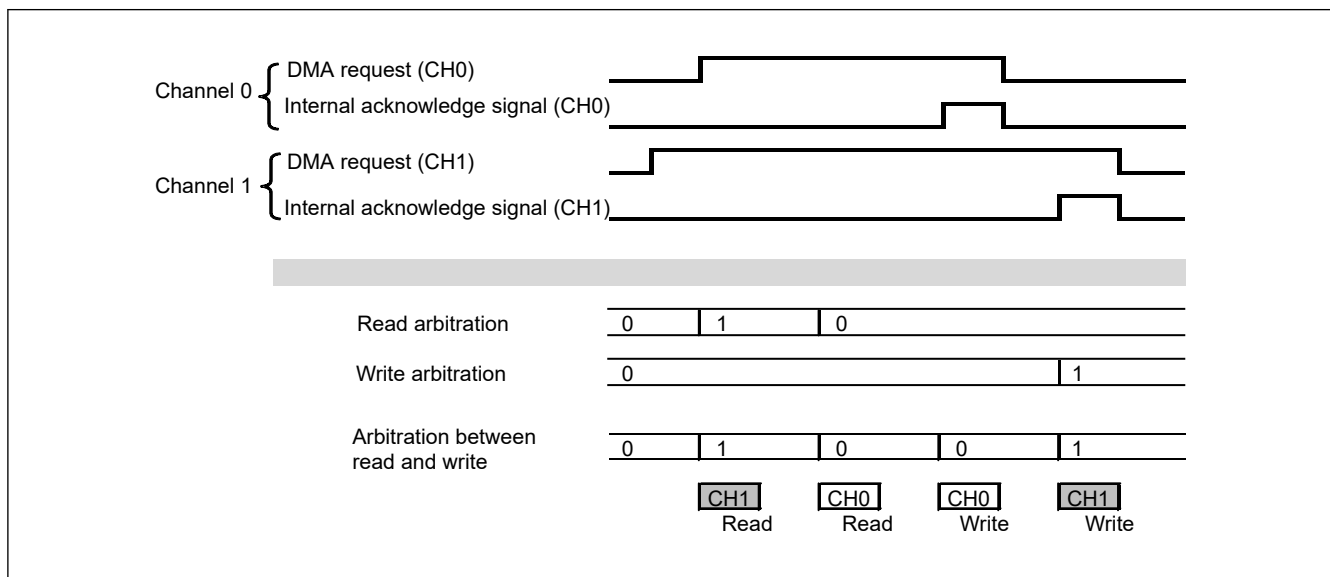


Figure 31.28 Round-robin mode (two channels, REQD = 1)

Note: In the DMAC, arbitration between read channels and arbitration between write channels are performed. As a result of these arbitrations, further arbitration is performed, and a bus access is issued.

31.15.3 Forced Ejection Request

When a forced ejection request is input, data that is not yet transferred in a buffer is transferred to the DMA transfer destination address. After data is flushed, the DMA transfer resumes.

The following describes notes on forced ejection requests:

- If a forced ejection request conflicts with a USB control DMA transfer request, the forced ejection takes precedence, and then the DMA transfer is performed.
- When the transfer destination is a system on the USB control side (the REQD bit in the CHCFG_n register = 1), the unit on the DMA transfer destination side receives data even if the USB control side does not request a DMA transfer. Therefore, a malfunction, such as buffer overflow, may occur. Because of this, when REQD = 1, no forced ejection is performed by hardware.
- Differences from flush mode described in (2) [Aborting a Transfer \(with Buffer Flush: SBE = 1\)](#) (The EN bit is cleared to 0 when the SBE bit in the CHCFG_n register = 1.)
 Flush mode: The DMAC stops operation after data in a buffer is written.
 Forced ejection request: A DMA transfer continues after flush operation ends.

31.15.3.1 Software Forced Ejection Request

For software forced ejection requests, use the SETSSWPRQ bit in the CHCTRL_n register. To make a forced ejection request, write 1 to the SETSSWPRQ bit. The DMAC outputs data in a buffer to the DMA transfer destination.

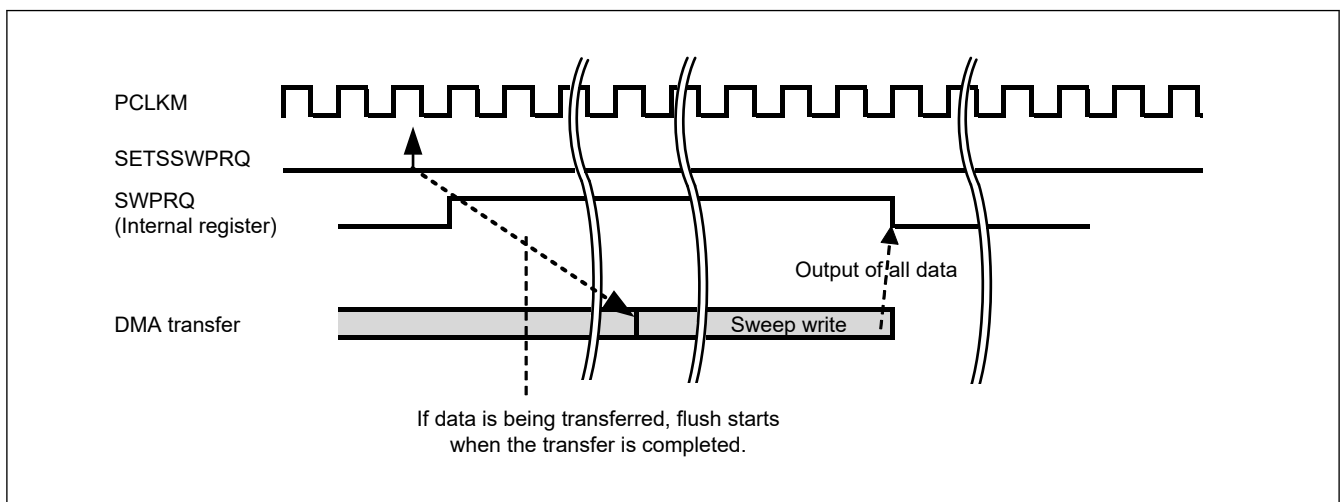


Figure 31.29 Software forced ejection timing

31.15.4 DMA Transfer Completion Interrupt (USB_FDMAn)

The USB_FDMAn[1:0] interrupt request signals indicate that a DMA transaction is completed.

Each bit of USB_FDMAn[1:0] corresponds to each channel.

When a transfer for the total number of transfer bytes loaded to the CRTB_n register is completed with an OKAY response, the END bit in the CHSTAT_n register is set to 1. If the DEM bit in the CHCFG_n register = 0 at this time, USB_FDMAn is asserted to high level (n = 1, 0). (When write-back is performed in link mode, this pin is asserted after the write-back operation.)

In addition, when the DRRP bit in the CHCFG_n register = 0 in link mode and the header of the read descriptor is LV = 0, the DER bit in the CHSTAT_n register is set to 1. If the DIM bit in the CHCFG_n register = 0 at this time, USB_FDMAn is asserted to high level.

Use these signals to detect a transfer completion interrupt in the interrupt controller.

Table 31.50 USB_FDMA_n assert conditions

Source	Condition	USB_FDMA _n mask signal
DMA transaction completed	When a transfer for the total number of transfer bytes loaded to the CRTB _n register is completed with an OKAY response (after write-back when write-back is performed in link mode)	DEM bit in the CHCFG _n register
Descriptor invalid	When the header of the read descriptor is LV = 0 while DRRP and DIM in the CHCFG _n register are 0 in link mode	DIM bit in the CHCFG _n register

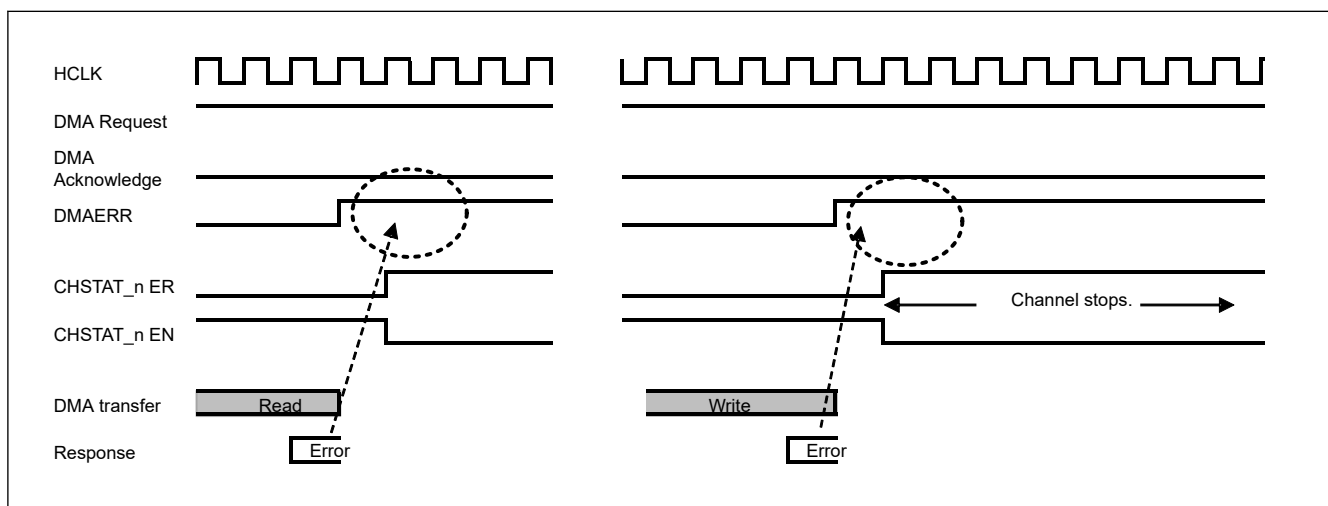
31.15.5 DMA Error Interrupt (USB_FDMAERR)

If an error response is received during a DMA transfer or descriptor access, this module determines it as an error and aborts the transfer. When an error response is received, the EN bit in the CHSTAT_n register for channel *n* where a transfer is in progress is cleared to 0 and the ER bit is set to 1 (*n* = 1, 0). In addition, high level is output from the USB_FDMAERR pin.

The USB_FDMAERR signal cannot be masked.

Data for a series of error transfers is not guaranteed. Be sure to perform the following procedure to retry the transfer from the beginning.

1. Set the SWRST bit in the CHCTRL_n register to 1.
2. Set each register again.

**Figure 31.30 Transfer stop timing due to an error response (ERROR)**

Note: The USB_FDMAERR pin output is not asserted even if an error occurs during an access to the slave of this controller.

31.15.6 Interval Count Function

By the setting for the ITVL field in the CHITVL_n register, the execution interval of DMA transfers can be adjusted. This function is used to avoid occupation of a bus by the DMAC. Until the count value becomes 0, no DMA transfer for the next DMA request is performed.

The following figure provides an operation example.

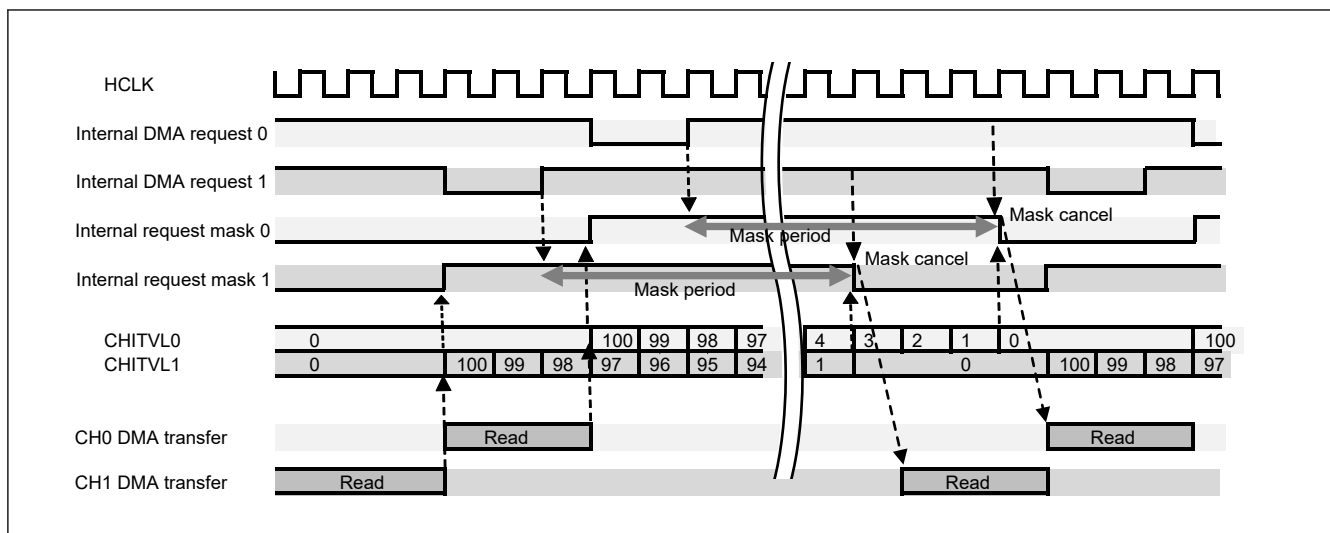


Figure 31.31 Interval count (REQD = 0, SDS < DDS)

An interval is inserted after a transfer specified by the REQD bit in the CHCFG_n register.

The following figure shows the relationship between the setting values for the REQD, SDS, and DDS bits in the CHCFG_n register and the cycles applied to intervals.

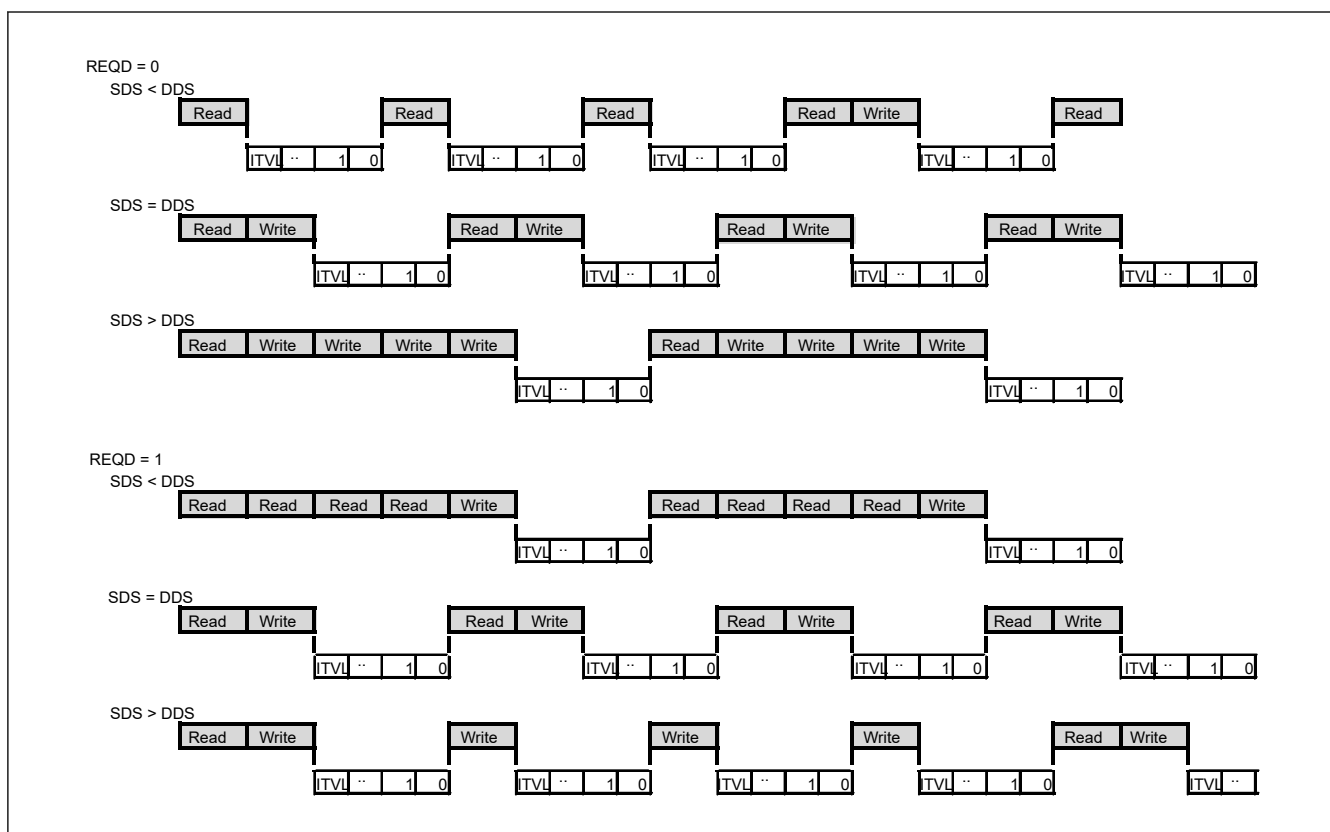


Figure 31.32 DMA transfer settings and interval counts

31.15.7 Differences in Operation According to the Transfer Data Size

31.15.7.1 When the Transfer Data Size on the Transfer Source Side is Small

When read operation for the transfer data size on the transfer destination side is completed, write operation to the transfer destination starts.

The following figure shows the timing chart (rising edge detection) when the transfer source data size is 8 bits and the transfer destination data size is 32 bits (SDS = 0 and DDS = 2 in the CHCFG_n register).

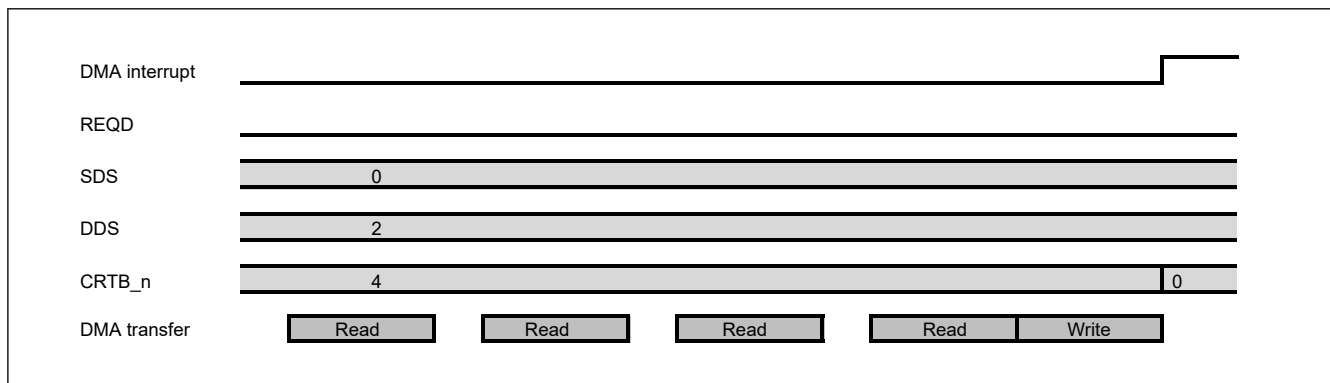


Figure 31.33 When the data size on the transfer source size is small (LVL = 0, HIEN = 1, REQD = 0, SDS < DDS in CHCFG_n)

31.15.7.2 When the Transfer Data Size on the Transfer Destination is Small

Because the transfer data size on the transfer source side is large, after a single read operation of the transfer source, write operation to the transfer destination is performed a few times. The following figure shows the timing chart (rising edge detection) when the transfer source data size is 64 bits and the transfer destination data size is 16 bits (SDS = 3 and DDS = 1 in the CHCFG_n register).

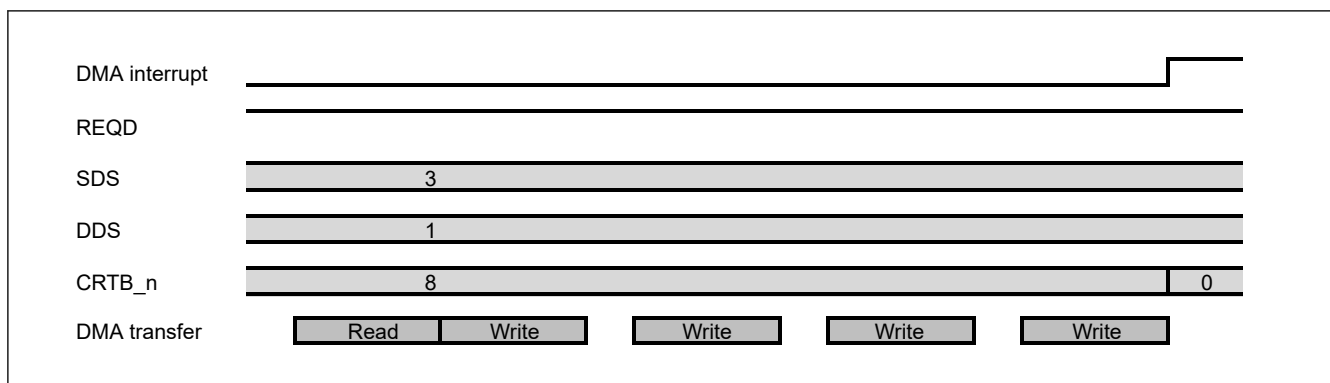


Figure 31.34 When the data size on the transfer destination side is small (LVL = 0, HIEN = 1, REQD = 1, SDS > DDS in CHCFG_n)

31.15.7.3 When Transfer Data Sizes on the Transfer Source Side and the Transfer Destination Side are Equal

Each time a DMA transfer request is detected, read operation is performed on the transfer source side and write operation is performed on the transfer destination side. The following figure shows the timing chart (rising edge detection) when the transfer source data size and the transfer destination data size are 8 bits (SDS = 0 and DDS = 0 in the CHCFG_n register).

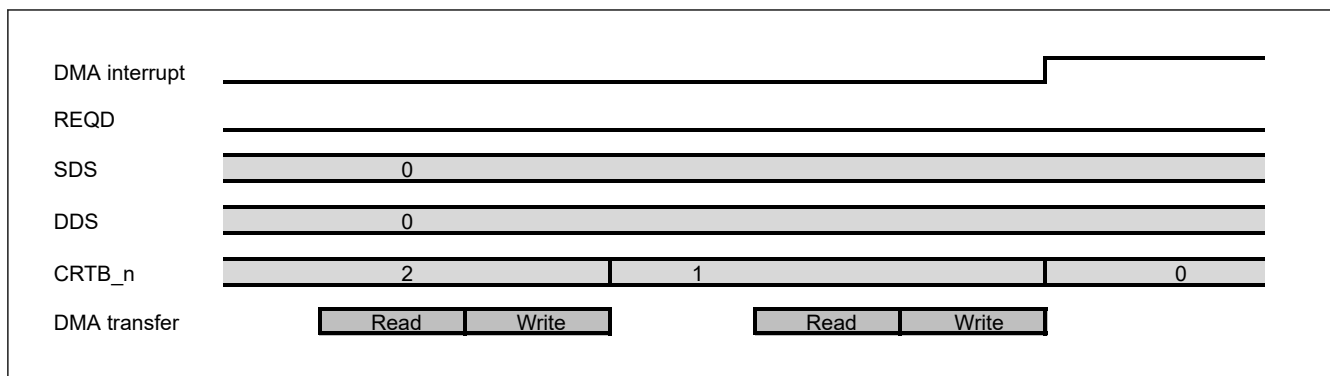


Figure 31.35 When transfer data sizes on the transfer source side and the transfer destination side are equal (LVL = 0, HIEN = 1, REQD = 0, and SDS = DDS in CHCFG_n)

31.15.8 Transfer Status

The CHSTAT_n register indicates the transfer status of each channel.

31.15.8.1 Transfer Status

The TACT bit in the CHSTAT_n register indicates that channel n is operating. Writing 1 to the SETEN bit in the CHCTRL_n register sets 1. The TACT bit remains 1 during an access to a descriptor or waiting for a DMA request.

The TACT bit is cleared when the EN bit in the CHSTAT_n register is cleared (for details on clearing conditions, see [section 31.2.17.1. CHSTAT_n : Channel Status Register n \(n = 0, 1\)](#)), and the ongoing DMA transfer is completed.

The TACT bit is not cleared when the EN bit is not cleared (when the REN bit in the CHCFG_n register = 1 in register mode or the next descriptor access is performed in link mode) even if a DMA transaction finishes.

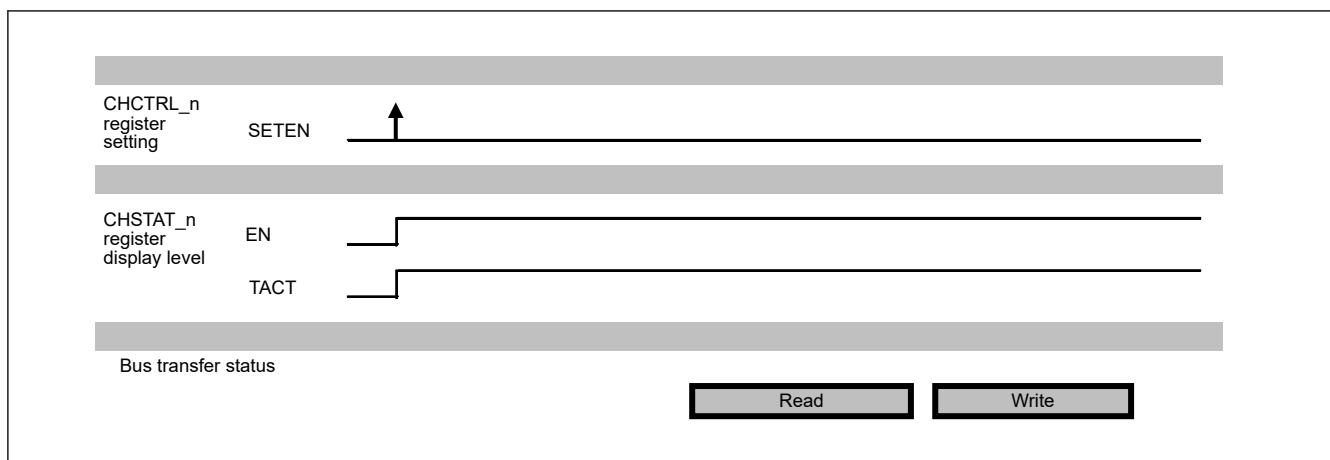


Figure 31.36 DMAC status example 1 (hardware request)

31.15.8.2 Suspending a Transfer

A DMA transfer can be suspended using the SETSUS bit in the CHCTRL_n register. At that time, if there is an active bus cycle, the DMA transfer is suspended after the bus cycle finishes. Writing 1 to the CLRSUS bit in the CHCTRL_n register resumes from the suspended state.

To check whether the transfer is suspended, set the SETSUS bit in the CHCTRL_n register, and then make sure that the SUS bit in the CHSTAT_n register or the SUS bit in the DSTAT_SUS register of the applicable channel is set to 1.

31.15.8.3 Aborting a Transfer

If 1 is written to the CLREN bit in the CHCTRL_n register during a DMA transaction, the DMA transaction of the channel can be aborted. As processing after aborting the transaction, you can use the SBE bit in the CHCFG_n register to select a mode to flush data remaining in a buffer when the transaction is aborted or a mode that does not flush buffer data. SBE = 0 (mode not to flush data) is selected by default.

When this mode (flush data) is enabled, if an ongoing transfer is aborted when the CLREN bit in the CHCTRL_n register = 1, data remaining in the buffer of the DMAC is flushed and the operation stops.

(1) Aborting a Transfer (without Buffer Flush: SBE = 0)

If 1 is written to the CLREN bit in the CHCTRL_n register during a DMA transfer, the DMA transfer is aborted and then stopped. The timing to stop the transfer depends on the set REQD bit value. After the transfer stops, write 1 to the SWRST bit in the CHCTRL_n register to clear the contents of the DMAC, and then make settings for the next transfer.

- When the TACT bit in the CHSTAT_n register is cleared to 0, you can confirm that the channel has completely stopped.
- If a DMA transfer is aborted, USB_FDMAn is not asserted.
- When the REQD bit in the CHCFG_n register = 0, the transfer stops when the next read operation is completed. (However, if data that can be written is remaining in the buffer, the data is written and then the transfer stops.)
- When the REQD bit in the CHCFG_n register = 1, the transfer stops when the next write operation is completed.

(2) Aborting a Transfer (with Buffer Flush: SBE = 1)

If 1 is written to the CLREN bit in the CHCTRL_n register during a DMA transfer, the DMA transfer is aborted. When the REQD bit in the CHCFG_n register = 0, already read data is flushed (written), and then the DMA transfer stops. When REQD = 1, flush mode cannot be used by hardware.

After the transfer stops, set the SWRST bit in the CHCTRL_n register to clear the contents of the DMAC, and then make settings for the next transfer.

- When the TACT bit in the CHSTAT_n register is cleared to 0, you can confirm that the channel has completely stopped.

(3) Checking Channel Stop

Even if 1 is written to the CLREN bit in the CHCTRL_n register, and the EN bit in the CHSTAT_n register is cleared to 0, if a transfer is already in progress on the bus, the DMAC cannot stop the transfer immediately. To check that DMA has completely stopped, make sure that the EN bit is 0 and the TACT bit in the CHSTAT_n register is 0.

(4) Transfer Aborting Procedure

The following describes the procedure for aborting a transfer:

1. Write 1 to the CLREN bit in the CHCTRL_n register.
2. When the SBE bit in the CHCFG_n register = 0, the transfer stops according to the REQD bit value in the CHCFG_n register. If SBE = 1 at this time, flush mode is enabled.
3. Read the CHSTAT_n register and confirm that the TACT bit is 0. When TACT = 0, DMA has completely stopped. When TACT = 1, continue polling until the TACT bit is cleared to 0.
4. If you want to perform the next DMA transfer after a transfer is aborted, be sure to set the SWRST (software reset) bit in the CHCTRL_n register to 1 before the next transfer starts.

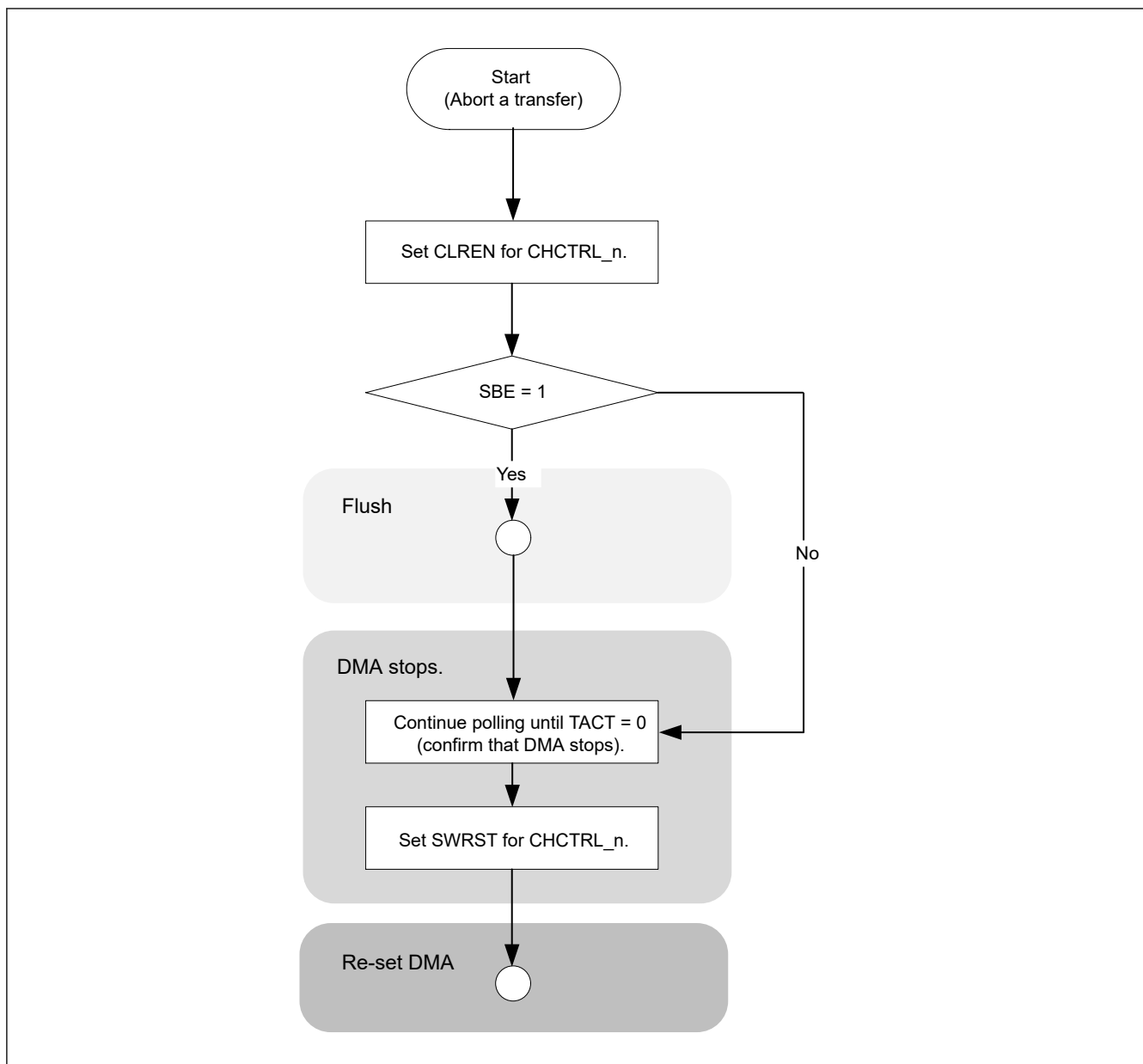


Figure 31.37 Transfer aborting flow

31.16 Access Type

31.16.1 AHB Master DMA Transfer Combinations

31.16.1.1 Read Access

The DMA read access issuance type has the following features.

- Read access is made to a beat-aligned space (including the source address indicated in the CRSA_n register) with the size specified by the SDS[2:0] bits in the CHCFG_n register. Beat-unaligned transfer is not performed for the bus. An unnecessary area may be read depending on the set values of the CRSA_n register and each SKIP register. In this case, necessary data in the read data is loaded to a buffer.
- The size and the burst length are determined by the value set for the SDS[2:0] field.
When the value set for the SDS[2:0] field is equal to or smaller than the bus width:

Size: SDS[2:0] set value

Burst type: Single

When the value set for the SDS[2:0] field is larger than the bus width:

Size: Bus width

Burst type: Fixed-length burst (burst length = SDS[2:0] set value/bus width)

The following table shows AHB bus access types.

Table 31.51 DMA read transfer combinations

SDS	Source address	AHB transfer					
		First transfer			Second transfer		
		Address	Size	Burst	Address	Size	Burst
0 (8 bits)	—	addr	8	Single	—	—	—
1 (16 bits)	2-byte align	{addr[31:1], 0b}	16	Single	—	—	—
	2-byte unalign				{addr[31:1], 0b} + 0x2	16	Single
2 (32 bits)	4-byte align	{addr[31:2], 00b}	32	Single	—	—	—
	4-byte unalign				{addr[31:2], 00b} + 0x4	32	Single
4 (128 bits)	16-byte align	{addr[31:4], 0x0}	32	INCR4	—	—	—
	16 bytes				{addr[31:4], 0x0} + 0x10	32	INCR4
5 (256 bits)	32-byte align	{addr[31:5], 0x00}	32	INCR8	—	—	—
	32 bytes				{addr[31:5], 0x00} + 0x20	32	INCR8
6 (512 bits)	64-byte align	{addr[31:6], 0x00}	32	INCR16	—	—	—
	64 bytes				{addr[31:6], 0x00} + 0x40	32	INCR16

Note: If the transfer becomes EBT during the burst transfer, remaining transfer is performed by 32-bit INCR.

31.16.1.2 Write Access

The DMA write access issuance type has the following features.

- Write access is made from the destination address (indicated in the CRDA_n register) to a beat-aligned boundary with the size specified by the DDS[2:0] bits in the CHCFG_n register.
- The size and the burst length are determined by the value set for the DDS[2:0] field.
When the value set for the DDS[2:0] field is equal to or smaller than the bus width:

Size: DDS[2:0] set value

Burst type: Single

When the value set for the DDS[2:0] field is larger than the bus width:

Size: Bus width

Burst type: Fixed-length burst (burst length = DDS[2:0] set value/bus width)

- Write access to any space other than the specified space is not performed. In the following cases, write access is made with a combination of sizes smaller than the value set for the DDS[2:0] field.
 - When the destination address is beat-unaligned with the value set for the DDS[2:0] field
 - When an access with the DDS field size is made over the SKIP boundary
 - When the DDS[2:0] size is larger than the remaining number of transfer bytes

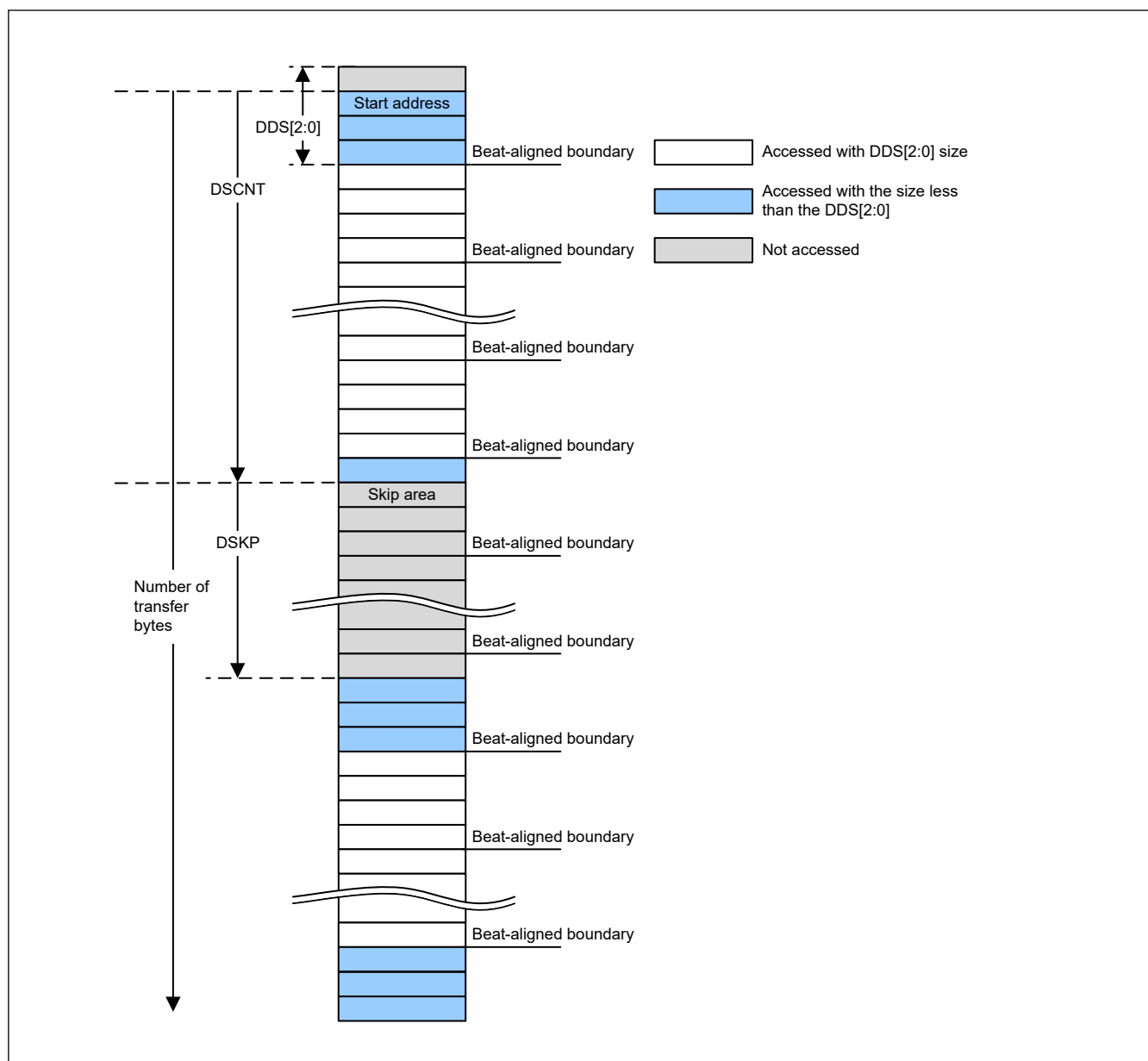


Figure 31.38 Example of DMA write access space and access type

32. Serial Communications Interface (SCI)

32.1 Overview

This LSI has six independent serial communications interface (SCI) channels. The SCI is configurable to five asynchronous and synchronous serial interfaces:

- Asynchronous interfaces (UART and Asynchronous Communications Interface Adapter (ACIA))
- 8-bit clock synchronous interface
- Simple I2C (master-only)
- Simple SPI
- Smart card interface

The smart card interface complies with the ISO/IEC 7816-3 standard for electronic signals and transmission protocol. Each SCI has FIFO buffers to enable continuous and full-duplex communication, and the data transfer speed can be configured independently using a baud rate generator.

[Table 32.1](#) lists the SCI specifications, [Figure 32.1](#) shows a block diagram of SCI channel n, and [Table 32.2](#) lists the I/O pins.

Table 32.1 SCI specifications (1 of 3)

Item	Specifications
Number of channels	6 channels (ch0 to ch4 in NONSAFETY and ch5 in SAFETY)
Serial communication modes	• Asynchronous • Clock synchronous • Smart card interface • Simple I2C • Simple SPI
Transfer speed	Bit rate specifiable with the baud rate generator
Full-duplex communications	• Transmitter: Continuous transmission possible using double-buffering • Receiver: Continuous reception possible using double-buffering
Half-duplex communication	Possible by using TXD pins
Data transfer	Selectable as LSB-first or MSB-first transfer
RXD/TXD polarity	Selectable polarity for TXD and RXD independently
Interrupt sources	• Transmit end, transmit data empty, receive data full, receive error, receive data ready, and address match • Completion of generation of a start condition, restart condition, or stop condition (for simple I2C mode)
RS-485 driver control function	Output DE pin to enable transmission mode of external transceiver
Loopback function	Self-diagnosis of communication function inside module supported
Synchronizing circuit bypass function	Bus clock (PCLKM) can be used as operation clock (PCLKSCIn) when bypass function is enabled.
Module-stop function	Module-stop state can be set for each channel